

SEMESTER 3

SAFETY & FIRE ENGINEERING

SEMESTER S3

MATHEMATICS FOR ELECTRICAL SCIENCE AND PHYSICAL SCIENCE – 3

(Common to B & C Groups)

Course Code	GYMAT301	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic knowledge in complex numbers.	Course Type	Theory

Course Objectives:

1. To introduce the concept and applications of Fourier transforms in various engineering fields.
2. To introduce the basic theory of functions of a complex variable, including residue integration and conformal transformation, and their applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fourier Integral, From Fourier series to Fourier Integral, Fourier Cosine and Sine integrals, Fourier Cosine and Sine Transform, Linearity, Transforms of Derivatives, Fourier Transform and its inverse, Linearity, Transforms of Derivative. (Text 1: Relevant topics from sections 11.7, 11.8, 11.9)	9
2	Complex Function, Limit, Continuity, Derivative, Analytic functions, Cauchy-Riemann Equations (without proof), Laplace's Equations, Harmonic functions, Finding harmonic conjugate, Conformal mapping, Mappings of $w = z^2$, $w = e^z$, $w = \frac{1}{z}$, $w = \sin z$. (Text 1: Relevant topics from sections 13.3, 13.4, 17.1, 17.2, 17.4)	9

3	Complex Integration: Line integrals in the complex plane (Definition & Basic properties), First evaluation method, Second evaluation method, Cauchy's integral theorem (without proof) on simply connected domain, Independence of path, Cauchy integral theorem on multiply connected domain (without proof), Cauchy Integral formula (without proof). (Text 1: Relevant topics from sections 14.1, 14.2, 14.3)	9
4	Taylor series and Maclaurin series, Laurent series (without proof), Singularities and Zeros – Isolated Singularity, Poles, Essential Singularities, Removable singularities, Zeros of Analytic functions – Poles and Zeros, Formulas for Residues, Residue theorem (without proof), Residue Integration- Integral of Rational Functions of $\cos\theta$ and $\sin\theta$. (Text 1: Relevant topics from sections 15.4, 16.1, 16.2, 16.3, 16.4)	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Determine the Fourier transforms of functions and apply them to solve problems arising in engineering.	K3
CO2	Understand the analyticity of complex functions and apply it in conformal mapping.	K3
CO3	Compute complex integrals using Cauchy's integral theorem and Cauchy's integral formula.	K3
CO4	Understand the series expansion of complex function about a singularity and apply residue theorem to compute real integrals.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	2	-	-	-	-	-	-	-	2
CO2	3	3	-	2	-	-	-	-	-	-	-	2
CO3	3	3	-	2	-	-	-	-	-	-	-	2
CO4	3	3	-	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Engineering Mathematics	Erwin Kreyszig	John Wiley & Sons	10 th edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Complex Analysis	Dennis G. Zill, Patrick D. Shanahan	Jones & Bartlett	3 rd edition, 2015
2	Higher Engineering Mathematics	B. V. Ramana	McGraw-Hill Education	39 th edition, 2023
3	Higher Engineering Mathematics	B.S. Grewal	Khanna Publishers	44 th edition, 2018
4	Fast Fourier Transform - Algorithms and Applications	K.R. Rao, Do Nyeon Kim, Jae Jeong Hwang	Springer	1 st edition, 2011

SEMESTER S3

FIRE ENGINEERING FUNDAMENTALS

Course Code	PCSFT302	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To get understanding about the fundamentals of fire phenomena, fire-fighting, various fire-fighting tools and equipment's.
2. To provide the students an illustration of significance of the Fire Engineering profession in the protection of life, property and environment.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction- temperature, heat, specific heat, flash point, fire point, ignition, combustion; Ignition- pilot ignition, spontaneous ignition, ignition sources; Types of combustion-rapid, spontaneous, physical explosion, chemical explosion; Smoldering combustion; Product of combustion harmful to human-flame, heat, smoke, fire gases; Special kinds of combustion- Flash fire, Pool fire, Deep seated fire, Spillover, Boil over, Dust explosion, BLEVE, VCE; Classification of fire based on material. Development of enclosure fire in terms of temperature-incipient, smoldering, flame and heat stages; factors influencing fire development in an enclosure. Diffusion flames-Zones of combustion, characteristics of diffusion flames , laminar and turbulent jet flames, flame from natural fires, buoyant plume; Premixed flames-structure, limits of flammability, burning velocity, explosion and expansion ratios, deflagration and detonation, characteristics of premixed flame; Spread of flames over liquid and solid surfaces.	11

2	Heat transfer in compartment fires- Fuels, Calorific value, fire load, heat release rate and factors controlling heat release rate, basic heat transfer equations, Convective heat transfer in fire, Radiative heat transfer in fire; Radiation from hot gases and non-luminous flames; Radiation from luminous flames and hot smoky gases. Fire plumes and flame heights- flame characteristics, mean flame height, flame height correlations, turbulent fire plume characteristics, ideal plume, line plume, bounded plume, ceiling jets, response of ceiling mounted fire detectors, interaction between sprinkler spray and the fire plume.	11
3	Use, operation and maintenance of fire service equipments and accessories- Suction and delivery Hose, Hose reel, Hose fittings-coupling, adapters, branches, branch holders, radial branches, collecting heads, stand pipe, monitors, hydrants; Introduction to fire fighting vehicles and appliances- crash tenders, rescue tenders, hose laying tenders, control vans, hydraulic platforms; Ladders- extension ladders, hook ladder, turntable ladders, snorkel; Fire pumps and primers; Uses and maintenance of small gear and miscellaneous equipments used during fire fighting; Lamps and lighting sets; Ropes and Lines- wire and rope lines used in fire service, use and testing of lines, knots, bends and hitches and general rope work.	11
4	Fire Hydraulics- path and range of fire stream, types of nozzles and calculation of discharge capacity, nozzle reaction; Pressure loss or gain because of elevation; back pressure; Flow in pipes and fire hoses-hydraulic and energy grade lines, friction loss in pipes, fire hoses and fixtures, parallel and series connection of fire hoses, branching lines; Water relay techniques; Estimation of fire protection water requirements, pump capacity and other parameters relating to fire hydraulics. Fire ground operations - preplanning, action on arrival and control, methods of rescue, methods of entry. Personnel safety. Control procedure and use of other safety equipment. Ventilation and salvage operations.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the theory of fire chemistry and the different kinds of combustion and their characteristics	K1,K2
CO2	Summarize the characteristics of diffusion and premixed flame	K1
CO3	Explain the theory of heat transfer mechanism in compartment fires.	K2
CO4	Describe the use and operation of fire service equipment, machineries and accessories.	K2,K3
CO5	Calculate the water requirement and the pump capacity for fire fighting	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3					2						
CO4	3					2						
CO5	3	2										

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Hand book of fire technology.	Gupta, R.S	Universities Press Pvt. Ltd., New Delhi .(2010).	2 nd (2010)

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Under downs practical fire precautions		Ron Hirst Gower publishing company Ltd England.	(1989).
2	Fire hydraulics.		Andrew L Simon, Prentice-Hall Inc., USA	(1983)
3	Enclosure Fire Dynamics.		Bjorn Karlsson and James G. Quintiere, CRC Press, USA	(2000).
4	An Introduction to Fire Dynamics,		Dougal Drysdale, John Wiley & Sons Ltd., England.	2 nd edn.1999
5	Fire safety in buildings		Jain V.K. (New Age International (P) Ltd., New Delhi.	2 nd edn.2010

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
2	https://youtu.be/0LNklcBhl_Q

SEMESTER S3

FLUID MECHANICS AND HYDRAULIC MACHINES

Course Code	PCSFT303	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-1-0-0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To establish fundamental knowledge of basic fluid mechanics and its simple applications.
2. To familiarize students with the relevance of turbomachines and find solutions to the associated engineering problems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Prerequisite: Properties of fluid: Specific gravity, Specific Weight, Specific Volume, Dynamic and Kinematic Viscosity Introduction to fluid mechanics - Types of fluids, Newton's law of viscosity Pressure Measurement: Fluid pressure, Pressure head, types of pressures. Piezometer, Simple, differential Manometers. Fluid statics: Pressure, density, height relationship. Hydrostatic force and pressure on plane and inclined surfaces, Centre of pressure. Buoyancy and Metacentre. Stability of immersed and floating bodies.	8
2	Fluid kinematics: Description of fluid motion – Types of flows, Material derivative velocity and acceleration – Streamlines, path lines and streak lines, Stream function and velocity potential function, flow net Fluid dynamics: Continuity equation, Euler's, and Bernoulli's equations. – Measuring instruments – Pitot tube, Orificemeter, Venturimeter, Rectangular and Triangular Notches-(notches Problems not required).	8

3	Pipe flow – laminar and turbulent flows, significance of Reynolds number, shear stress and velocity distribution in a pipe flow.– Hagen-Poiseuille equation, Darcy-Weisbach equation and Chezy's equation , Moody's chart for estimating frictional losses, Major and minor energy losses, hydraulic gradient, and total energy line. Navier-Stokes equation and explanation (without proof) Dimensional analysis using Buckingham's π theorem. Boundary layer theory: Qualitative comparison between laminar and turbulent boundary layer. Boundary layer separation.	8
4	Impact of jets: Impact of jet on fixed vertical, moving vertical flat plates. Impact of jet on curved vanes – fixed and moving. Velocity triangles. Classification of Turbines and pumps Comparison and examples. Pelton, Francis and Kaplan Turbines: Principle and working, head, work done, efficiencies (Problems using velocity triangles not required) Centrifugal Pumps: Principle and working, head, work required, efficiencies, Priming and cavitation.(Problems using velocity triangles not required) Reciprocating Pump: Principle and working – slip, negative slip, work required and efficiency.	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify the fundamental fluid properties, their relationships and apply them to estimate the fluid pressure and hydrostatic forces on bodies.	K3
CO2	Classify the fluid flow and apply the principles kinematics and dynamics using the conservation of mass and momentum equations.	K3
CO3	Analyse viscous flow through pipes and estimate the major and minor losses associated with piping network.	K3
CO4	Understand the basic concept of dimensional analysis	K3
CO5	Select suitable turbo machine for specific application by identifying the pertinent parameters.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	-	-	-	-	-	-	-	-	-	-
CO2	3	3	2	-	-	-	-	-	-	-	-	-
CO3	3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	2										
CO5	2	2	-	-	-	2	-	-	-	-	-	

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fluid Mechanics	Cengel Y. A. and J. M. Cimbala	Tata McGraw Hill	2013
2	Introduction to Fluid Mechanics and Fluid Machines	Som S K	McGraw Hill Education India	2011
3	Fluid Mechanics and Hydraulic Machines	Bansal R K	Laxmi Publications	2005

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fluid Mechanics	White F.M.	Tata McGraw Hill	2003
2	Engineering applications of fluid dynamics	Fisher and Henley	Willford Press	2023

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Fluid Statics - https://www.youtube.com/watch?v=rY7bvZn75Do&list=PLwdnzlV3ogoWrAmpEcsPXayfsXnFfYY1O&index=4 Bouyancy, Metacentre and stability - https://www.youtube.com/watch?v=gMuucNxc7eI&list=PLwdnzlV3ogoV-ATGY2ptuLS9mwLFOJoDw&index=7&pp=iAQB
2	Fluid kinematics - https://www.youtube.com/watch?v=rY7bvZn75Do&list=PLwdnzlV3ogoWrAmpEcsPXayfsXnFfYY1O&index=4
3	Internal viscous flow - https://www.youtube.com/watch?v=qLx7ip0eBps&list=PLCoE5wxWtHFYiVGswvsWRaHjv18vxZzE2&index=17
4	Introduction to turbo machines - https://www.youtube.com/watch?v=ocVzrn4DLj8&list=PLbMVogVj5nJQQp3QLuzbcHrt0XncZZTiE&index=2

SEMESTER S3

CHEMICAL PROCESS PRINCIPLES

Course Code	PBSFT304	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To teach various techniques used for Stoichiometry
2. To learn thermodynamic principles and calculations
3. To learn the particle technology and related equipments.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to chemical engineering, basic chemical calculations-mole concept, methods of expressing composition-mole fraction, weight fraction, volume fraction, concentration of liquid solutions- molarity, molality, normality, ppm. Ideal gases and gas mixtures- ideal gas law, Amagat's law, Dalton's law, Henry's law, average molecular weight, density of gases, partial pressure and partial volume calculations. Material balance involving chemical reactions and not involving chemical reactions, simple calculations involving recycle, bypass and purge streams	9
2	Energy balance- heat capacity, specific heat and enthalpy, heat capacity of gases at constant pressure, heat capacity of gaseous mixtures, latent heats, enthalpy changes accompanying chemical reactions- standard heat of formation and standard heat of combustion, standard heat of reaction	9

3	Chemical thermodynamics, fundamental concepts and definitions- types of thermodynamic systems and properties- closed, open and isolated system- intensive and extensive properties- path and state functions, first law of thermodynamics, second law of thermodynamics, entropy, change in entropy, Maxwell relations, heat capacity in terms of entropy, equation of state of gases, the principle of corresponding states, compression and expansion of fluids – Joule Thomson expansion. Gibbs free energy change, equilibrium constant, effect of temperature on equilibrium constant.	9
4	Solids: Properties of solids, screening, screening equipment, effectiveness of screens, sieve analysis, average diameter and specific surface. Size reduction, types of equipment used in the various stages of reductions. Laws of crushing & grinding power requirements. Belt conveyer, bucket conveyer, pneumatic conveyers. Capacity and power requirements of conveyers. Flow of solids through fluids, terminal settling velocity & hindered settling. Laboratory batch sedimentation, Kynch theory, calculation of area and depth for continuous thickeners. Filtration: Filtration theory, equipment for filtration, constant pressure and constant rate filtration, filter calculations, optimum cycle time & filter aids.	9

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16marks)	- Each question carries 6 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Exemplify material balance for various unit operations or processes with or without chemical reaction.	K1,K2,K6
CO2	Solve energy balance on processes with and without reaction.	K1,K2,K4
CO3	Explain the properties of thermodynamic systems and interpret the significance of the laws of thermodynamics	K2,K4
CO4	Summarize the working principle of the major equipments for screening, size reduction, and filtration	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2		2								
CO2	3	2		2								
CO3	3	2		2								
CO4	3											

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Stoichiometry and process calculations,	Narayanan, K.V. and Lakshmikutty, B	Prentice-Hall of India Pvt. Ltd, New Delhi	2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Unit operations of chemical engineering	McCabe, W.L., J.C. Smith, J.C. and Peter Harriott	McGraw-Hill Education, New York	7(2004)
2	Stoichiometry	Bhatt, V.I. and Vora, S.M	Tata McGraw Hill, New Delhi	2004
3	A text book of chemical engineering thermodynamics	Narayanan, K.V	Prentice Hall of India Pvt Ltd, New Delhi	2013
4	Introduction to chemical engineering	Badger, W.L. and Banchero, J.T	Prentice Hall India Pvt Ltd, New Delhi.	1993

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/-Zrvfgfi0Zs
2	https://youtu.be/WgAaVHWEj3w
3	https://youtu.be/9GMBpZZtjXM
4	https://youtu.be/BoaPIMM4ymc

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation / Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

Project Assessment and Evaluation criteria (30 Marks)

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

• Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

4. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

5. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S3

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Course Code	GNEST305	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Demonstrate a solid understanding of advanced linear algebra concepts, machine learning algorithms and statistical analysis techniques relevant to engineering applications, principles and algorithms.
2. Apply theoretical concepts to solve practical engineering problems, analyze data to extract meaningful insights, and implement appropriate mathematical and computational techniques for AI and data science applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to AI and Machine Learning: Basics of Machine Learning - types of Machine Learning systems-challenges in ML- Supervised learning model example- regression models- Classification model example- Logistic regression-unsupervised model example- K-means clustering. Artificial Neural Network- Perceptron- Universal Approximation Theorem (statement only)- Multi-Layer Perceptron- Deep Neural Network- demonstration of regression and classification problems using MLP.(Text-2)	11
2	Mathematical Foundations of AI and Data science: Role of linear algebra in Data representation and analysis – Matrix decomposition- Singular Value Decomposition (SVD)- Spectral decomposition- Dimensionality reduction technique-Principal Component Analysis (PCA). (Text-1)	11

3	Applied Probability and Statistics for AI and Data Science: Basics of probability-random variables and statistical measures - rules in probability- Bayes theorem and its applications- statistical estimation-Maximum Likelihood Estimator (MLE) - statistical summaries- Correlation analysis- linear correlation (direct problems only)- regression analysis- linear regression (using least square method) (Text book 4)	11
4	Basics of Data Science: Benefits of data science-use of statistics and Machine Learning in Data Science- data science process - applications of Machine Learning in Data Science- modelling process- demonstration of ML applications in data science- Big Data and Data Science. (For visualization the software tools like Tableau, PowerBI, R or Python can be used. For Machine Learning implementation, Python, MATLAB or R can be used.)(Text book-5)	11

Course Assessment Method

(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concept of machine learning algorithms including neural networks and supervised/unsupervised learning techniques for engineering applications.	K3
CO2	Apply advanced mathematical concepts such as matrix operations, singular values, and principal component analysis to analyze and solve engineering problems.	K3
CO3	Analyze and interpret data using statistical methods including descriptive statistics, correlation, and regression analysis to derive meaningful insights and make informed decisions.	K3
CO4	Integrate statistical approaches and machine learning techniques to ensure practically feasible solutions in engineering contexts.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3								
CO2	3	3	3	3								
CO3	3	3	3	3								
CO4	3	3	3	3								

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Linear Algebra	Gilbert Strang	Wellesley-Cambridge Press	6 th edition, 2023
2	Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow	Aurélien Géron	O'Reilly Media, Inc.	2 nd edition, 2022
3	Mathematics for machine learning	Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong	Cambridge University Press	1 st edition. 2020
4	Fundamentals of mathematical statistics	Gupta, S. C., and V. K. Kapoor	Sultan Chand & Sons	9 th edition, 2020
5	Introducing data science: big data, machine learning, and more, using Python tools	Cielen, Davy, and Arno Meysman	Simon and Schuster	1 st edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data science: concepts and practice	Kotu, Vijay, and Bala Deshpande	Morgan Kaufmann	2 nd edition, 2018
2	Probability and Statistics for Data Science	Carlos Fernandez-Granda	Center for Data Science in NYU	1 st edition, 2017
3	Foundations of Data Science	Avrim Blum, John Hopcroft, and Ravi Kannan	Cambridge University Press	1 st edition, 2020
4	Statistics For Data Science	James D. Miller	Packt Publishing	1 st edition, 2019
5	Probability and Statistics - The Science of Uncertainty	Michael J. Evans and Jeffrey S. Rosenthal	University of Toronto	1 st edition, 2009
6	An Introduction to the Science of Statistics: From Theory to Implementation	Joseph C. Watkins	chrome-extension://efaidnbmnnnibpcajpcglclef indmkaj/https://www.math.arizo	Preliminary Edition.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/106/106106198/
2	https://archive.nptel.ac.in/courses/106/106/106106198/ https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/resources/lecture-29-singular-value-decomposition/
3	https://ocw.mit.edu/courses/18-650-statistics-for-applications-fall-2016/resources/lecture-19-video/
4	https://archive.nptel.ac.in/courses/106/106/106106198/

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)

SEMESTER S3

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism , Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution -Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places -accessibility and social impacts, Managing conflict , Collective bargaining, Confidentiality , Role of confidentiality in moral integrity, Codes of Ethics . Basic concepts in Gender Studies - sex, gender, sexuality, gender	6

	spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education , employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives .	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and	6

	human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
--	---	--

Course Assessment Method
(CIE: 50 marks , ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks			50	

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut,

bamboo or rubber-based product) and present findings on its sustainability.

- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India

highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S3

FLUID MECHANICS AND HYDRAULIC MACHINES LAB

Course Code	PCSFL307	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To familiarize the applications of fluid mechanics and dynamics.
2. To get acquainted with the practical implication of viscous flow and discharge measuring equipment in both closed & open channel flow.
3. To gain practical experience in handling various hydraulic machines.

Expt. No.	Experiments
1	Determination of coefficient of discharge and calibration of Notches
2	Determination of coefficient of discharge and calibration of Orifice meter.
3	Determination of coefficient of discharge and calibration of Venturi meter.
4	Determination of hydraulic coefficients of orifices and mouthpieces with constant and varying head.
5	Determination of Chezy's constant and Darcy's coefficient on pipe friction apparatus.
6	Determination of the minor losses in pipe.
7	Experiments on hydraulic ram.
8	Reynolds experiment.
9	Bernoulli's experiment.
10	Determination of metacentric height and radius of gyration of floating bodies.
11	Performance test on Positive displacement pumps.
12	Performance test on Centrifugal pumps and determination of operating characteristics.

13	Performance test on Gear pump.
14	Performance test on Pelton turbine and determination of operating characteristics.
15	Performance test on reaction turbines (Francis and Kaplan Turbines) and determination of operating characteristics
16	Speed variation test on Pelton turbine.

Minimum 10 experiments should be completed

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

1. Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.

2. Endorsement by External Examiner: The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the fundamental principles of fluid mechanics to understand the flow of fluid through pipes, notches and associated losses.	K3
CO2	Select a pump/turbine based on the given operating conditions and determine the performance of a given Turbo machine under various operating conditions.	K4
CO3	Demonstrate the ability to work in groups and present results. Also, understand the ethical issues with decision making and professional conduct.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										
CO2	3	3	2	2		2						2
CO3								3	3	3		

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fluid Mechanics	Cengel Y. A. and J. M. Cimbala	Tata McGraw Hill	2013
2	Introduction to Fluid Mechanics and Fluid Machines	Som S. K.	McGraw Hill Education India	2011
3	Fluid Mechanics and Hydraulic Machines	Bansal R. K	Laxmi Publications	2005

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fluid Mechanics	White F.M.	Tata McGraw Hill	2003
2	Engineering applications of fluid dynamics	Fisher and Henley	Willford Press	2023

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Fluid Statics - https://www.youtube.com/watch?v=rY7bvZn75Do&list=PLwdnzlV3ogoWrAmpEcsPXayfsXnFfYY1O&index=4 Bouyancy, Metacentre and stability - https://www.youtube.com/watch?v=gMuucNxc7eI&list=PLwdnzlV3ogoV-ATGY2ptuLS9mwLFOJoDw&index=7&pp=iAQB
2	Fluid kinematics - https://www.youtube.com/watch?v=rY7bvZn75Do&list=PLwdnzlV3ogoWrAmpEcsPXayfsXnFfYY1O&index=4
3	Internal viscous flow - https://www.youtube.com/watch?v=qLx7ip0eBps&list=PLCoE5wxWtHFYiVGswvsWRaHjv18vxZzE2&index=17
4	Introduction to turbomachines - https://www.youtube.com/watch?v=ocVzrn4DLj8&list=PLbMVogVj5nJQQp3QLuzbcHrt0XncZZTiE&index=2

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S3

MACHINE SHOP

Course Code	PCSFL308	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To understand the parts of various machine tools and impart hands-on experience on lathe, drilling, shaping, milling, grinding, tool and cutter grinding machines.
2. To study process parameters and practice arc and gas welding technologies.

Expt. No.	Experiments
1	Exercises on lathe: - Plain and step turning.
2	Exercises on lathe: - Ball & curve, and Taper turning.
3	Exercises on lathe: - Thread cutting.
4	Exercises on lathe: - Measurement of cutting forces in turning Process.
5	Exercises on shaping machine: - flat surfaces.
6	Exercises on shaping machine: - Grooves and key ways.
7	Exercises on drilling machine: - drilling, boring, reaming, tapping and counter sinking etc.
8	Exercises on drilling machine: - Measurement of cutting forces in drilling process.
9	Exercises on cylindrical grinding machine: - Grinding of a plain cylindrical surface.
10	Exercises on surface grinding machine: - Grinding of a flat surface.
11	Exercises on tool and cutter grinding machine: - Grinding of a single-point cutting tool.
12	Exercises on milling machine: - Plane and pocket milling operations.
13	Exercises on milling machine: - Spur gear cutting operation.
14	Exercises on arc welding: - butt welding and lap welding of M.S. sheets.

15	Exercises on gas welding: - butt welding and lap welding of M.S. sheets.
16	Study and preparation of program, simulation and exercise on CNC lathe:- turning, step turning, taper turning, thread cutting, ball and cup turning etc.
17	Study and preparation of program, simulation and exercise on CNC milling machine: surface milling, pocket milling, contour milling etc.
18	Metallurgy: - Specimen preparation, etching & microscopic study of Steel, Cast iron and Brass and grain size measurement.

A minimum of 12 sets of experiments are mandatory but both experiments mentioned for programming and experiments on CNC machines are mandatory.

Course Assessment Method **(CIE: 50 marks, ESE: 50 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.

Endorsement by External Examiner: The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	The students can operate different machine tools with understanding of work holders and operating principles to produce different part features to the desired quality.	K3
CO2	Apply cutting mechanics to metal machining based on cutting force and power consumption.	K3
CO3	Programming and manufacturing of complex profiles in CNC machines with high precision.	K3
CO4	Fabricate and assemble various metal components by welding and students will be able to visually examine their work and that of others for discontinuities and defects.	K3
CO5	Gain knowledge on the structure, properties, testing and applications of ferrous and non ferrous metals.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3			1								
CO2	3			3								
CO3	3			2	3							
CO4	3			1								
CO5	3			1								

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Production Technology	HMT	Tata McGraw Hill	2017
2	Workshop Technology Part I	W.A. J. Chapman	ELBS & Edward Arnold Publishers	1972
3	Numerical Control of Machine Tools	Yoram Koren	McGraw-Hill	2014

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 4

SAFETY & FIRE ENGINEERING

SEMESTER S4

MATHEMATICS FOR PHYSICAL SCIENCE – 4

(C GROUP)

Course Code	GCMAT401	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic calculus.	Course Type	Theory

Course Objectives:

1. To familiarize students with the foundations of probabilistic and statistical analysis mostly used in varied applications in engineering and science.
2. To provide the students with the basics of various numerical methods to develop problem solving skills used in various engineering disciplines.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Random variables, Discrete random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Binomial distribution, Poisson distribution, Poisson distribution as a limit of the binomial distribution, Joint pmf of two discrete random variables, Marginal pmf, Independent random variables, Expected value of a function of two discrete variables. [Text 1: Relevant topics from sections 3.1 to 3.4, 3.6, 5.1, 5.2]	9
2	Continuous random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Uniform, Normal and Exponential distributions, Joint pdf of two Continuous random variables, Marginal pdf, Independent random variables, Expectation value of a function of two continuous variables. [Text 1: Relevant topics from sections 3.1, 4.1, 4.2, 4.3, 4.4, 5.1, 5.2]	9

3	Confidence Intervals, Confidence Level, Confidence Intervals and One-side confidence intervals for a Population Mean for large and small samples (normal distribution and t -distribution), Hypotheses and Test Procedures, Type I and Type II error, z Tests for Hypotheses about a Population Mean (for large sample), t Test for Hypotheses about a Population Mean (for small sample), Tests concerning a population proportion for large and small samples. [Text 1: Relevant topics from 7.1, 7.2, 7.3, 8.1, 8.2, 8.3, 8.4]	9
4	Newton-Raphson Method, Gauss Elimination Method, Gauss - Jordan Method, Numerical solution of ordinary differential equations-Euler's method, Modified Euler's method, Runge - Kutta method of 2 nd Order, Numerical solution of Laplace equation –Jacobi's Method, Curve Fitting by Method of Least Squares - Straight lines, Parabola. (Text 2: Relevant topics from sections 2.5, 4.2, 7.5, 8.4, 8.5, 9.4)	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the concept, properties and important models of discrete random variables and to apply in suitable random phenomena.	K3
CO2	Understand the concept, properties and important models of continuous random variables and to apply in suitable random phenomena.	K3
CO3	Estimate population parameters, assess their certainty with confidence intervals, and test hypotheses about population means and proportions using <i>z</i> -tests and the one-sample <i>t</i> -test.	K3
CO4	Apply numerical methods to find solutions of linear system of equations, ordinary differential equations and Laplace equations.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	-	-	-	-	-	-	2
CO2	3	3	2	2	-	-	-	-	-	-	-	2
CO3	3	3	2	2	-	-	-	-	-	-	-	2
CO4	3	3	2	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability and Statistics for Engineering and the Sciences	Devore J. L	Cengage Learning	9 th edition, 2016
2	Introductory Methods of Numerical Analysis	S S Sastry	PHI Learning Pvt Limited	5 th edition, 2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability, Random Variables and Stochastic Processes,	Papoulis, A. & Pillai, S.U.,	McGraw Hill.	4 th edition, 2002
2	Introduction to Probability and Statistics for Engineers and Scientists	Ross, S. M.	Academic Press	6 th edition, 2020
3	Numerical methods for Engineers	Steven C. Chapra, Raymond P. Canale	McGraw Hill Education	8 th edition, 2021

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/117/105/117105085/
2	https://archive.nptel.ac.in/courses/117/105/117105085/
3	https://archive.nptel.ac.in/courses/117/105/117105085/
4	https://archive.nptel.ac.in/courses/111/107/111107105/

SEMESTER S4

STRENGTH OF MATERIALS

Course Code	PCSFT402	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To acquaint with the basic concepts of stress and deformation in solids.
2. To practice the methodologies to analyse stresses and strains in simple structural members, and to apply the results in simple design problems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to analysis of deformable bodies. Types of external loads - Normal, Shear, Bending and Bearing stress- Linear and Shear strains. Hooke's law - Stress-Strain diagrams - concepts of Isotropy, Orthotropy, Anisotropy. Young's Modulus, Bulk Modulus and Rigidity Modulus. Poisson's ratio - Relationship between elastic constants. Deformation in axially loaded bars –uniform cross section, varying cross section, dissimilar materials, principle of superposition. Thermal effects – simple, composite bars.	11
2	Torsion: Shafts - torsion theory of elastic circular bars – assumptions and limitations – polar modulus - torsional rigidity – shaft design for torsional load. Beams- Classification - Diagrammatic conventions for supports and loading – Differential equations between load, Shear Force and Bending Moment- Shear Force and Bending Moment Diagrams of Cantilever and Simply supported beam with Point load/UDL. Point of Inflection.	11

3	Stresses in Beams: Pure Bending – Flexure formula for beams -assumptions and limitations – Section Modulus - Flexural Rigidity Shearing Stress formula for beams – Shear stress distribution in beam section - derivation and problems for rectangular section only -assumptions and limitations Deflection of Beams: Moment-Curvature relation – assumptions and limitations - Double Integration method – Macaulay’s method	11
4	Stress on an inclined plane due to Uniaxial stress- Stress on an inclined plane due to Biaxial stress- Stress on an inclined plane due to two Normal Stresses accompanied by Shear stresses- principal planes and stresses. Mohr’s circle of stress. Buckling and stability of long columns-Euler’s buckling/crippling load for columns with different end conditions- Euler equation derivation for both ends hinged only- Rankine’s formula	11

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Determine the stresses and strains in deformable bodies subjected to different types of external loads and thermal effects	K3
CO2	Analyse the torsion of circular bars and draw the shear force and bending moment diagrams for beams	K4
CO3	Determine the stresses and deflections in beams subjected to transverse loads	K3
CO4	Determine analytically and graphically the principal stresses and planes for structural members subjected to loads and analyse the strength of columns	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2									
CO2	3	3	2									
CO3	3	3	2									
CO4	3	3	2									

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanics of Solids	R.K.Bansal	Laxmi Publications	2012
2	Mechanics of Solids	S. S. Bhavikatti	New Age International	2013
3	Strength of Materials	Surendra Singh	S. K. Kataria & Sons	2013
4	Strength of Materials	Rattan	McGraw Hills	2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanics of materials	R. C. Hibbeler	Pearson Higher Education	2018
2	Engineering Mechanics of Solids	Popov E	PHI	2002
3	Mechanics of Materials	Beer & Johnston	McGraw Hills	2017
4	Mechanics of Materials	Pytel A. and Kiusalaas J.	Cengage Learning India Private Limited,	2015

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc22_ce46/preview
2	https://onlinecourses.nptel.ac.in/noc22_ce46/preview
3	https://onlinecourses.nptel.ac.in/noc22_ce46/preview
4	https://onlinecourses.nptel.ac.in/noc22_ce46/preview

SEMESTER S4

ELECTRICAL TECHNOLOGY AND SAFETY

Course Code	PCSFT403	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To teach various techniques used for attaining electrical safety and application of electrical equipments

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Construction and principle of operation of dc machines, e.m.f equation of a generator, use of interpoles. Characteristics of shunt, series and compound generators – starting and speed control – losses and efficiency, application of motor for industrial drive. Construction and principle of operation of single phase transformers, e.m.f equation, phasor diagrams, equivalent circuit, regulation, losses and efficiency. Instrument transformers CT, PT.	11
2	Induction Motor; construction and principle of operation, torque slip characteristics, method of starting, applications. Fundamentals of motors used in EVs, Motor control for BLDC motors , Analyse EVs based on power sources and calculate range of an EV. - Perform motor power and torque calculations to select a motor to build their own EV. Circuit breakers - SF6, vacuum, air blast. Function of switch gear, circuit breakers principle of operation, arc phenomenon, arc interruption. MCB and ELCB. Faults in power systems – causes – types ,SCADA system application	11
3	Fuses, types and selection of fuses, advantages and disadvantages. Grounding, neutral grounding, solid grounding, resistance grounding, arc	11

	suppression coil grounding. Equipment grounding for safety, spark gap, surge protection, lighting arrester, grounding of line structure. Effect of electric and magnetic fields, human safety aspects, effect of current and voltage on human beings. Electrical accident- safety precaution, electric shocks and their prevention. Insulation, FRLS insulation, continuity test. Protective relays, requirement of relay, types of protection, distance relay, differential relay, state relays and digital relay. Transmission line protection.	
4	Safety during installation of plant and equipment. Safe sequences in installation, risk during installation. Safety during testing and commissioning. Test on relays, protection and interlock systems for safety. Hazardous zones, classification of hazardous zones. Intrinsically safe and explosion proof electrical apparatus. Selection of equipments in hazardous area. Electrical fires, hazards of static electricity. Safe procedures for electrical maintenance. Maintenance principle and safety precautions of VFD (Variable-frequency Drive). Electrical fires, hazards of static electricity. Safe procedures for electrical maintenance - Statutory requirements. Safety provisions in Indian Electricity Act & Rules	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the constructional features and applications of electrical machines	K2
CO2	Solve numerical problems related to the operation of electrical machines	K4
CO3	Explain the working principle of various electrical safety equipments	K2
CO4	Summarise the safety precautions to be taken during installation of plants and	K1.K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3					3						
CO4	3					3						
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electrical technology	Cotton, H	CBS Publishers and	Electrical technology

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Basic electrical engineering	Kothari, D.P. and Nagrath, I.J	Tata McGraw Hill Publishing Company Ltd., New Delhi	3(2009)
2	National Safety Council. Accident prevention manual for industrial operations		Chicago	1982
3	Electrical safety, fire Engineering and safety Management	Fordham-Cooper, W	Khanna Publishers, Delhi	2012
4	Electrical safety, fire Engineering and safety Management	Rao, S. and Saluja, H.L	Khanna Publishers, Delhi	2012
5	Electric Vehicle Engineering	Per Enge, Nick Enge, Stephen Zoepf,	McGraw Hill Inc., New Delhi	2021

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.dvidshub.net/video/863793/electrical-safety-training
2	
3	https://www.necanet.org/programs/safety/safety-videos-resources
4	https://safeelectricity.org/public-education/educational-lesson-plans-and-activities-for-teachers

SEMESTER S4

HAZARD IDENTIFICATION AND RISK ASSESSMENT & MITIGATION

Course Code	PBSFT404	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. To teach various techniques used for hazard identification, Risk assessment and reliability analysis,
2. To learn estimation of frequency of occurrence of hazards consequence analysis, Risk quantification and human reliability analysis..

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Hazard and risk, Types of hazards – fire, explosion and toxic gas release, Structure of hazard identification and risk assessment. Identification of hazards : Inventory analysis, Fire and explosion hazard rating of process plants - The Dow Fire and Explosion Hazard Index, The Mond Index, Plant layout and unit hazard rating, Preliminary hazard analysis, Hazard and Operability study (HAZOP), What If analysis, Case studies.	9
2	Plant availability and process reliability: ways of improving plant availability, MTBF and MTTF, the reliability function, failure rate, bathtub curve, probability relationships, simple reliability estimation. Estimation of frequency of occurrence of a hazard: The logic tree approach, set theory and Boolean algebra, application to probability, Boolean manipulation. Fault tree analysis – logic symbols, minimal cut set, logic gates, fault tree quantification. Event tree analysis – notation, event tree construction, advantages and disadvantages of ETA.Process Hazard Analysis (PHA), Failure mode and Effect Analysis (FMEA) – methodology, criticality	9

	analysis, corrective action and follow-up.	
3	Consequence modelling: Source models – discharge rate models, flash and evaporation, dispersion models. Explosions and fires – vapour cloud explosions, flash fires, physical explosions, BLEVE and fire ball, confined explosions, pool fires, jet fires. Effect models –dose-response functions, probit functions, toxic gas effects, thermal effects, explosion effects – Software application for effect and damage calculations.	9
4	Quantification of risk: QRA, Vulnerability analysis, accepted and imposed risk, perception of risk, risk indices, individual risk and societal risk, acceptance criteria for risk, ALARP, Presentation of measures of risk – risk contour, F-N curve. Calculation of individual risk and societal risk. Human reliability analysis (HRA): factors leading to human error, characteristics of HRA techniques, Technique for Human Error Rate Prediction (THERP), Accident Sequence Evaluation Program (ASEP), Techniques using expert judgment, Operator Action tree (OAT).	9

Suggestion on Project Topics

Course Assessment Method (CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the hazard identification and risk assessment procedure and apply suitable tools for the identification of hazards in a process	K2,K3
CO2	Create HAZOP Worksheet after analysing a real-time scenario.	K6
CO3	Explain PHA, What if analysis, ETA and FTA	K2
CO4	Predict the consequences of fire, explosion and toxic gas release using suitable models.	K6
CO5	Compute the individual and societal risk involved in a process industry	K3
CO6	Describe the methods of quantification of risk, Human reliability analysis, THERP	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3	3							2			
CO3	3	2							2			
CO4	3	2			2				2			
CO5	3	2										
CO6	3	2							2			

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	AICHE/CCPS, Guidelines for Chemical Process Quantitative Risk Analysis,	Centre for Chemical Process Safety,	American Institute of Chemical Engineers, New York	Edition 2 1992
2	AICHE/CCPS, Guidelines for Chemical Process Quantitative Risk Analysis,	Centre for Chemical Process Safety,	American Institute of Chemical Engineers, New York	Edition 2 2000
3	Process Systems Risk Management, Volume 6	Ian Cameron R. Raman	Elsevier Academic Press	E 3,2005
4	Chemical Process Safety: Fundamentals with Applications,	Crowl, D.A., and Louvar, J.F.	Prentice Hall Publication Inc., 723 pages.	Third Edition, (2011).

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	System safety engineering and risk assessment: a practical approach.	. Bahr, N. J.	CRC Press, 443 pages.	(2014).
2	Fire, explosion, and toxic gas dispersion	Assael, M J, and Kokosimos, K E.	, CRC Press, New York, 327 pages	(2010).
3	Risk Analysis in Engineering, Techniques, Tools, and Trends,	Modarres, Mohammad	CRC Press, NY, 401 pages.	(2006).
4	Risk Analysis: A quantitative guide,	Vose, D.	John Wiley, Sussex, UK, 417 pages.	(2000).
5	Introduction to Reliability and Maintainability Engineering,	Ebeling, C.	McGraw Hill Publication, Boston,	(2010)
6	Loss Prevention in the Process Industries,	Lees F.P,	Butterworth's, London,	2e,1996
7	Lee's Loss Prevention in the Process Industries.	Sam Mannan (Editor).	Butterworth-Heinemann Ltd., UK.	(Fourth edition). (2012).
8	Process safety analysis: An Introduction,	Bob Skelton.	Institution of Chemical Engineers.	(1997)
9	Reliability Engineering,	L.S.Srinath,	Affiliated East-West Press, NewDelhi	, (2005).

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/n1s1UB80JgA
2	https://youtu.be/6vR50hzznZk

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project Creativity in solutions and approaches

SEMESTER 4

LEGAL ASPECTS OF HEALTH SAFETY AND ENVIRONMENT

Course Code	PESFT411	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. To familiarize the various laws related to HSE.
2. To understand the various provisions in the relevant Acts and Rules.
3. To know application of various legal provisions to different types of industries.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Factories Act– Definitions, Preliminary, Inspecting staff, Health, Safety, Provisions relating to hazardous processes, Welfare, Working hours of adults, Employment of young persons, Special provisions. Workmen's Compensation Act: Definitions, Employer's liability for compensation, Calculation of amount of compensation. ESI Act and Rules: Applicability, Definitions and Benefits. Duties of Safety Officers, Reporting of accidents, Emergency Action Plan, Safety Committee.	9
2	Explosives Act: Definitions, Categories of Explosives, General Safety Provisions, Use of Explosives, Grant of license, Notice of Accidents, Inquiry into ordinary and more serious accidents, Extension of definition to other explosive substances. Important technical provisions in the Explosives Rules, SMPV Rules and Gas Cylinder Rules. Public Liability Insurance Act and Rules- Definitions, Calculation of amount of relief, Environmental Relief Fund, Advisory Committee, Powers of	9

	District Collector, Extent of Liability, Contribution to Relief Fund.	
3	Petroleum Act & Rules - Definitions, Control over Petroleum import, transport, storage, production, refining and blending, Need for license, exemption, Notice of Accidents and Inquiries. Water Act- Definitions, Powers and Functions of Boards, Provisions regarding prevention and control of water pollution, Power to make rules, Rules on Consent for Establishment and Operation. Dock Workers (Safety, Health and Welfare) Act and Regulations (in brief) - Definitions, Powers of inspectors, Power of Govt. to direct inquiry, Obligation of dock workers.	9
4	Air Act - Definitions, Power & Functions of Boards, Prevention & Control of Air Pollution, Consent as per Air Pollution Rules. Environment (Protection) Act and Rules- Definitions, general powers of central government, prevention, control and abatement of environmental pollution, standards for emission, prohibition and restrictions on siting and operation of industries. MSIHC Rules and Definitions, Duties of Authorities, Notification of Major Accidents Safety Reports, Safety audit, MSDS, On-site & Off-site Emergency Plan, Giving safety information to public.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the importance of Factories Act and Workmen's Compensation Act.	K2
CO2	Understand the legal aspects and procedures related to Explosives and Public Liability.	K2
CO3	Understand Petroleum Act and Rules in detail and Water related legal aspects.	K2
CO4	Understand Air and Environmental Protection Acts and legal aspects.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3	2	3				3
CO2	3					3	2	3				3
CO3	3					3	2	3				3
CO4	3					3	3	3				3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Occupational Health and Safety Legal and operational guide		PRIA International Academy	2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Legal Aspects of Health and Safety	Bridgit Dimond	Mark and Allen Group	3 rd 2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-4	https://onlinecourses.nptel.ac.in/noc24_lw09/unit?unit=17&lesson=18

SEMESTER S4
DESIGN FOR SAFETY

Course Code	PESFT412	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. Describe the concept of design for safety and the pro-active management of risk at source.
2. Point out safe designs around the environment that we work and live in.
3. Develop Building and Facility Management Professionals

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Safety in process design and pressure system design : Design process, conceptual design and detail design, assessment, inherently safer design - chemical reactor, types, batch reactors, reaction hazard evaluation, assessment, reactor safety, operating conditions, unit operations and equipments, utilities. Pressure system, pressure vessel design, standards and codes- pipe works and valves heat exchangers- process machinery- over pressure protection, pressure relief devices and design, fire relief, vacuum and thermal relief, special situations, disposal- flare and vent systems- failures in pressure system.	9
2	Plant design: synthesis, Characteristics of design problems, design as equipment selection: emergency isolation valves, classification activities, designs with shallow structure, designs with deep structure; plant design: analysis, hazop, fault tree analysis; operating procedures; process monitoring; fault administration; malfunction detection, computer aids, Project Aids, Aids for representing regulatory and code requirements, Other process design aspects.	9

3	Instrumentation in safety applications- Applications of Advanced equipment and Instruments, Thermo Calorimetry, Differential Scanning Calorimeter(DSC),Thermo Gravimetric Analyzer(TGA), Accelerated Rate Calorimeter(ARC), Reactive Calorimeter(RC), Reaction System Screening Tool(RSST) - Principles of operations, Controlling parameters, Applications, advantages. Explosive Testing, Deflagration Test, Detonation Test, Ignition Test, Minimum ignition energy Test, Sensitiveness Test, Impact Sensitiveness Test(BAM) and Friction Sensitiveness Test (BAM), Shock Sensitiveness Test, Card Gap Test.	
4	Plant commissioning and inspection: Commissioning phases and organization, pre-commissioning documents, process commissioning, commissioning problems, post commissioning documentation Plant inspection, pressure vessel, pressure piping system, non-destructive testing, pressure testing, leak testing and monitoring- plant monitoring, performance monitoring, condition, vibration, corrosion, acoustic emission-pipe line inspection.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop safer design process and apply codes for safety.	K3
CO2	Apply computer aids and other process design aspects.	K4
CO3	Understand instrumentation facilities for safer designs in various processes.	K2
CO4	Understand plant commissioning procedures.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3									2
CO2	3	2	3		2							2
CO3	3	2	3		2							2
CO4	3	2	3									2

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Loss Prevention in Process Industries	Lees F.P.	Butterworth publications, London,	3rd edition, 2005

Reference Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial hazards and plant safety.	Sanoy Banerjee,	Taylor & Francis, London	2003
2	High Risk Safety Technology	Green, A.E.	John Wiley and Sons	1984
3	Methodologies for Risk and Safety Assessment in Chemical Process Industries	Raghavan K. V., Asad Ali Khan	Commonwealth Science Council, UK,	1990
4	Safety and Accident Prevention in Chemical Operations	Fawcett, H. and Wood,	Wiley inters,	2nd Edition 1984.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/NbqytWyLwds
2	https://youtu.be/IJGV3LgNq-U
3	https://youtu.be/OuarJA9n8PQ
4	https://youtu.be/NADSBVIT7Rk

SEMESTER S4

PROCESS INSTRUMENTATION AND CONTROL ENGINEERING

Course Code	PESFT413	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide an insight into various measuring parameters and sensors used in process industries.
2. To make students aware of basic concepts of safety instrumented system.
3. To understand the various control schemes associated with process industries

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Elements of measurement – Fundamental standards, Quality of measurement, Meaning of measurement, Errors in measuring instruments, Precision and accuracy, Calibration principle, Static and dynamic characteristics of measuring instruments. active and passive transducers. General features and classification of transducers, Resistive Transducers – Potentiometer, Unbounded strain gauge, Bonded strain gauge-Load cell, Inductive transducers – Differential output transducers – LVDMeasurement of temperature – Bimetallic and pressure thermometers, Thermocouples, Resistance thermometers, Pyrometry, Calibration. Pressure and vacuum measurement – Manometers, Measuring element, Absolute pressure measurement, Static accuracy of pressure gauges	9
2	Flow measurement - Orifice installation, Pitot tube, Area flow meters, Open channel meters. Level measurement – Direct method, Measurement of level in open and pressure vessels. Measurement of pH and humidity. Noise	9

	pollution – measurement of sound, tolerable levels of sound. Measurement of sound level. Recording Instruments, Indicating and signaling instruments, Signal transmission, and codes	
3	Introduction to frequency response of closed-loop systems, control system design by frequency response techniques, Bode diagram, stability criterion, tuning of controller settings. Introduction to advanced control systems, cascade control, feed forward control, Smith predictor controller, Control valves: final control operation-electrical and pneumatic Signal conversion actuators	9
4	Open loop and close loop systems – elements of feedback control loop - Transfer function modeling – block diagram representation of mechanical, thermal and liquid level systems. Transient response analysis – Time response of first and second order system for impulse and step inputs – Effect of damping factors on transient response – Characteristics of proportional, integral, derivative, PI, PD and PID controllers.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the measurement principles and techniques of measuring temperature and pressure.	K1
CO2	Compare the methods for the measurement of flow, level, pH and humidity.	K1
CO3	Interpret the operation of different types of control system	K2
CO4	Analyse the stability of a control system	K2,K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3	2	2									
CO4	3	2	2									

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanical Measurement and Instrumentation	A.K.Sawhney	Dhanpat Raj and Sons, New Delhi	1999

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Control Systems.	Benjamin C. Kuo.,	Oxford University Press,	1992
2	Principles of Industrial Instrumentation	D Patranabis	Tata McGraw-Hill Publishing Company Ltd, New Delhi	1996
3	Chemical Process Control : An Introduction to Theory and Practice	George Stephanopolous	Prentice Hall of India Pvt. Ltd	6 th 1990
4	Modern Control Engineering, Prentice Hall	Ogata, K	Prentice Hall of India Pvt. Ltd	1995
5	Industrial Instrumentation	Eckman D P	Wiley Eastern Ltd, New Delhi	1990
6	Mechanical and Industrial Measurements	R.K.Jain	, Khanna Publishers, New Delhi	1999

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
3	https://archive.nptel.ac.in/courses/103/105/103105064/
4	https://archive.nptel.ac.in/courses/103/105/103105064/

SEMESTER S4

ENERGY MANAGEMENT AND CONSERVATION

Course Code	PESFT414	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. To understand the current energy consumption and future needs
2. To create awareness about sustainability requirements in power sector

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Energy Scenario :Commercial and Non-commercial energy, primary energy resources, commercial energy production, final energy consumption, Indian energy scenario, Sectoral energy consumption (domestic, industrial and other sectors), energy needs of growing economy, energy intensity, long term energyscenario, energy pricing, Energy security, energy conservation and its importance, energy strategy for the future, Energy Conservation Act 2001 and its features. Basics of Energy its various forms and conservation :Electricity basics - Direct Current and Alternative Currents, electricity tariff, Thermal Basics-fuels, thermal energy contents of fuel, temperature and pressure, heat capacity, sensible and latent heat, evaporation, condensation, steam, moist air and humidity and heat transfer.	9
2	Evaluation of thermal performance: calculation of heat loss - heat gain, estimation of annual heating & cooling loads, factors that influence thermal performance, analysis of existing buildings setting up an energy management programme and use management - electricity saving techniques Energy Management & Audit: Definition, energy audit, need, types of	9

	energy audit. Energy management (audit) approach-understanding energy costs, Bench marking, energy performance, matching energy use to requirement, maximizing system efficiencies, optimizing the input energy requirements, fuel and energy substitution, energy audit instruments and metering	
3	Energy Monitoring and Targeting: Defining monitoring & targeting, elements of monitoring & targeting, data and information-analysis, techniques - energy consumption, production, cumulative sum of differences (CUSUM). Energy Management Information Systems (EMIS)	9
4	Energy Efficiency in Thermal Utilities and systems: Energy efficiency in thermal utilities like boilers, furnaces, pumps and fans , compressors, cogeneration (steam and gas turbines), heat exchangers ,lighting system, Motors belts and drives, refrigeration system. Heat Recovery and Co-generation: Heat recovery from ventilation, air co-generation of heat and electricity, heat recovery and bottoming cycles.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Calculate the efficiency of various thermal utilities	K2
CO2	Design suitable energy monitoring system to analyse and optimize the energy consumption in an organization	K3
CO3	Design suitable systems for heat recovery and co-generation to improve thermal efficiency	K3
CO4	Relate energy audit methods learnt to identify the areas deserving tighter control to save energy expenditure	K4
CO5	Explain to the employees of the organization about the need and the methods of energy conservation	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3	2	2									3
CO3	3		2			3						3
CO4	3	2					2			2		3
CO5	3					3		3		3		3

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Energy Conservation in Process Industry	Kenny W.F.	Academic Press	2012
2	Energy Engineering and Management	Amlan Chakrabarti	Prentice Hall India	2011
3	Energy Management Principles	Smith C.B. and Parmenter K	Elsevier	2 nd Edition 2015
4	Handouts	Bureau of Energy Efficiency	Bureau of Energy Efficiency New Delhi	
5	Energy Management Hand Book	Turner W. C. and Steve Doty	The Fairmount Press Inc	6 th Edition 2007

SEMESTER S4

LIFE SAFETY IN BUILDING FIRE

Course Code	PESFT416	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the importance of life safety in building fire.
2. To understand the method of evacuation.
3. To design the buildings with fire protection

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Process of emergency evacuation- special features of personnel movement, Parameter characteristics of the movement of people; Evacuation exits and routes- stages of evacuation, importance of different stages of evacuation, planning and design of evacuation routes, passages, and exits; calculation of building evacuation time; Planning of seating arrangements in premises where large number of people gather. Classification of buildings based on occupancy and type of construction according to fire resistance as per NBC; Fire zone; General fire safety requirements applicable to all individual occupancies. General exit requirements as per NBC; Internal staircases; horizontal exits; fire tower; ramps; fire lifts; external fire escape ladders; Planning of location and calculation of capacity, number and width of exit as per NBC for different occupancy classification	9
2	Production of smoke particles, constituents of smoke, quantity and rate of production of smoke, quality of smoke, smoke density, visibility in smoke, smoke movement in buildings; Smoke control in buildings-natural and	9

	mechanical ventilation; Design principles of smoke control using pressurization technique; Principles of smoke and explosion vent design; Introduction to zone modeling of compartment fires- pre-flashover and post-flashover fires. National Building Code (NBC) guidelines for general smoke control in buildings; and smoke, fire and explosion venting requirements in industrial buildings	
3	Fire prevention measures in buildings as per NBC- Classification of buildings based on occupancy and type of construction, Fire zone; General fire safety requirements applicable to all individual occupancies. Life safety measures in buildings as per NBC- General exit requirements; Requirements for different types of exit access and exits-doorway, corridors and passageways, horizontal exits, internal staircases, exit passageways, external staircases, ramps; Planning of location and calculation of capacity, number and width of exit as per NBC for different occupancy classification; requirements for compartmentation; requirements of fire detection and alarm system in different occupancies; , firefighting shaft, Fireman's lift. Fire protection measures in buildings as per NBC- requirements for portable and fixed fire fighting installations in buildings of different occupancy.	9
4	. Fire and life safety requirements in different groups of buildings, Hotel, Schools & Colleges, Hospitals, Theatres, Shopping malls, etc.; Fire protection and prevention in high rise buildings; Fire protection in underground structures and in buildings under construction. Siting of detectors as per relevant Indian standard specifications; Selection and planning of alarm system as per relevant standards (BIS). General requirements and guidelines for the installation of fire detection and alarm system in buildings of different occupancy classification. Selection and distribution of portable extinguishers (for class A and B fires) and other fire protection equipments and systems for different occupancy classification as per NBC; Planning of fixed fire fighting installation for different occupancy classification, sprinkler system; total flooding system; CO2 system; foam system; Fire Investigation; Detection of arson; Fire training and education- fire drill, fire order; - Fire safety audits; Fire risk assessment	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Plan and design evacuation route and exits based on the concepts of human behaviour under emergency movement	K2
CO2	Interpret the mechanism of production of smoke due to fire, its harmful effect and movement within a building, and the control measures to be adopted in buildings	K2
CO3	Summarize the fire safety requirements for buildings of different occupancy as per the National Building Code of India.	K1
CO4	Plan, design and distribute various fire safety measures required in buildings of different occupancy type	K2,K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					2						
CO2	3					2						
CO3	3					2						
CO4	3					2						

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of fire safety standards for building construction..	Roytman, M. Ya.	Amerind Publishing Co. Pvt. Ltd., New Delhi	1975
2	Fire safety in buildings .	Jain, V.K.	New Age International (P) Ltd., New Delhi.	2 nd 2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Designing of fire safety	Butcher, E. G. and Parnell, A. C	John Wiley and Sons Ltd., New York, U.S.A	1983
2	Fire protection and prevention the essential handbook	Barendra Mohan Sen.	UBS Publishers and Dist., New Delhi	2013
3	An Introduction to Fire Dynamics	Dougal Drysdale	John Wiley & Sons Ltd., England	1999
4	Relevant Indian Standard Codes of Practice	(BIS)		

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
3	https://youtu.be/Gmi1dTP4DMo
4	https://youtu.be/G4ovKEsCUaI

SEMESTER S4

NUCLEAR SAFETY

Course Code	PESFT417	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. To bring out the importance of nuclear engineering and its safety.
2. To impart the safety design principles of nuclear reactors.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Binding energy – fission process – radio activity – alpha, beta and gamma rays radioactive decay – decay schemes – effects of radiation – neutron interaction – cross section – reaction rate – neutron moderation – multiplication – scattering – collision – fast fission – resonance escape – thermal utilization – criticality. Characteristics of the fission reaction, neutron moderation, practical fission fuels-Reactor power, fuel burn up, and fuel consumption, Neutron chain-reacting systems-Homogeneous and heterogeneous cores, reflectors Reactor kinetics and dynamics, reactivity feedback Core composition changes during reactor operation, nuclear system lifetime.	9
2	Control requirements in design considerations – means of control – control and shut down rods – their operation and operational problems – control rod worth – control instrumentation and monitoring – online central data processing system. Boiling water reactors – radioactivity of steam system – direct cycle and dual cycle power plants- pressurized water reactors and pressurized heavy water reactors – fast breeder reactors and their role in	9

	power generation in the Indian context – conversion and breeding – doubling time – liquid metal coolants – nuclear power plants in India and important plants across the globe.	
3	Safety design principles – engineered safety features – site related factors – safety related systems – heat transport systems – reactor control and protection system – fire protection system – quality assurance in plant components – operational safety – safety regulation process – public awareness and emergency preparedness. Accident Case studies- Three Mile island accident and Chernobyl accident.	9
4	Radiation shielding – radiation dose – dose measurements – units of exposure – exposure limits – barriers for control of radioactivity release – control of radiation exposure to plant personnel – health physics surveillance – nuclear safety in medical treatment - waste management and disposal practices –environmental releases.	9

Course Assessment Method

(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic principles of nuclear reactions which can be used for Power Generation and Medical purposes	K2
CO2	Detail about various nuclear power plants and control systems	K2
CO3	Understand safety design principles and regulations in Nuclear Power plants	K2
CO4	Analyse the accidents in nuclear power plants as case study	K4
CO5	Understand the exposure control and waste management in nuclear power and medical purpose operations	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3											3
CO4	3											
CO5	3					3	2					

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Nuclear Safety Principles and Applications	Paul F. Coley	Whittles Publishing	2022

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Nuclear Safety	Gianni Petrangeli	Elsevier – Butterworth Heinemann	2 nd Edition 2006

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-4	https://onlinecourses.swayam2.ac.in/cec24_ma13/unit?unit=6&lesson=9 https://onlinecourses.swayam2.ac.in/cec24_ma13/unit?unit=6&lesson=21

SEMESTER S4
ORGANISATIONAL BEHAVIOUR

Course Code	PESFT418	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. To develop an understanding of the individuals and groups behaviour inside organizations should further enhance the skills in understanding and appreciating individuals, interpersonal, and group process for increased effectiveness both within and outside of organizations.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to organization, Organization and managers, Manager' roles and skills, Behaviour at work, Introduction to organization behaviour, Major behavioural science disciplines contributing to OB, Challenges and opportunities managers have in applying OB concepts, OB models Motivation models and levels of OB model.	9
2	Introduction to individual behaviour, Values, Attitudes, Job satisfaction, Personality, Personality Traits, Perception and individual decision making, Importance of Perception in OB, Learning, Motivation at work, Managing emotions and stress, Meaning and Definition of Stress and job performance relationship, Work Stressors - Approaches to stress management and Coping with stress. Impression Management. Emotions and Moods in workplace.	9
3	Groups in organizations – Influence – Group dynamics – Interpersonal Communication: Importance, Types, Barriers to communication, Communication as a tool for improving Interpersonal Effectiveness, Team building - Interpersonal relations - Johari Window – Group decision making techniques. Meaning of conflict and its types, Conflict Redressal process, Leadership & management, Theories of leadership- Trait theory, Behavioural Theory, Contingency Theory, Leadership & Followership, How to be an	9

	Effective Leader, Introduction to Transactional Analysis. – Ego states, Types of transactions, Life positions.	
4	Foundations of organization structure, Organization design, Organizational culture, Factors affecting organizational climate, Organization change, Managing across cultures, Diversity at work. Introduction to Human Resource Management: Policies and Practices - Selection, Orientation, Training & Development, Performance Appraisal, Incentives, Importance of Organisational Change, Planned Change & OB Techniques, International OB: An Introduction to Individual & Interpersonal Behaviour in Global Perspectives. Organizational effectiveness – Benchmarking - TQM and Six Sigma.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To explain the concept of Organisation Design and determine the factors that affect Organisation Design.	K2
CO2	To identify the components of Individual Behaviour and apply the concept of Learning, Perception, Attitudes and values.	K3
CO3	To analyse the behaviour of individuals and groups in organisations in terms of the key factors that influence organisational behaviour and demonstrate skills required for working in groups (team building)	K4
CO4	To justify how organizational change and conflict affect working relationships within organizations and demonstrate how to apply relevant theories to solve problems of change and conflict within organizations	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3		3				2
CO2	3					3		3	3			3
CO3	3					3		3	3	3	2	3
CO4	3					3		3		3		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Organizational Behaviour	Stephen Robbins	Pearson Education	18 th Edition 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human Behaviour at work	Keith Devis	McGraw-Hill Inc.USA	Edition 4 1972
2	Organizational Behavior; Concepts, Skills and Practices	Kinicki Kreitner	McGraw-Hill Higher Education	Edition 3 2007
3	Dimension of Organizational Behavior	T. Herbert	Collier Macmillan Ltd	1976

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_mg87/unit?unit=17&lesson=18
2	https://onlinecourses.nptel.ac.in/noc24_mg87/unit?unit=17&lesson=20
3	https://onlinecourses.nptel.ac.in/noc24_mg87/unit?unit=17&lesson=21
4	https://onlinecourses.nptel.ac.in/noc24_mg87/unit?unit=17&lesson=22

SEMESTER S4

COMPUTATIONAL FLUID DYNAMICS

Course Code	PESFT415	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. The finite difference methods and finite volume methods as a means of solving different types of differential equations that arise in fluid dynamics and heat transfer.
2. The fundamentals of numerical analysis, ordinary differential equations and partial differential equations related to fluid mechanics and heat transfer
3. The error control and stability considerations associated with numerical solutions.
4. A class of methods used in computational fluid dynamics for numerically solving the Navier-Stokes equations for 2D incompressible flows.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Computational Fluid Dynamics, Governing Equations of fluid flow and heat transfer, Conservative form of Navier-Stokes equation, General transport equation. Physical and mathematical classifications of partial differential equations. Elliptic, parabolic and hyperbolic equations, Comparison of experimental, theoretical and numerical approaches; applications of CFD.	9
2	Discretization-converting derivatives to their finite difference forms- Taylor's series approach and polynomial fitting approach. Central difference, backward difference, and forward difference of first and second	9

	order derivatives. Solution of partial differential equations using finite difference equations. Discretization error, truncation error, round off error. Consistency and numerical stability. Iterative convergence, condition for convergence, rate of convergence. Termination of iteration. Boundary and Initial conditions	
3	Introduction to finite volume method. Finite volume method for steady one-dimensional conduction problems. one-dimensional transient heat conduction problems -explicit, implicit, Crank- Nicholson schemes, under and over relaxations, handling of boundary conditions; dealing with Dirichlet, Neumann, and Robins type boundary conditions; two-dimensional steady state conduction problems; point-by-point and line-by-line method of solution; tri-diagonal matrix algorithm.	9
4	Finite volume method for steady-state diffusion and convection-diffusion problems; Central difference and Upwind schemes for convection and diffusion problems. Two dimensional incompressible viscous flows. Staggered grid. Pressure correction methods. Solution algorithm for pressure-velocity coupling in steady flows- SIMPLE algorithm to solve Navier - Stokes equations. Computer graphics techniques to present CFD results.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To understand the governing equations of fluid flow and heat transfer.	K2
CO2	To apply finite difference methods to simple partial differential equations	K3
CO3	To demonstrate the use of finite volume method for simple 1D/2D problems	K3
CO4	To understand different solution techniques for convection diffusion equation	K2
CO5	To apply the knowledge of CFD to interpret the graphical results	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	-	1	-	-	-	-	-	-	-	-
CO2	3	3	-	2	-	-	-	-	-	-	-	-
CO3	3	3	3	3	-	-	-	-	-	-	-	-
CO4	2	2	-	2	-	-	-	-	-	-	-	-
CO5	3	3	-	3	-	-	-	-	-	-	-	-

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computational Fluid Dynamics	John D Anderson Jr	McGraw-Hill Book Company	2012
2	Numerical Heat Transfer and Fluid Flow	S V Patankar,	McGraw-Hill	2017
3	An Introduction to Computational Fluid Dynamics: The Finite Volume Method	H. Versteeg, W. Malalasekera	Pearson	2nd, 2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to computational fluid dynamics	Anil W. Date	Cambridge University Press	2005
2	Introductory methods to numerical analysis	S SSastry	PHI learing Private Ltd.	2012
3	Heat transfer	S P Venkatesh	Ane books Pvt Ltd	2009

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/112105045
2	https://onlinecourses.nptel.ac.in/noc20_me64/preview
3	https://nptel.ac.in/courses/112105045 https://onlinecourses.nptel.ac.in/noc21_me126/preview
4	https://onlinecourses.nptel.ac.in/noc21_me126/preview https://onlinecourses.nptel.ac.in/noc20_me64/preview

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)

SEMESTER S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism , Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution -Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places -accessibility and social impacts, Managing conflict , Collective bargaining, Confidentiality , Role of confidentiality in moral integrity, Codes of Ethics .	6

	Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education, employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and	6

	climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	6

Course Assessment Method
(CIE: 50 marks , ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.

- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks			50	

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO1 0	PO1 1	PO1 2
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S4

STRENGTH OF MATERIALS LABORATORY

Course Code	PCSFL407	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. Characterize the mechanical behaviour of materials under various loading conditions.
Relate material properties and microstructure to engineering applications.
2. Relate material properties and microstructure to engineering applications.

Expt. No.	Experiments
1	Conduct compressions test on a ductile material (mild steel, tor-steel, or high-strength steel) using a Universal Testing Machine (UTM) equipped with an extensometer.
2	Determine the tensile properties of cast iron (a brittle material) using a Universal Testing Machine (UTM) equipped with an extensometer.
3	Determine the shear strength of a mild steel rod using a shear test.
4	Perform Brinell/Vickers/Rockwell hardness tests on a given material
5	Determine the torsional rigidity of mild steel/copper/brass rods.
6	Evaluate the flexural stiffness (flexural rigidity) of mild steel/copper/brass specimens using a three-point bend test on a Universal Testing Machine (UTM)
7	Determine the notch toughness of the material at room temperature using Izod and Charpy impact testing.
8	Investigate the effect of coil type (close-coiled vs. open-coiled) and arrangement (series vs. parallel) on spring stiffness.
9	Microstructure of mild steel/copper/ brass/aluminium using opticalmicroscope, double disc polishing machine, emery papers and etchant.
10	Analyse the fracture surface morphology of a ductile or brittle material using an optical microscope for fractographic characterisation.

11	Evaluate the fracture toughness of a material with a Universal Testing Machine (UTM)
12	To study the procedure for plotting S-N curve using Fatigue testing machine
13	Perform stress analysis using photoelasticity.
14	Measure the deformation (strain) of an object using strain gauges.
15	Perform a bending test on a wooden beam to assess its load-carrying capacity.
16	Conduct compressions test on a ductile material (mild steel, tor-steel, or high-strength steel) using a Universal Testing Machine (UTM) equipped with an extensometer.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Evaluate the mechanical properties of different materials under various loading conditions.	K3
CO2	Relate material microstructure to its mechanical behaviour.	K4
CO3	Analyse the effect of design features on the performance of mechanical components.	K4
CO4	Utilize experimental techniques to determine material properties.	K3
CO5	Apply fundamental engineering principles to analyse the behaviour of structures under load.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3			2								
CO2	3			2								
CO3	2		3									
CO4	2			3	2							
CO5	3	2			2							

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Callister's Materials Science and Engineering	D. Wayne Callister and David G. Rethwisch	Wiley	10th Ed (2018)
2	Mechanical Testing and Evaluation	Howard Kuhn; Dana Medlin	ASM International	Volume 8 (2000)

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanics of Materials	James M. Gere and Barry J. Goodno	Cengage Learning	9th Ed (2022)
2	Introduction to Materials Science for Engineers	James F. Shackelford	Pearson	8th Ed (2022)

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc23_mm38/preview
2	https://archive.nptel.ac.in/courses/112/107/112107146/
3	https://archive.nptel.ac.in/courses/112/106/112106293/

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.

- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S4

ELECTRICAL MACHINES LABORATORY

Course Code	PCSFL408	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. The Electrical Machines Laboratory is designed to support the theory study for Electromechanical Devices.
2. The objective of this course is to supplement the theory with suitable practical experiments.

Expt. No.	Experiments
1	Verification of Kirchoff's Laws
2	Verification of Superposition Theorem
3	Study of B.H. Curve on C.R.O
4	Measurement of power in an A.C. circuit by 3 ammeter and 3 voltmeter method
5	Load test on a d.c. series motor
6	Speed characteristics of d.c. shunt motor
7	Regulation of a Transformer
8	Load characteristics of a 3 phase induction motor
9	Study of protective relays and circuit breakers
10	Study of insulation testing and ground testing

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the laws governing electric circuits	K3
CO2	Interpret the load and speed characteristics of d.c motors	K3
CO3	Draw the load and speed characteristics of a.c motors	K3
CO4	Summarize the functions of protective relays and circuit breakers	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3				2							
CO2	3				2							
CO3	3											
CO4	3											

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Basic electrical engineering	Kothari, D.P. and Nagrath, I.J	Tata McGraw Hill Publishing Company Ltd., New Delhi	3(2009)
2	Electrical safety, fire Engineering and safety Management	Fordham-Cooper, W	Khanna Publishers, Delhi	2012

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.dvidshub.net/video/863793/electrical-safety-training https://www.necanet.org/programs/safety/safety-videos-resources https://safeelectricity.org/public-education/educational-lesson-plans-and-activities-for-teachers/
2	
3	
4	

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.

- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 5

SAFETY & FIRE ENGINEERING

SEMESTER S5

STRUCTURAL FIRE SAFETY AND DESIGN OF FIRE PROTECTION SYSTEMS

Course Code	PCSFT501	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Explain the working principle of different types of fire detectors and alarm system and plan and distribute them in buildings of different occupancy.
2. Interpret the concept of fire extinguishment with different extinguishants and the working principle and operation of portable fire extinguishers
3. Plan and distribute appropriate portable extinguishers in buildings of different occupancy.
4. Explain the behaviour of compartment fire at the pre-flashover and post-flashover stages.
5. Interpret the effect of fire on structural and non-structural materials used for construction.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Compartment fire-factors controlling fire severity, ventilation controlled and fuel controlled fires. Pre-flashover compartment fire- growth period and definition of flashover, conditions necessary for flashover, fuel and ventilation conditions for flashover, factors affecting time to flashover, factors affecting growth. Post flashover compartment fire- regimes of burning, fully developed fire behavior, temperatures achieved in fully developed fire, fire resistance and fire severity, methods of calculating fire resistance, projection of flames from burning compartments, spread of fire from compartments. Compartment temperature-time response at pre-flashover and post flashover periods; Equivalence of fire severity of compartment fire and furnace fire;	8

2	Effect of temperature on the properties of structural materials- Thermal data for concrete, steel, aluminium, masonry and wood; Mechanical properties of concrete, steel, aluminium, masonry and wood exposed to elevated temperatures(Material data); Constitutive stress-strain models for steel and concrete.; Fire protection of structural members made of steel, RCC and wood. Fire area- calculation of building fire area, subdivision of fire areas in Industrial, Residential and Public buildings; Fire separation between building-principle of calculation of safe distance. Design principles of fire resistant walls and ceilings; Fire resistant screens-solid screens and water curtains; Local barriers; Fire stopped areas-in roof, in partitioned fire areas, and in connecting structures; Fire doors- Low combustible, Non-combustible and Spark-proof doors; method of suspension of fire doors; Air-tight sealing of doors; Specification, test and performance criteria of Plate, Metal covered and rolling type fire doors as per BIS.	8
3	Fire detection- Need and importance of automatic fire detection system, principle of detection, classification of detectors; Heat detectors – fixed temperature, rate of rise , thermistor rate of rise and rate compensated type detectors; Smoke detectors- optical and ionization type, photo electric light scattering and light obstruction type detectors; Flame detectors – infra red and ultra violet detectors; Flammable gas detection- Pellistor and laser detectors; Testing of fire detection devices as per relevant Indian standards specifications; Performance characteristics of detectors; Lag time associated with fire detection; Comparison of detectors. Principles of Fire Extinguishments-extinction of premixed flames, diffusion flames and burning metals; fire triangle, fire tetrahedron. Basic concept of fire fighting with water, carbon dioxide, powders, foams, inert gases, halons; Need for halon replacement and halon substitutes; Extinguishant performance- flame extinguishing concentration, inerting concentration, fire trials. Description, working principle, method of operation of different types of portable fire extinguishers-water type, foam type, dry powder type, CO2 type, vapourizing liquid type; Selection and distribution of portable extinguishers for different occupancy classification as per NBC;	10
4	Automatic water sprinkler system- requirement and source of water supply, automatic pumps; Sprinkler heads-Quartzoid type, fusible link type, modern types; mounting and protection of sprinkler heads; Sprinkler pipe works-standard and staggered lay out, hangers; Control valves for wet and dry	10

	installations; deluge valve; Drenchers; High velocity and medium velocity spray system. Performance and effectiveness of water sprinkler system Fixed firefighting system using CO₂, Dry chemical powder, and Foam - concept of total flooding and local application, advantages and disadvantages of each system; Basic system components and design principles as per relevant BIS. Design of water sprinkler and spray system as per relevant BIS-application to illustrative problems. Design of fixed foam type firefighting systems for total flooding and for local application as per relevant BIS -Application to typical hydrocarbon storage tank protection problems.	
--	---	--

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the factors affecting fire severity and growth in compartment fires.	K3
CO2	Apply knowledge of material behavior under elevated temperatures to fire-resistant design.	K2
CO3	Select and evaluate fire detection systems based on various detection technologies.	K2
CO4	Design and assess fire suppression systems, including sprinklers and fixed firefighting systems.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									
CO2	3	3	3	2								2
CO3	3	2	3	2								
CO4	3	3	3	3	2	2					2	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design of buildings for fire safety.	Smith, E.E. and Harmathy, T.Z. (Editors),	ASTM Special Publication 685, American Society for Testing and Materials, Boston, U.S.A.	1979
2	Designing of fire safety.	Butcher, E. G. and Parnell, A. C.	John Wiley and Sons Ltd., New York, U.S.A.	1983
3	Fire safety engineering design of structures (2nd edn.),	John A. Purkiss	Butterworth Heinemann, Oxford, U.K. .	2009
4	Underdowns practical fire precautions,	Ron Hirst.	Gower publishing company Ltd., England.	1989
5	Fire safety in buildings (2nd edn.)	Jain V.K.	New Age International (P) Ltd., New Delhi.	2010
6	Fire protection and prevention the essential handbook,	Barendra Mohan Sen	UBS publishers and Dist., New Delhi.	2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Structural design for fire safety,	Andrew H Buchanan.	John Wiley & Sons Ltd., England.	2001
2	An Introduction to Fire Dynamics	Dougal Drysdale	John Wiley & Sons Ltd	Third Edition 2011
3	Principles of fire safety standards for building construction.	Roytman, M. Ya.	Amerind Publishing Co. Pvt. Ltd., New Delhi.	1975
4	Evaluation of Fire Safety,	Rasbash D., Ramachandran G., Kandola B., Watts, J., Law M.,	John Wiley & Sons Ltd., England,	2004
5	Design of special hazard and fire alarm systems, (2nd edn.),	Robert M Gagnon,	Thomson Delmar Learning, New York.	2008

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_me147/unit?unit=16&lesson=35
2	https://onlinecourses.nptel.ac.in/noc24_me147/unit?unit=19&lesson=66
3-4	https://onlinecourses.nptel.ac.in/noc24_me147/unit?unit=20&lesson=71

SEMESTER S5

QUALITATIVE AND QUANTITATIVE APPROACH IN SAFETY

Course Code	PCSFT502	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Develop Foundational Knowledge in Safety and Risk Assessment
2. Techniques
3. Enhance Skills in Hazard Identification and Reliability Engineering
4. Foster Analytical and Quantitative Competence in Safety Engineering

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Safety domain terminologies, Hazard, Risk, Mishap, Incident, Hazard triangle, Risk Assessment & Control, Qualitative Perspective, Qualitative Perspective, Safety Engineering & Accident causing mechanisms, Safety quantification, Preliminary Hazard List, Preliminary Hazard Analysis, Hazard and operability study (HAZOP), Failure Modes and Effects Analysis (FMEA)- Identification of Failure Modes, Application of Hazard Identification Techniques, Fault Tree Analysis (FTA)- Construction, Gate by Gate method, Cut-set method, Importance measures.	11
2	Event Tree Analysis (ETA), Bowtie Tool, Common Cause Cut Sets, Cut-sets for Accident Scenarios, Identification of Safety Barriers, Risk Assessment, Consequence Assessment, Energy Control Model and Hazard Control Hierarchy, Safety Function Deployment, Ranking of Design Solutions: AHP approach, Quantification of Basic Events for Non-repairable Components, Hazard Rate, Exponential Distribution, Weibull Distribution.	11

3	Reliability engineering, Basic concepts – Evaluation of network Reliability / Unreliability – Series systems, Parallel systems, Series - Parallel systems, partially redundant systems – Types of redundancies - Evaluation of network Reliability / Unreliability using conditional probability method. Time Dependent Probability Basic concepts Failure, Fault, Error, Failure mode, Failure classification, Markovian models, Common cause and common mode failures, – Reliability functions $f(t)$, $F(t)$, $R(t)$, $h(t)$ – Relationship between these functions	11
4	Baths tubs curve – Exponential failure density and distribution functions - Expected value and standard deviation of Exponential distribution – Measures of reliability – MTTF, MTTR, MTBF – Evaluation of network reliability / Unreliability of simple Series, Parallel, Series-Parallel systems - Partially redundant systems - Evaluation of reliability measure – MTTF for series and parallel systems – Examples. Monte Carlo simulation, Application to uncertainty analysis, Simple integration, Trivial integration.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand Safety Domain Ontology and Risk Assessment Techniques	K2
CO2	Apply Hazard Identification and Safety Engineering Methods	K3
CO3	Conduct Event Tree Analysis and Use Bowtie Tool for Risk Management	K2
CO4	Implement Reliability Engineering Concepts and Network Reliability Evaluation	K3
CO5	Analyse Failure Data and Apply Reliability Measures	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											2
CO2	3											
CO3	3	3										
CO4	3	2										
CO5	3	3		2								

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	System Reliability Concepts	V. Sankar,	Himalaya Publishing House,	2015.
2	Probabilistic Risk Assessment for Engineering and Scientists: A Systems Approach	Komamoto and Henley	IEEE Press,	1995
3	Techniques for safety management - A systems approach	Petersen D	ASSE	1998
4	Industrial Accident Prevention	Heinrich et al	McGraw Hill	1980

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Reliability Evaluation of Engineering Systems	Roy Billinton and Ronald N. Allan	B. S. Publications,	2007
2	Reliability Engineering	E. Balagurusamy,	Tata McGraw Hill,	2003.
3	Reliability and Maintainability Engineering	Charles E. Ebeling,	Tata McGraw Hill,	2000.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-2	https://onlinecourses.nptel.ac.in/noc23_mg98/unit?unit=32&lesson=33
3-4	https://onlinecourses.nptel.ac.in/noc23_mg98/unit?unit=46&lesson=48

SEMESTER S5

OCCUPATIONAL HEALTH, HYGIENE AND FIRST AID

Course Code	PCSFT503	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. To understand the basics of reactive chemicals in industry.
2. To study the various storage and transportation methods of toxic, inflammable and reactive chemicals.
3. To study the various chemical reactions and variables affecting the reactions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Concept and spectrum of health- functional units and activities of occupational health services- occupational and work related disease- Levels of prevention of diseases – notifiable occupational diseases such as silicosis, asbestosis, pneumoconiosis, siderosis, anthracosis, aluminosis and anthrax - Lead-Nickel, chromium and manganese toxicity- gas poisoning (such as CO, ammonia, coal and dust), their effects and prevention - Industrial toxicology – local and systemic and chronic effects, temporary and cumulative effects – threshold limit values, calculation of TLVs – carcinogens, mutagens, teratogens. Instruments for Radiation detection and measurement. Early recognition of radiation hazard. Personal monitoring devices, Medical support. Hazards associated with the following radiations and preventive measures- Laser, infra-red, ultra violet and ELF.	11
2	Recognition, evaluation and control of physical hazards. Vibration – description and measurement of vibration. Vibration control methods. Effects of whole body vibration on human body and control measures.	11

	Noise- noise measurement, evaluation, noise control methods -hearing loss – causes - Biological effects of noise exposure. Thermal stress – heat disorders and health effects such as heat exhaustion, heat cramp etc. WBGT index, acclimatization. Ventilation systems - purpose of ventilation-general principles ventilation requirements. Physiological and comfort level. Natural ventilation - Dilution ventilation - Mechanical ventilation - Local exhaust ventilation - Ventilation measuring instruments. Fundamentals of hood and duct designs. Standards on ventilation. Industrial Hygiene: Anticipation and Identification, Classification of atmospheric contaminants. Contamination reduction or removal methods.. Hygiene Evaluation, Hygiene Control Purpose of lighting. Advantages of good illumination. Lighting and the work. Sources and kinds of artificial lighting principles of good illumination. Design of lighting installation. Maintenance. Lighting and colour. Standards on lighting and illuminations.	
3	First Aid principles-Role of the first aider-sequence of action on arrival at scene. Vital signs-breathing -pulse. Biomechanics – Structure and functions of musculoskeletal systems, tendons, ligaments, fascia, bone, muscles, joints and basic mechanisms. The respiratory system-respiratory failure - asphyxia- abdominal thrust in Heimlich manoeuvre. Chest injuries-types-fractured ribs -pneumothorax-haemothorax. Unconsciousness-causes-level of consciousness-management of unconscious casualty-problems of unconsciousness. Fainting-recognition-management-aftercare. Diabetes -hypoglycaemia - hyperglycaemia- management Seizures (epileptic fits, convulsions) features- management, stroke. Head injuries-fractures of the base-vault and sides of skull.	11
4	The circulatory system-heart attack-chest compression- CPR. Shock -causes - signs and symptoms - management of shock.Eye-eye injuries-foreign body in eye-eye trauma-corrosive chemical in eye-arc eye. Wounds-bleeding-classification-types of wounds-case of wounds-bleeding from special sites.Fractures- classification of fractures-principles of immobilisation-sprains and dislocation. Broad and narrow fold bandages-hand bandages-slings.The skin. Burns-rule of nines-pure thermal burns. Electric burns. Chemical burns. Radiation burns. Cold burns. Poisoning. Physical fitness. Lifting - casualty handling. Use of stretchers.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub-divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course, students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the concept and spectrum of health -functional units and activities of occupational health service.	K1,K2
CO2	Interpret physical chemical and biological hazards in the work environment and its control measures.	K1,K2
CO3	Recognize the principles of first aid.	K1
CO4	Summarize the anatomy and functions of different human systems.	K1

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3					3						
CO4	3					3						

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Patty's Industrial Hygiene and Toxicology	Clayton, C.D. Clayton F	Wiley Inderscience New York	1981
2	First aid to the injured	Cantlie, James	St John Ambulance Association	1932
3	Basic Principles of Industrial Hygiene	Vivek Kanjiangat Manikandan H	Notion Press Chennai	1 st Edition 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Environmental, Safety and Health Engineering,	Gayle Woodside Dianna Kocurek	John Wiley & Sons New York	1997
2	Encyclopaedia of occupational health and safety. (Four volumes).	Jeanne Mager Stellman	International Labour Office Geneva	1998
3	The industrial environment - Its evaluation and control	-	DHHS (NIOSH) publication number 74-117	1973
4	Accident first aid	Yudenich V.V	Mir Publishers Moscow	1986

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/UFs8Eg8Ll2s
2	https://youtu.be/VlsLoqtep6k
3-4	https://youtu.be/ErxKDbH-iiI

SEMESTER S5

HEAT AND MASS TRANSFER OPERATIONS

Course Code	PBSFT504	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	PBL

Course Objectives:

1. To familiarize the concept of heat conduction, convection and radiation
2. To describe the application of different types of heat exchangers used in industries.
3. To illustrate with examples the different types of evaporators used in chemical industries.
4. To understand the principle of mass transfer, theories and application.
5. To describe absorbers with the help of relevant examples.
6. To compute number of stages in a distillation column with the help of McCabe Thiele method.
7. To describe drying and extraction principles with their application.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Importance of heat transfer in Chemical Engineering operations - Modes of heat transfer -Concept of heat conduction in solids - Fourier's law of heat conduction - Steady state heat conduction- Concept of heat transfer coefficient, Individual and overall heat transfer coefficient. Heat Transfer by Convection and Radiation Convection: Concept of heat transfer by convection concept of boundary layers- Natural and forced convection - Heat transfer from condensing vapours, Heat transfer to boiling liquids Radiation: Concept of thermal radiation -Black body concept – Laws of radiation -concept of grey body, radiation between surfaces	9
2	Evaporators and Heat Exchangers Heat Exchangers: Parallel and counterflow heat exchangers Single pass and multipass heat exchangers; plate heat exchangers; - Log mean temperature difference - heat exchangers effectiveness; number of transfer unit	9

	Evaporators: Types of evaporation -single effect and multiple effect evaporator, natural and forced circulation evaporator	
3	Principles of Mass Transfer: Principles of mass transfer-Fick's law of molecular diffusion, diffusion in solids and liquids. Concept of mass transfer coefficients, theories of mass transfer and their applications, interphase mass transfer and overall mass transfer coefficients in binary systems Absorption: Types of absorbers- plate column packed column Operating characteristics of stage wise and differential contactors.	9
4	Distillation: Relative volatility-simple distillation, steam distillation, distillation with reflux, principle of azeotropic and extractive distillation. McCabe Thiele method of calculation of number of theoretical stages, total, minimum and optimum reflux. Drying and Extraction Drying: Equilibrium moisture and free moisture, critical moisture content, bound and unbound water, rate of drying curves, drying equipments-tray dryers, tower dryers, rotary dryers, fluid-bed dryers, spray dryers Extraction & Leaching: Liquid extraction liquid-liquid equilibrium, equipment for liquid extraction - mixer settlers, spray towers, Solid- liquid extraction- simple leaching, Bollmann extractor	9

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. - Each question carries 6 marks. (4x6 = 24 marks)	40

Project based learning (8 Hours)

1. Direct the students to select a heat exchanger or evaporator. The faculty may ensure that it is suitable for the course. The working should have been analysed for heat transfer calculations, energy balance and applicability of laws.
2. Students should prepare the analysis report before completion of the I module.
3. Students should use appropriate software for the simulation of the operation parameters.
4. Design the process modifications if any.
5. Design the heat exchanger.
6. Prepare a report.
7. Demonstrate/present the results.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the modes of heat transfer	K1,K2
CO2	Analyse heat exchanger performance	K1
CO3	Understand gas- liquid ,liquid- liquid, gas-solid and solid –liquid operations	K2
CO4	Understand drying equipment, extraction and leaching operations	K2,K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3					2						
CO4	3					2						

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Essentials Of Management	Harold Koontz, Heinz Weihrich and Mark V. Cannice	McGraw Hill	11th Edition, 2020
2	Unit Operations of Chemical Engineering,	W.L. McCabe, J.C. Smith and P. Harriott	McGraw Hill,	7th Edition, 2005..

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Process Heat Transfer	Donald Q Kern	Tata McGraw Hill Education Pvt. Ltd.,	2001
2	Mass Transfer Theory and Applications	K. V. Narayanan & B. Lakshmikutty	, CBS, New Delhi	
3	Mass Transfer Operations,	R.E. Treybal,	McGraw Hill	
4	Introduction to Chemical Engineering	Badger & Banchero	Tata McGraw Hill Education Pvt. Ltd	

Module No.	Video Link(NPTEL)
1	https://youtu.be/qa-PQOjS3zA
2	https://youtu.be/qa-PQOjS3zA
3	https://youtu.be/EyREi7I5020 , https://youtu.be/wLfmhIO2sRY
4	https://youtu.be/u0titKpYl8w , https://youtu.be/R3JSdpRR1ls

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S5

NON – DESTRUCTIVE TESTING METHODS

Course Code	PESFT521	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To comprehend the fundamental ideas, methodologies, tools, applications and constraints of NDT approach.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Visual Inspection: Fundamentals of visual testing, tools, applications and limitations. Vision, lighting, material attributes, environmental factors. Visual perception, direct and indirect methods, mirrors, magnifiers, boroscopes, fibroscopes, closed circuit television, light sources special lighting, a systems, computer enhanced system. Liquid penetrant Testing: properties required for a good penetrants and developers - Types of penetrants and developers. LPI technique/ test procedure interpretation and evaluation of penetrant test indications, false indication and safety precaution required in LPI.	9
2	Magnetic Particle Testing: Methods of magnetization, magnetization techniques such as head shot technique, cold shot technique, central conductor testing, and magnetization using yokes. Direct and indirect method of magnetization, continuous testing of MPI, residual technique of MPI, system sensitivity, checking devices in MPI. Eddy Current Testing: physics aspects of ECT .Field factor and lift of effect, edge effect, end effect, impedance plane diagram in brief, depth of penetration of ECT, relation between frequency and depth of penetration in ECT. Equipment and accessories, Various	9

	application of ECT such as conductivity measurement, hardness measurement, defect detection coating thickness measurement.	
3	Ultrasonic Testing: UT testing methods, contact testing and immersion testing, normal beam and straight beam testing, angle beam testing, dual crystal probe, ultrasonic testing techniques, resonance testing, through transmission technique, pulse echo testing technique, instruments used UT, accessories such as transducers, types, frequencies, and sizes commonly used. Reference blocks with artificially created defects, calibration of equipments. Radiography Testing (RT): Electromagnetic radiation sources. Inspection techniques like SWSI, DWSI, DWDI, panoramic exposure, real-time radiography, films used in industrial radiography, types of film, speed of films, qualities of film screens used in radiography, quality of a good radiograph, film processing, interpretation, evaluation of test results, safety aspects required in radiography.	11
4	Advanced NDI Techniques: Principle and Procedure of Digital Signal and image Processing & Digital Image correlation, Acoustic emission Inspection ,Thermography, Computed Tomography	7

**Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Have a basic knowledge of NDT Techniques which enables to carry out various inspections in accordance with the established procedures.	K2
CO2	Familiarize with basic principles of electromagnetic NDT methods	K2
CO3	Apply the principles of signal processing of ultrasonic signals and image processing of radiographic images.	K3
CO4	Have a better knowledge in the field of advanced techniques in NDT	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Practical Non- destructive testing	Baldev Raj	Alpha Science International	2008
2	Non - destructive testing	Hull V and V John	McMillan	2012
3	Non Destructive testing Techniques	Ravi Prakash	New Academic Science	2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Recent developments in the field of non-destructive testing, safety and material science	Elena Lysenko, Alexander Rogachev, Oldrich Stary	Springer	2022
2	New Technologies in electromagnetic non-destructive Testing	Songling Huang & Shen Wang	Springer	2016
3	Recent Advances in Non -Destructive Inspection	Carosena Meola	Nova Science publishers	2010

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-4	https://archive.nptel.ac.in/courses/113/106/113106070/

SEMESTER S5

SAFETY IN MATERIAL HANDLING

Course Code	PESFT522	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Hazards involved in manual and mechanical material handling
2. Selection, testing, usage, inspection and maintenance of material handling equipment and PPE's.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Preventing common injuries, lifting by hand, team lifting and carrying, handling specific shape machines and other heavy objects – accessories for manual handling, hand tools, jacks, hand trucks, dollies and wheel barrows – storage of specific materials - problems with hazardous materials, liquids, solids – storage and handling of cryogenic liquids - shipping and receiving, stock picking, dock boards, machine and tools, steel strapping and sacking, glass and nails, pitch and glue, boxes and cartons and car loading – personal protection – ergonomic considerations.	9
2	Fiber rope, types, strength and working load inspection, rope in use, rope in storage - wire rope, construction, design factors, deterioration causes, sheaves and drums, lubrication, overloading, rope fitting, inspection and replacement – slings, types, method of attachment, rated capacities, alloy chain slings, hooks and attachment, inspection.	9

3	Hoisting apparatus, types - cranes, types, design and construction, guards and limit devices, signals, operating rules, maintenance safety rules, inspection and inspection checklist – conveyors, precautions, types, applications. Powered industrial trucks, requirements, operating principles, operators selection and training and performance test, inspection and maintenance, electric trucks, gasoline operated trucks, LPG trucks	9
4	Power elevators, types of drives, hoist way and machine room emergency procedure, requirements for the handicapped, types- Escalator, safety devices and brakes, moving walks – man lifts, construction, brakes, inspection. Respirator, Breathing Apparatus, Particle Filters – Nanomaterials, specifications, standards selection, inspection, testing and maintenance - Face-pieces, Half and Full Facemasks, Hoods, Helmets, Visors, Blouses, Suits - , specifications, standards selection, inspection, testing and maintenance	9

**Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recognize the practical solutions to eliminate and/or minimize hazards in material handling	K2
CO2	Administer a crane and sling safety to operation.	K3
CO3	Disseminate the basic safety concepts and techniques in mechanical material handling.	K4
CO4	Recognise the safe use, inspection and maintenance of PPE's.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2					2						3
CO2	3				2	2						2
CO3	3					2	3			3		2
CO4	3				2		3					2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Accident Prevention Manual for Industrial Operations	NSC, Chicago	NSC, Chicago	Third Edition 2008
2	Conveyors and related Equipment Vol.I& II	Spivakosky	Peace Pub. Moscow	1982
3	Material Handling Equipment	Alexandrov, M.P	Mir Publishers Moscow	1981

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Material Handling Equipments	Rudenko N	Mir Publishers Moscow	1981
2	Material Handling Hand Book - II	Reymond A.Kulwice	John Wiley and Sons NewYork	1985

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1 -2	https://weeklysafety.com/product/materials-handling-safety-video
3-4	https://donnegan.com/materials-handling-isda-video/

SEMESTER S5

MARKETING MANAGEMENT

Course Code	PESFT523	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2:30
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To evaluate the Marketing concepts and ideas.
2. To analyse the consumer behaviour in the market.
3. To interpret the ideas in pricing of products.
4. To identify modern day advertisement and marketing methods.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	INTRODUCTION: Definition of Marketing- Evolution-Marketing concept- Marketing mix- 4 Ps Frame work-Marketing orientation and philosophies. : Types of Markets-Different Market segmentation- Non segmented markets- Benefits- Limitations - Marketing Research & Environment. Stages in Marketing research- Types of Research Methods- Exploratory- Descriptive- Experimental -Survey Methods - Marketing Environment- Micro& Macro environment -Factors affecting- Economic-Technological- political - Competitive Environment-Green Marketing concept.	9
2	CONSUMER BEHAVIOUR :- Consumer Psychology- Choice criteria- order management cycle – Buying situation- Personal and social influence .PRODUCT DECISION: Concept of a Product – Types of Products- Business, Consumer , Service product-.Commodity- Technology and	9

	Customised product .New product development- Product idea-Product Life Cycle. Brand- Brand attributes- Building a brand name -strategies of corporate Branding.	
3	PRICING STRATEGIES: setting the price of a product-pricing policies and constraints-factors influencing pricing decision-Methods of pricing- cost oriented -competitor oriented- marketing oriented pricing -tactics of price adjustment. price wars- price sensitivity CHANNEL DECISION- Nature of Marketing Channels –. Types of Channel flows –Consumer- Industrial-Service channels. Functions of Distribution Channel – Structure and Design of Marketing Channels -Channel co-operation, conflict and competition – Channel Intermediaries-Franchising Retailers and wholesalers-Theory of retailing.	9
4	ADVERTISEMENTS: Advertisements- Identifying audience-Types of Advertisements-Impact of advertisements. Role of Media in advertisements- Advertisement restrictions & legal actions. DIRECT & INTERNET MARKETING: Direct marketing Techniques- Direct mail-Tele marketing- catalogues- direct response .Internet Marketing- Types of Networks- e business practices in post covid era. B2B-B2C-C2C- C2B exchanges.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To familiarise with the basic terms of marketing.	K2
CO2	Evaluate the marketing concepts and ideas	K3
CO3	Identify the consumer concepts in buying	K2
CO4	Understand the method of channelling the product	K2
CO5	Analysis of various pricing strategies in the market	K4
CO6	Analyse the modern day advertising methods	K4
CO7	Understand the digital marketing methods	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2					2	2					2
CO2	2											2
CO3	2					2	2					2
CO4	2	2	2	2								2
CO5	2	2										2
CO6	2						2					2
CO7	2	2										2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Marketing Management	Phillip Kotler, Kevin Lane Keller	Pearson Publication	15th Edition (2018)
2	Marketing Management	Arun Kumar, N Meenakshi	Vikas Publishing house	2nd Edition (2013)
3	Research for Marketing decisions	Paul E Green, Donald S Tull, Gerald Albaum	PHI learning	5th Edition (2010)
4	Managing Marketing	Noel Capon, Sidharth Shekhar Singh	Wiley Publications	1 ST Edition (2014)

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Marketing Analytics	Wayne L Winston	Wiley publication	2nd Edition 2018
2	Strategic Market Management- Global perspective	David A Aaker, Damien McLoughlin	Wiley Publications	3rd Edition 2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
Module 1	https://archive.nptel.ac.in/courses/110/104/110104068/
Module 2	https://archive.nptel.ac.in/courses/110/104/110104068/
Module 3	https://archive.nptel.ac.in/courses/110/104/110104068/
Module 4	https://archive.nptel.ac.in/courses/110/104/110104068/

SEMESTER S5

HUMAN AND ORGANIZATIONAL FACTORS IN SAFETY

Course Code	PESFT524	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand the fundamental concepts and theories of human and organizational factors in safety.
2. Analyze the impact of human behavior, decision-making, and cognitive processes on safety performance.
3. Evaluate the influence of organizational culture, leadership, and management practices on safety outcomes.
4. Develop strategies for designing and implementing effective safety management systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Occupational safety and industrial safety, Human and organizational factors approach to industrial safety, Definition and scope of human and organizational factors (HOF) in safety. Historical development of HOF, rule-based safety, managed safety, definition and model of safety culture, Elements of the safety culture model, Organisational behaviour, Safety outcomes,	8
2	Human Error Types and Models: Types of human errors, Causes and consequences, Errors and Violations, Models of human error (Swiss Cheese Model, SHEL Model). Techniques for human error identification., Error producing Conditions, Human error in system design and operations. Techniques for managing and mitigating human error. Error-tolerant design and resilient systems. Case studies on human error in engineering incidents (e.g., Chernobyl, Deepwater Horizon). Principles of ergonomics in engineering design (physical, cognitive, organizational ergonomics). Human-	9

	machine interface design, usability, and accessibility.	
3	Safety Management Systems (SMS), Components and principles of SMS (policy, risk management, assurance, promotion). Integration of human factors into SMS. Behavior-Based Safety (BBS), Principles of BBS and its implementation in engineering environments, Role of observation, feedback, and reinforcement in safety improvement, Case studies on BBS applications., Training and Competency Management, Role of simulation, drills, and scenario-based training. Overview of key safety standards (ISO 45001, ISO 31000)	8
4	Human Reliability Analysis (HRA), Overview of HRA methodologies (THERP, HEART, SPAR-H), Application of HRA in complex systems and high-risk industries., Evaluation and mitigation of human errors through HRA. Automation and human factors challenges (AI, robotics, autonomous systems)., Digitalization and safety management (wearables, digital twins, big data analytics). Balancing productivity, cost, and safety	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the impact of human and organizational factors on industrial safety.	K3
CO2	Analyze types of human errors and apply techniques to reduce them.	K2
CO3	Apply safety management systems (SMS) and behavior-based safety (BBS) strategies.	K2
CO4	Use Human Reliability Analysis (HRA) and address challenges in automation and digitalization.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3			3				3				3
CO2	3	3		2				3				
CO3	2	2	3	3			2	2	2	2		3
CO4	2	2	2	3	2		2	2	2		3	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Safety-I and Safety-II: The Past and Future of Safety Management	Erik Hollnagel	CRC Press	2014
2	Introduction to Human Factors and Ergonomics for Engineers	Mark R. Lehto and Steven J. Landry	CRC Press	2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human Factors Methods: A Practical Guide for Engineering and Design	Neville A. Stanton, Paul M. Salmon, Guy H. Walker, and Chris Baber	CRC Press	2017
2	Managing the Risks of Organizational Accidents	James Reason	Ashgate Publishing	1997
3	The Field Guide to Understanding Human Error	Sidney Dekker	CRC Press	2014

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=5eBGRnBdy0o
2	https://www.youtube.com/watch?v=HD9V4NzUg7U
3-4	https://www.youtube.com/watch?v=pGJ0ZZPbgZw

SEMESTER S5

SAFETY IN CONSTRUCTION

Course Code	PESFT526	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Recognize the safety issues at different stages of construction activity.
2. Interpret the safety requirements in various construction operations in framing guidelines to ensure safety at construction site.
3. Interpret the safety requirements in material handling in framing guidelines to ensure safety at construction site.
4. Interpret the safety requirements in the operation and handling of different tools and vehicles in framing guidelines to ensure safety at construction site.
5. Summarize the legal provisions with respect to the health and welfare of workers at construction site.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to construction industry and safety issues in construction – Human factors in construction safety management – Roles of various groups and stake-holders in ensuring safety in construction industry – Framing of contract conditions on safety and related matters – Ergonomic risks in construction.	9
2	Safety in the construction of various elements of buildings– Foundations-excavation, piling and other deep foundations; Wall construction; Roof construction; Erection of concrete framed structures; Erection of structural steel work; construction of Temporary Structures; Safety in various construction operation- General safety requirements at work place; Underpinning & Shoring; Working at height-Use of Ladders and Scaffolds;	9

	Tunnelling; Blasting; working in confined space; Under-water works; Safety in demolition of buildings; Familiarization with relevant Indian Standards and the National Building Code provisions on construction safety.	
3	Safety in storage, stacking and handling of common construction materials- Cement, Lime, Masonry units, floor, wall and roof tiles, Aggregates, Timber, steel, Doors, windows and ventilators, Glass sheets, Flammable materials; Safety in the use of construction equipment- hoists and lifts, lifting gears, wire ropes, chain and pulley blocks pneumatic and hydraulic tools, hand held power tools; Safety in the use of construction vehicles/machinery– Movement of people and vehicles, excavators, graders, dozers, mobile and tower cranes, concrete mixers Different hazards at construction site and their control- Electrical hazard; Fire hazard; Chemical health hazard; Biological health hazard; Physical health hazard; Psychological health hazard.	9
4	Requirements of habitat for construction workers- site selection, minimum facilities, sanitary facilities, drinking water and first aid facilities Contract Labour (R&A) Act and Central Rules: Definitions, Registration of Establishments, Licensing of Contractors, Welfare and Health provisions in the Act and the Rules, Penalties, Rules regarding wages. Building & Other Construction Workers (RE & CS) Act, 1996 and Central Rules, 1998: Applicability, Administration, Registration, Welfare Board & Welfare Fund, Training of Building workers, General Safety, Health & Welfare provisions, Penalties.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recognize the safety issues at different stages of construction activity.	K2
CO2	Interpret the safety requirements in various construction operations in framing guidelines to ensure safety at construction site.	K3
CO3	Interpret the safety requirements in material handling in framing guidelines to ensure safety at construction site.	K3
CO4	Interpret the safety requirements in the operation and handling of different tools and vehicles in framing guidelines to ensure safety at construction site.	K3
CO5	Understand the legal provisions with respect to the health and welfare of workers at construction site.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3					3					2	2
CO3	3					3		2			2	2
CO4	3					3						2
CO5	3							3				3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to health and safety at work	Phil Hughes Ed Ferret	Butterworth-Heinemann UK.	(2nd edn.) (2007)
2	Occupational health and safety in construction project management	Helen Lingard Steve Rowlinson	Spon Press Taylor & Francis Group London	(2005)
3	Construction safety management	Vaid, K.N	National Institute of Construction Management and Research Mumbai	(1988)
4	Relevant Indian standard codes of practice.			

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Construction safety handbook	Davies, V. J Tomasin, K	Thomas Telford Publishing London	(1996)
2	Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Act, and Central Rules			1996
3	The Contract Labour (Regulation and Abolition) Central Rules			(1971)

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.swayam2.ac.in/nou24_ce10/unit?unit=12&lesson=13
2	https://onlinecourses.swayam2.ac.in/nou24_ce10/unit?unit=12&lesson=73
3	https://onlinecourses.swayam2.ac.in/nou24_ce10/unit?unit=6&lesson=70
4	https://onlinecourses.swayam2.ac.in/nou24_ce10/unit?unit=16&lesson=78

SEMESTER S5

TRANSPORTATION SYSTEMS AND SAFETY

Course Code	PESFT527	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Safe transport, handling and storage of hazardous goods
2. Elements road safety, features of roads and vehicles, derive safety programme

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Transportation of Hazardous goods - Transport emergency card (TREM) – driver training-parking of tankers on the highways-speed of the vehicle – warning symbols – design of the tanker lorries -static electricity-responsibilities of driver – inspection and maintenance of vehicles-check list-loading and decanting procedures – communication. Road transport - Introduction – factors for improving safety on roads – causes of accidents due to drivers and pedestrians-design, selection, operation and maintenance of motor trucks-preventive maintenance-check lists-motor vehicles act – motor vehicle insurance and surveys.	10
2	Safety in Driving - Driver safety programme – selection of drivers – driver training-tacho-graph-driving test driver's responsibility-accident reporting and investigation procedures-fleet accident frequency-safe driving incentives-slogans in driver cabin-motor vehicle transport workers act driver relaxation and rest pauses – speed and fuel conservation – emergency planning and Haz mat codes	8
3	Road Safety - Road alignment and gradient-reconnaissance-ruling gradient-maximum rise per kilometre.- factors influencing alignment like tractive	10

	resistance, tractive force, direct alignment, vertical curves breaking characteristics of vehicle-skidding-restriction of speeds-significance of speeds Pavement conditions – Sight distance – Safety at intersections – Traffic control lines and guide posts-guard rails and barriers – street lighting and illumination overloading-concentration of driver. Plant railway: Clearance-track-warning methods-loading and unloading-moving cars-safety practices.	
4	Shop Floor and Repair Shop Safety - Transport precautions-safety on manual, mechanical handling equipment operations-safe driving-movement of cranes-conveyors etc., servicing and maintenance equipment-grease rack operation-wash rack operation-battery charging-gasoline handling-other safe practices-off the road motorized equipment.	8

**Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recognize the safety aspects in design of tanker lorries	K2
CO2	Summarize the loading and decanting procedures of hazardous goods	K3
CO3	Identify the factors for improving safety on roads	K2
CO4	Analyze the causes of road accidents	K4
CO5	Relate road safety with the features of roads and vehicles	K2
CO6	Administer a driver safety programme	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2		2									3
CO2	3	2			2							2
CO3	3					3		3				2
CO4	3	2				3						2
CO5	3	2						2				3
CO6	3		2							3		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Traffic Control and Road Accident Prevention	Popkes, C.A	Chapman and Hall Limited	1986
2	Road Conditions and Traffic Safety	Babkov, V.F	MIR Publications Moscow	1986
3	Traffic Engineering and Transport Planning	Kadiyali	Khanna Publishers New Delhi	1983

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Motor Vehicles Act		Government of India	1988
2	Road Safety guide for drivers of heavy vehicle	Pasricha	Nasha Publications Mumbai	1999
3	Accident Prevention Manual for Industrial Operations		NSC Chicago	1982
4	Safer Roads – A guide to Road Safety Engineering	K.W.Ogden		

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-2	https://nptel.ac.in/courses/105107210
3-4	https://youtu.be/hCGXgy5xydE

SEMESTER S5

THEORY OF FIRE PROPAGATION AND DYNAMICS

Course Code	PESFT528	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to apply the knowledge of combustion to fire propagation and control
2. To emphasize the basics of the subject of fire, its physical characteristics and modelling.
3. To explain the mathematics involved in transport processes during fire propagation as well as the empirical correlations for fire spread and fuel burning rates in detail.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basics of Fire, Fire Triangle, Classification of Fires, Properties of Combustible Materials, Calorific Value, Stoichiometric air, Standard Heat of Combustion, Standard Enthalpy, Adiabatic Flame Temperature, Liquid Fuel Properties - Flash Point, Fire Point, Boiling Point, Latent heat of vaporization, Surface Tension, Marangoni Convection, Solid Fuel Properties - Proximity Analysis, Heat of Pyrolysis, Latent Heat of Melting, Latent Heat of Sublimation, Ignition Temperature, Critical Heat Flux, Thermal Inertia, Thermal Response Parameter, Smoldering and factors affecting smoldering, Backdraft, Smoke, Flame heat flux, Flame spread rate, Flame propagation index, Heat release rate,	8
2	Estimation of- convective heat release, radiative heat release, smoke point, Non-dimensional numbers for fire study, Heat calculations, Fire loading, Standard heat of reaction, Oxygen calorimetry, Carbon-dioxide calorimetry, Sensible enthalpy, Heat calculation for multicomponent fuel, Standard heat of reaction of LPG, Enthalpy of formation of solid fuel, Equilibrium flame	8

	temperature, Analysis of systems- Reaction at constant pressure, constant volume, Chemical kinetics- global reaction, elementary reaction. Molecularity of elementary reaction, Law of mass action, Molecularity and Order, Rate of a reaction, Activated complex, rate coefficient, Equilibrium constants, Net reaction rates, Global reaction rate,	
3	Premixed and diffusion flames, Bunsen burner, Adiabatic flame temperature vs. equivalence ratio, Laminar flame speed vs. equivalence ratio, Laminar flame speed for various fuels, Structure of premixed flames, Preheat and reaction zones, Formation of diffusion flames, Flammability limits, Correlations for flammability limits, Premixed flame theories- thermal theory, diffusion theory, comprehensive theory, Laminar flame speed, Flame thickness, Premixed flame quenching, Quenching distance, Quenching distance for various fuels, Laminar diffusion flame, Candle flame, Structure of diffusion flame, Flame colour, Flame length, Soot formation, Soot oxidation, Flame tip oscillations, Smoke point of saturated, unsaturated and aromatic hydrocarbons.	10
4	Scaling analysis of diffusion flames, Flame Froude number- Momentum and Buoyancy controlled flames, Roper's flame height correlation, Laminar diffusion flame height correlations, Jet flame heights, Turbulent jet diffusion flames, Flame life-off and blow off, Turbulent mass diffusivity, Eddy viscosity, Turbulent mixing length, Turbulent intensity, Turbulent diffusion flame height-Thomas and Heskestad' correlations, Flame height of wall fires, Mass diffusivity, Kinematic viscosity, Lewis number, Prandtl number, Schmidt number, Chemical time, Residence time, Damkohler number, Kinetically controlled and transport controlled fires, Classification of liquid fuels based on flash point values, Liquid fuel evaporation- Steady evaporation using Sfetan's problem, Fick's law, Burning of solid fuels Analysis of fire plumes, Enclosure fires, Introduction to dust ignition, dust explosion and forest fires, Fire safety aspects.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To apply the knowledge of combustion to fire propagation and control	K3
CO2	To understand the basics of the subject of fire, its physical characteristics and modelling.	K2
CO3	To explain the mathematics involved in transport processes during fire propagation	K2
CO4	To explain the empirical correlations for fire spread and fuel burning rates in detail.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3						3					2
CO2	3		2		2							2
CO3	3	2					2					3
CO4	3	2					3					2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanism of Fires: Chemistry and Physical Aspects	Ali S. Rangwala V. Raghavan	ANE Publishers Springer	2021 2022
2	Enclosure Fire Dynamics	Björn Karlsson James G. Quintiere	CRC Press	1999
3	Fundamentals of Fire Phenomena	James G. Quintiere	John Wiley & Sons Ltd	2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fire Dynamics	Gregory E. Gorbett James L. Pharr Scott Rockwell	Pearson Education	2017
2	An Introduction to Fire Dynamics	Dougal Drysdale	John Wiley & Sons Ltd	Third Edition 2011

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_me147/unit?unit=16&lesson=35
2	https://onlinecourses.nptel.ac.in/noc24_me147/unit?unit=19&lesson=66
3-4	https://onlinecourses.nptel.ac.in/noc24_me147/unit?unit=20&lesson=71

SEMESTER S5

RELIABILITY ENGINEERING

Course Code	PESFT525	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Recall probability fundamentals and describe different hazard models, simple and complex systems and tools for assessing reliability.
2. Define probability rules & theorems and exemplify standard distribution functions and simple & complex systems and reliability assessing techniques.
3. Compare different hazard models, systems and compute the reliability of complex systems using different tools.
4. Determine reliability, MTTF, MTBF from failure data, implement methods for reliability improvement and differentiate the reliability assessing techniques.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Reliability: Definition; Probability Concept; Addition of Probabilities; Complimentary Events; Kolmogorov Axioms. Failure Data Analysis: Introduction, Mean Failure Rate, Mean Time to Failure (MTTF), Mean Time between Failures (MTBF), Graphical Plots, MTTF in terms of Failure Density, MTTF in Integral Form.	8
2	Hazard Models: Introduction, Constant Hazard; Linearly Increasing Hazard, The Weibull Model, Density Function and Distribution Function, Reliability Analysis, Important Distributions and their Choice, Standard Deviation and Variance. Conditional Probability: Introduction, Multiplication Rule, Independent Events, Vern Diagram, Hazard Rate as conditional probability, Bayes Theorem.	10

3	System Reliability: Series. Parallel and Mixed Configurations, Complex Systems, Logic Diagrams, Markov Models. Reliability Improvement & Repairable Systems: Redundancy, Element, Unit and standby Redundancy, Optimization; Reliability – cost trade- off, Introduction to Repairable Systems, Instantaneous Repair Rate, MTTR, Reliability and Availability Functions, Important Applications.	10
4	Fault-Tree Analysis and Other Techniques: Fault-tree Construction, Calculation of Reliability, Tie- set and Minimal Tie-set. Maintainability and Availability: Introduction, Maintenance Planning, Reliability and Maintainability trade – off.	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recall probability fundamentals and describe different hazard models, simple and complex systems and tools for assessing reliability.	K1
CO2	Define probability rules & theorems and exemplify standard distribution functions and simple & complex systems and reliability assessing techniques.	K2
CO3	Compare different hazard models, systems and compute the reliability of complex systems using different tools.	K3
CO4	Determine reliability, MTTF, MTBF from failure data, implement methods for reliability improvement and differentiate the reliability assessing techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2				2						2
CO2	3		2									3
CO3	3	3			3		2			2		2
CO4	3	3		2			2					3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Reliability Engineering	Srinath L.S	Affiliated East-West Press, New Delhi.	(2005)
2	Reliability Engineering	Govil A.K	Tata Mc-Graw Hill, New Delhi.	(1983)
3	Reliability Engineering	Rao S.S	Pearson Education	(2014)

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Reliability Engineering	Balagurusamy L	Tata Mc-Graw Hill New Delhi	(2002)
2	Reliability in Engineering Design	Kapur K.C Lamberson L.R	Wiley Publications	(2009)

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1 -4	https://onlinecourses.nptel.ac.in/noc21_ce41

SEMESTER S5

FIRE ENGINEERING LAB

Course Code	PCSFL507	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To summarize the effect of temperature on the viscosity of hydrocarbon liquids
2. To test the quality and hence the acceptability of DCP and AFFF in fighting fire.
3. To test the performance and hence the acceptability of portable fire extinguishers in fighting fire of Class A B C D and Electrical Fire Hazards.
4. To explain the effect of temperature on the strength of building materials.

Expt. No.	Experiments
1	Determination of flash Point of hydrocarbon liquids.
2	Determination of Fire Point and Pour Point of hydrocarbon liquids.
3	Study on the effect of temperature on the viscosity of hydrocarbon liquids using redwood viscometer.
4	Physical tests on Dry Chemical Powder as per relevant Indian standard specifications
5	Physical tests on AFFF as per relevant Indian standard specifications
6	Performance Tests on Portable Fire Extinguishers – AFFF, Water and DCP type
7	Determination of flame spread rate of materials
8	Study on the effect of temperature on strength of building materials – Concrete and masonry units.
9	Rebound hammer test on concrete exposed to elevated temperatures.
10	Test of non-combustibility of Building Materials.
11	Study on Class D Fire Extinguisher.
12	Study of Extinguishers in Electrical Fire.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the effect of temperature on the viscosity of hydrocarbon liquids	K3
CO2	Test the quality and hence the acceptability of DCP and AFFF in fighting fire.	K3
CO3	Test the performance and hence the acceptability of portable fire extinguishers in fighting fire of Class A B C D and Electrical Fire Hazards.	K3
CO4	Explain the effect of temperature on the strength of building materials	K2
CO5	Interpret the method of determination of the flame spread rate of materials.	K4
CO6	Explain the method of determination of the combustibility of building materials.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										2
CO2	3						2					3
CO3	3						2					3
CO4	3											2
CO5	3	3					2					2
CO6	3											2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High)

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Hand book of fire technology	Gupta, R.S	Universities Press Pvt. Ltd., New Delhi . (2010).	2 nd (2010)

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Underdowns practical fire precautions		Ron Hirst Gower publishing company Ltd England.	(1989).
2	Fire safety in buildings		Jain V.K. (New Age International (P) Ltd., New Delhi.	2 nd edn.2010

Video Links (NPTEL, SWAYAM...)	
Exp No.	Link ID
1-12	https://youtu.be/0LNkIcBhl_Q

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S5

CHEMICAL ENGINEERING LAB

Course Code	PCSFL508	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To understand the surface and core characteristics of various materials.
2. To understand the physical chemical and biological features of materials.

Expt. No.	Experiments
1	Sieve Analysis – To analyse a given sample using a set of standard sieves and thus to determine the specific surface area, the volume surface mean diameter and the mass mean diameter by differential analysis and cumulative analysis.
2	Verification of the laws of crushing in a Ball mill and calculation of critical speed.
3	Study of the working of Plate and frame filter press.
4	Free settling – To find out the drag coefficient of a falling sphere in a fluid and verification of Stoke's law.
5	Sedimentation – To study batch sedimentation of slurry and to determine the area of the continuous thickener.
6	Heat transfer from steam to air – Determination of overall heat transfer coefficient.
7	Verification of material balance equation and Rayleigh's equation for simple distillation.
8	Steam distillation.
9	Leaching – leaching a mixture of salt and sand.
10	Study of the kinetics of chemical reaction in a batch reactor.
11	Adsorption isotherms.
12	Frequency response of first and second order systems.
13	Determination of pH, turbidity, total hardness, total solids and dissolved oxygen of water samples.
14	Determination of BOD and COD of waste water samples.
15	Jar test for determining the optimum coagulant dose for water treatment.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Determine the surface characteristics of solid particles	K2
CO2	Determine the energy requirements for size reduction equipment	K2
CO3	Calculate the area required for a continuous thickener	K4
CO4	Compare the efficiency of different types of distillation	K3
CO5	Determine the physical, chemical and biological characteristics of water and waste water	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3						2					2
CO2	3											2
CO3	3	2										2
CO4	3											2
CO5	3					3	3					3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Dryden's outlines of chemical technology for the 21 st century	Gopal Rao, M. and Sittig, M	Affiliated East West Press, New Delhi	2010
2	Chemical reaction engineering	Levenspiel, O	Wiley India Pvt. Ltd., New Delhi.	3 rd 2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Text book of Chemical Technology - Vol.I	Shukla, S.D. Pandey, G.N		1977
2	Elements of chemical reaction engineering	Scott Fogler, H.	Prentice-Hall of India, New Delhi.	4 th 2005
3	Shreve's chemical process industries	George T. Austin	McGraw- Hill Book Co Inc., New York	5 th 1984

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/xKCYYZiqxu0
2	https://nptel.ac.in/courses/103/106/103106139/

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 6

SAFETY & FIRE ENGINEERING

SEMESTER S6

HAZARD CONTROL IN INDUSTRIES

Course Code	PCSFT601	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. Describe the various manufacturing processes- hot & cold working of metals, metal cutting, welding & cutting, heat treatment methods, and material handling methods and the equipment and machineries employed in engineering industries.
2. Summarise the various hazards associated with different manufacturing processes, heat treatment methods and material handling methods.
3. Evaluate the various hazards, relate to the hazard control principles and determine the different hazard control measures.
4. Differentiate various hazards identified and suggested hazard control methods.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction - Classification of Engineering Industry –Manufacturing Processes - Hot Working-Foundry operations-furnace and equipment, health hazard, safe methods of operation. Forging operations, heat radiation, maintenance of machines, shop equipment and hand tools - safe work practice. Operations in hot and cold rolling mills.	11
2	Machinery safeguard-Point-of-Operation, Principle of machine guarding - breakdown of machine guarding - types of guards and devices. Cold Working-Safety in Power Presses, primary & secondary operations - shearing -bending - rolling – drawing. Metal Cutting- safety in turning, boring, milling, planning and grinding. Maintenance of machine tools -	11

	health hazards and prevention.	
3	Welding and Cutting-Safety Precautions of Gas welding and Arc Welding, Cutting and Finishing. Gas Cylinders and Equipment's. Heat Treatment-Furnaces and Salt baths-operations and maintenance -safety in handling and storage of salts- disposal of effluents - health precautions, exposure to hazardous fumes, source of fumes, ventilation and fume protection.	11
4	Material Handling-Classification-safety consideration- manual and mechanical handling. Handling assessments and techniques- lifting, carrying, pulling, pushing, palletizing and stocking. Material Handling equipment-operation & maintenance. Maintenance of common elements-wire rope, chains slings, hooks, clamps.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub-divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course, students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the various manufacturing processes- hot & cold working of metals, metal cutting, welding & cutting, heat treatment methods, and material handling methods and the equipment and machineries employed in engineering industries.	K1,K2
CO2	Summarise the various hazards associated with different manufacturing processes, heat treatment methods and material handling methods.	K2
CO3	Evaluate the various hazards, relate to the hazard control principles and determine the different hazard control measures.	K5
CO4	Differentiate various hazards identified and suggested hazard control methods.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					2			2	2		3
CO2	3					2			2	3		3
CO3	3					3	2		2	2		3
CO4	3					3			2	3		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Safety.	Ronald P. Blake.	Prentice Hall, New Delhi	(3rd edition) (2000)
2	National Safety Council	Chicago Accident prevention manual for industrial operations.		(1982)

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Health and Safety in Welding and Allied process	Balchin, N.C.	Jaico Publishers, New Delhi.	(2005).
2	Manufacturing Engineering and Technology,	Kalpakistan, S Schmid S.R	Pearson Education Asia	(Sixth edition)(2009)

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.swayam2.ac.in/nou24_ge81/unit?unit=10&lesson=46
2	https://onlinecourses.swayam2.ac.in/nou24_ge81/unit?unit=10&lesson=102
3-4	https://onlinecourses.swayam2.ac.in/nou24_ge81/unit?unit=10&lesson=47

SEMESTER S6

ENVIRONMENTAL ENGINEERING AND TOXIC RISKS

Course Code	PCSFT602	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Summarize the most suitable technique for air pollution monitoring and control for a given application.
2. Recognise the type of unit operations and unit processes involved in wastewater treatment plants
3. Explain the techniques for the disposal and management of urban solid wastes and hazardous wastes
4. Exemplify the tools for environmental management in industries.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Air pollution- Sources of air pollution, effects of air pollution, classification of pollutants, Atmospheric transport of pollutants-wind profiles, atmosphere stability, inversion, turbulence, dispersion and diffusion of air pollutants, Gaussian plume dispersion model. Principles and techniques of ambient air and stack emission monitoring; Particulate matter control equipment-working principles of gravity settlers , cyclones, wet scrubbers, fabric filters and electrostatic precipitators; Gaseous control methods- an overview of absorption, adsorption and combustion methods; Biological methods for VOC and odour control.	8
2	Physical, chemical and biological characteristic of waste water; Effects of pollutants on water quality and aquatic life; Physical unit operations in waste water treatment- flow equalization, sedimentation, and flotation; Chemical unit processes in waste water treatment- coagulation and flocculation,	8

	chemical precipitation and adsorption; Biological unit processes- kinetics of microbial growth, Aerobic treatment systems: working principle and design parameters of trickling filter, activated sludge process, and rotating biological contactor; Anaerobic treatment systems: mechanism of anaerobic process, low rate and high rate digesters, working principle and applications of anaerobic filters and UASB; Biological nitrification –denitrification; Characteristics and treatment methods for the waste water from fertilizer plants, petroleum refineries, pulp and paper mills and distilleries.	
3	Solid wastes- environmental ,aesthetic and health risk; Sources, quantities and composition of solid wastes; Storage, collection and transportation of urban solid waste, disposal options sanitary landfills, composting and its variations, anaerobic digestion, incineration and pyrolysis; Vermi composting; Recovery alternative; Monitoring of solid wastes. Hazardous wastes definition and classification, health and environmental effects, treatment, disposal and management of hazardous wastes, legal frame work for hazardous waste management in India.	10
4	Environmental auditing and auditing for waste minimization; Environmental impact assessment description of the environmental setting, prediction and assessment of impacts, methods of impact analysis, Strategies for pollution prevention – recycle and reuse, cleaner technologies. Life cycle assessment – principle and methodology. The concept of industrial ecology. Principles of Environmental Toxicology and the Dose-Response Relationship Toxicity Testing, and ADME I (Absorption, Distribution, Metabolism and Excretion) ADME II (Toxicokinetics and Metabolism), and Classes of Environmental Chemicals, Biochemical Effects and Mechanisms of Toxicity of Pollutants, Environmental Toxicology, Environmental Fate and Sources of Pollutants, Ecotoxicology, Biomarkers and Biomonitoring, Developmental Origins of Disease and Endocrine Disruption, Carcinogenic & Genotoxicity Pollutants and Their Effects, Specialized Toxicology and Case Studies, Ecological and Human Health Risk Assessment Neurotoxicological Effects of Environmental Pollutants.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand and apply the principles of air pollution control, including the monitoring and management of pollutants using various techniques and equipment.	K3
CO2	Analyze the characteristics of wastewater and design appropriate treatment systems using physical, chemical, and biological processes.	K2
CO3	Assess the sources, risks, and management strategies for solid and hazardous wastes, including their disposal and recovery methods.	K2
CO4	Evaluate environmental impacts and toxicological effects of pollutants, employing techniques like environmental impact assessment, life cycle assessment, and risk assessment.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	2	2	3					2
CO2	3	3	3	3	2	2						2
CO3	3	3	3	3	2	3	2					2
CO4	3	3	3	3	3	3	3	2	2		2	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Environmental pollution control engineering.	Rao, C.S.	New Age International (P) Ltd Publishers, New Delhi.	2007
2	Wastewater treatment,	Rao, M.N and Dutta, A.K.	Oxford & IBH Publishing Co. Pvt. Ltd, New Delhi.	1987
3	Wastewater engineering: treatment and reuse. (Fourth edition).	Metcalf and Eddy Inc.	Tata McGraw-Hill Publishing Company Limited, New Delhi.	2003

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Environmental impact assessment. (Second Edition).	Canter, L.W.	Irwin / McGraw – Hill, New York.	1996
2	Environmental engineers handbook. (Second Edition).	David, H.F. Liu, I (Ed).	Lewis Publishers, New York. (1997).	1997

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_me147/unit?unit=16&lesson=35
2	https://onlinecourses.nptel.ac.in/noc24_me147/unit?unit=19&lesson=66
3-4	https://onlinecourses.nptel.ac.in/noc24_me147/unit?unit=20&lesson=71

SEMESTER S6

SUSTAINABLE MANUFACTURING

Course Code	PESFT631	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. This course provides an overview of the Sustainability through Green Manufacturing Systems; various methodologies and its application in improving the eco-efficiency are focused.
2. Students will also learn about the commonly used Sustainable manufacturing tools such as Environmentally Conscious Quality Function Deployment (ECQFD) and Life Cycle Assessment (LCA).

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Concept of sustainability, manufacturing operations, resources in manufacturing. Concept of triple bottom line, environmental, economic and social dimensions of sustainability. Relation between green, lean and sustainable manufacturing.	9
2	Environmental conscious- quality function deployment-R3 and R6 cycles- Environmental impact assessment methods - CML, EI 95 and 99, ISO 14001, EMS and PAS 2050 standards, environmental impact parameters. Sustainability assessment-concept models and various approaches, product sustainability and risk assessment-corporate social responsibility.	10
3	Life cycle analysis-Remanufacture and disposal, tools for LCA, optimization for achieving sustainability in manufacturing, Environmentally Conscious Quality Function Deployment (ECQFD) Extended Producer Responsibility - Value analysis, Analysis for carbon footprint-software packages for sustainability analysis	9

4	Indicators of sustainability - Environmental, Economic, Societal and Business indicators - Concept Models and Various Approaches. Lifecycle of products/services, from the development, to the end of life stages	8
----------	---	----------

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Students will be able to have a basic knowledge about sustainable manufacturing	K2
CO2	Students will be able to use the tools required for implementing sustainable manufacturing	K2
CO3	Students will be able to do calculations required for implementing sustainable manufacturing	K3
CO4	Manufacturing Students will know about practical applications of sustainable manufacturing	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					2	3	3				3
CO2	3					2	3					3
CO3	2	3		3		3	3					3
CO4	2				2	2	3	2			3	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Handbook of Sustainable Manufacturing	G. Atkinson S. Dietz E. Neumayer	Edward Elgar Publishing Limited	2007
2	An Introduction to Sustainable Development	Rogers P.P Jalal K.F Boyd I A	Earth scan London	2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Development for the 21st Century: Sustainable Development Perspectives	D. Rodick	UN New York	2007
2	Sustainable Development Indicators in Ecological Economics	P. Lawn	Edward Elgar Publishing Limited	
3	The Economics of Sustainable Development	S. Asefa	W.E. Upjohn Institute for Employment Research	2005

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc23_me138/unit?unit=18&lesson=19
2	https://onlinecourses.swayam2.ac.in/aic19_ge05/unit?unit=7&lesson=13
3-4	https://onlinecourses.swayam2.ac.in/aic19_ge05/unit?unit=7&lesson=14

SEMESTER S6

INDUSTRIAL NOISE AND VIBRATION CONTROL

Course Code	PESFT632	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To imbibe knowledge on, concepts of vibration and noise and their causation factors
2. To imbibe knowledge on, measurement and control of vibration and noise in industrial environments.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic definitions and terminology used in Vibrations and acoustics – Mathematical concepts and degrees of freedom in vibratory systems – Natural frequencies and vibration modes – continuous systems and wave theory concept – wave equation and relation to acoustics - theory of sound propagation and terminology involved – Plane wave and spherical waves – Concepts of free field and diffuse field, nearfield and far field – frequency analysis and vibration and noise spectrum – Signature analysis and condition monitoring.	8
2	Sensors used in vibration and measurements – Frequency and spectrum analysers – Weighting networks – Hearing mechanism – relation between subjective and objective sounds – Auditory effects of noise and audiometric testing – Speech interference levels and its importance.	8
3	Mechanism of noise generation and propagation in various machinery and machine components, vehicles etc. – Directivity index – Concept of Leq and estimation – Noise ratings and standards for various sources like industrial, construction, traffic, aircraft community etc. – industrial safety and OSHA regulations – Noise legislations and management.	10

4	Energy transferring and dissipating devices Source: Structure borne and flow excited. Vibration isolation and absorption. Spring and damping materials, Dynamic absorbers, Mufflers and silencers, Path: Close filter and loosely covered enclosures – Acoustic treatment and materials – Transmission loss and absorption coefficient of materials and structures and their estimation – Reverberation time and room constant – Design of rooms / industrial halls/ auditorium for minimum noise. Receiver: Measure to control at the receiver end – use of enclosures, ear muffs and other protective devices. Active noise attenuators and scope for abatement of industrial noise.	10
----------	--	-----------

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts of vibration	K2
CO2	Select and apply suitable instrumentation techniques to measure vibration and noise in industrial environment	K3
CO3	Identify sources of vibration and employ actions to reduce or eliminate them at source	K2
CO4	Describe the concepts of noise and methods of reducing the noise	K2
CO5	Demonstrate actions to eliminate or reduce industrial noise at source	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3							3				3
CO2	3				3	3		2	2	2		3
CO3	3					3		2				3
CO4	3					3			2	3		3
CO5	3					3				2		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Noise and Vibration Control,	. Irwin, J.D Graf, E. R,	Prentice Hall Inc. New Jersey,	1979.
2	Theory of Vibrating Systems and Sound,	. Irwing B Crandall,	D. Vannostrand Company, New Jersey,	1974.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	, Hand Book of Noise Control,	Cyril M. Harris	McGraw Hill Book Company, New York,	1971.
	“Noise and Vibration”,	White R. G. Walker J. G,	John Wiley and sons New York,	1982

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I-II	https://onlinecourses.nptel.ac.in/noc24_me83/preview
III-IV	https://onlinecourses.nptel.ac.in/noc24_ae16/preview

SEMESTER S6

HEALTH SAFETY AND ENVIRONMENTAL ASPECTS OF FERTILIZER INDUSTRY

Course Code	PESFT633	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

3. Identify engineering problems in fertilizer manufacturing.
4. Suggest measures for occupational health management in fertiliser industry
5. Identify hazards associated with the various activities in fertiliser manufacturing and suggest loss prevention and control measures
6. Summarise the environmental problems in fertiliser industry and identify measures for the prevention and control of these problems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fertiliser an overview, Types of Fertilisers: (Chemical, Organic & Bio fertiliser), Nitrogenous fertilisers, Phosphatic & Potassic fertilisers, Complex fertilisers - Major Raw materials for fertiliser manufacture - Brief Manufacturing Processes for Major fertilisers: Urea, Ammonium Sulphate, DAP, SSP, Complex Fertilisers, Organic Fertiliser (Composting), Bio fertiliser - Manufacturing process for major intermediates: Ammonia, Nitric Acid, Sulphuric acid, Phosphoric acid.	8
2	Occupational Health problems in fertiliser industries - Measures for preventing the occurrence of occupational health issues - Requirements of ISO 9000, 14000 and OHSAS 1800 - Process hazard analysis and risk management - work controls through health protocols - effective occupational health management.	8

3	Safety in Fertiliser industry - Hazards associated with various fertiliser plant operations - Storage, handling and transport of raw materials and intermediates - Reaction Hazards – Measures for loss control in storage, handling and transport. International and Indian standards for the safe storage, handling, and transport of hazardous chemicals. Disaster preparedness and risk reduction in fertiliser industry. Behavioural safety through mind-set change. Management of hazards associated with highly hazardous chemicals - comprehensive management program that integrates technology, procedures and management practices - PSM assurance through asset integrity - Comprehensive “Management of Change” process - Contractor safety management	10
4	Fertiliser industry - Environmental Challenges: Major Pollutants generated in fertiliser manufacturing processes. Pollution Control measures in fertiliser industry. Pollution Prevention strategies in fertiliser industry. Environmental standards applicable to fertiliser industry. Phosphogypsum Challenge. Meeting the challenge of minimizing the pressures on natural resources, reducing the environmental footprint across the value chain and innovating to meet demands for more environmentally sustainable solutions. Global emission legislation. Climate Change: GHG emissions & Reducing the CO2 footprint.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify engineering problems in fertilizer manufacturing	K2
CO2	Suggest measures for occupational health management in fertiliser industry	K3
CO3	Identify hazards associated with the various activities in fertiliser manufacturing and suggest loss prevention and control measures	K2
CO4	Summarise the environmental problems in fertiliser industry and identify measures for the prevention and control of these problems	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3						3
CO2	3		2			3	3	2		3		3
CO3	3					3						3
CO4	3					3	3	3				3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	. Fertilizer Technology I: Synthesis.	Shishir Sinha, K K Pant Shailendra Bajpai, J N Govil(Acquisition Editor)	Stadium Press LLC.	2015.
2	Fertilizer Technology-II (Biofertilizers).	Shishir Sinha, K. K. Pant.	Stadium Press LLC.	2015.
3	Safety in Chemical Plants/Industry and Its Management.	B. K. Bhaskara Rao, R. K. Jain, Vineet Kumar.	Khanna Publishers.	2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fertilizer Manual.	UN Industrial Development Organization, Int'l Fertilizer Development Center.	Springer Science & Business Media	, 1998.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I-IV	https://onlinecourses.nptel.ac.in/noc24_ag12/preview

SEMESTER S6

LEADERSHIP AND MANAGING CHANGE

Course Code	PESFT634	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Develop a comprehensive understanding of leadership concepts and theories,
2. Learn to manage organizational change using key frameworks, while addressing resistance and shaping organizational culture.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Leadership: Concept, Characteristics, roles, motives, skills and functions. Leadership Vs Management. Effective leadership behaviours and attitudes. Impact of Leadership on organizational performance.	8
2	Leadership styles and theories: popular leadership styles, entrepreneurial and super leadership. transactional Vs Transformational leadership. Trait and Path – Goal theories of leadership. Organisational change: types and forces of change, framework for change management. Proactive change and reactive change. Elements of planned change. Action research model. Individual and organisational barriers to change management and overcoming the barriers to change management	8
3	Influence Of Change: Six belief changers that Influence change, organisational change through influencing individual change. Approaches : Kotter’s eight step plan, Greiner’s Change process model. Four key drivers of organisational change. Factors contributing to resistance to change. Best practices to overcome resistance to change	10

4	Organisational culture and change management: Organisational Culture and Leadership, Types of Cultures, Primary and Secondary ways to influence culture, elements of organisational culture. Diagnosing, creating and managing organisation culture.	10
----------	--	-----------

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand key leadership concepts, styles, and their influence on organizational performance.	K3
CO2	Differentiate between leadership and management, and analyze various leadership theories.	K2
CO3	Gain knowledge of change management frameworks and apply them to organizational scenarios.	K2
CO4	Develop skills to manage and influence organizational culture while overcoming resistance to change.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2			2	2			2	2	2
CO2	3	3	2	2		3	3		2	2	3	2
CO3	2	3	3	3	2	3	3	2	2	2	3	3
CO4	2	2	2	2	2	3	3	2	2	3	3	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Organisational Behaviour	S. Chandan	Vikas Publishing House Pvt. Ltd.	3rd Edition
2	Managing Organizational Change	Nilakant V and Ramnarayan	Response Books,	2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Theory and Behavior	Herbert G. Hicks and C.Ray Gullett	McGrew Hill	-
2	Research Findings, Practice and Skills	Andrew 1 Dubrin	Houghton Mifflin Company,	Third Edition.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I	https://www.youtube.com/watch?v=ryuJTouFXjU
II	https://www.youtube.com/watch?v=loXgscmKcc4
III-IV	https://www.youtube.com/watch?v=z_cZz7avHO8

SEMESTER S6

AUTOMOBILE ENGINEERING AND SAFETY

Course Code	PESFT636	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Describe the construction of all the sub-systems of an automobile: engine, fuel system, electrical system, transmission system and lubricating system.
2. Exemplify the various sub-systems of an automobile system.
3. Summarise the working of the various sub-systems of an automobile system and explain the function of the automobile system.
4. Relate the various provisions of CMV (Central Motor Vehicles) rules pertaining to limiting dimensions, proto-type testing, emission standards, lighting, brakes and safety devices.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Types of automobiles. Limiting Dimensions as per Central Motor Vehicles Rules. Engines – Classification, Construction, Materials of engine components. Prototype Testing as per Central Motor Vehicles Rules. Fuel System – Fuel tank, Fuel filter, Types of Fuel system. Carburettor – Simple and Modern, Fuel injection System. Emission Standards as per CMV Rules.	8
2	Electrical System – Storage Battery Operations and Maintenance. Ignition System – Coil and Magneto Ignition System. Starting System, Lighting System, Horn System – Wind Shield Wiper Motors, Fans, Heaters, and Traficators. Automobile air conditioning. Central Motor Vehicles Rules regarding Lighting, Windshields, Wipers.	8

3	Transmission System – Clutches – operation and fault finding of clutches, Fluid Flywheel, Gear Box-types, Steering Systems, Chassis Springs, and Suspension. Differential, Dead and Live axles, Rims, Tyre etc. Brakes – Types, construction and fault finding. CMV Rules – Brakes, Steering & Tyre.	10
4	Lubrication Systems – Types, Components, Lubricating oil, Cooling system – Details of components, Study of Systems, Types. Miscellaneous – Special gadgets and accessories for fire fighting vehicles. Automobile accidents. CMV Rules regarding Safety devices for drivers, passengers.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the construction of all the sub-systems of an automobile: engine, fuel system, electrical system, transmission system and lubricating system.	K2
CO2	Exemplify the various sub-systems of an automobile system.	K3
CO3	Summarise the working of the various sub-systems of an automobile system and explain the function of the automobile system.	K3
CO4	Relate the various provisions of CMV (Central Motor Vehicles) rules pertaining to limiting dimensions, proto-type testing, emission standards, lighting, brakes and safety devices.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		2									
CO2	3					3						2
CO3	3						3					
CO4	3				2	3	2					3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Automobile Engineering	Narang GBS	Khanna Publishers New Delhi	(2014)
2	Automotive Electrical Equipments	Kohli P.L	Mcgraw Hill Education New Delhi.	(1993)
3	The Central Motor Vehicles Rules			1989

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Automobile Engineering Vol.I &II	Kirpal Singh	Standard publishers- distributors New Delhi	13th edition (2012)
2	Automotive Mechanics - Principles & Practices	Joseph Heitner	CBS Publishers New Delhi	(2 nd edition) (2006)

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
2	https://onlinecourses.swayam2.ac.in/nou24_ec05/unit?unit=6&lesson=7

SEMESTER S6

CHEMICAL TECHNOLOGY AND REACTION ENGINEERING

Course Code	PESFT637	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBSFT304 –Chemical Process Principles	Course Type	Theory

Course Objectives:

1. To understand the basics of reactive chemicals in industry.
2. To study the various storage and transportation methods of toxic, inflammable and reactive chemicals.
3. To study the various chemical reactions and variables affecting the reactions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fuel gases-Natural gas, LPG. Sulphur and sulphuric acid. Phosphorous industries - phosphoric acid, single super phosphate and triple super phosphate, Chloride - alkali industries-soda ash, caustic soda-chlorine-hydrochloric acid Nitrogen industries- ammonia, nitric acid, urea, fertilizer industries- ammonium nitrate and nitrophosphates, mixed and complex fertilizers.	9
2	Natural product industries- soaps and detergents, Pulp and Paper, oils and fats, hydrogenation of vegetable oils, Food processing industry-sugar, industrial alcohol, absolute alcohol, beers, wines. Agrochemicals, Pharmaceuticals-Antibiotics, Basic classification of polymers, Synthesis of polyethylene, polystyrene and PVC, synthetic rubbers- SBR, Plastic recycling	9
3	Classification of chemical reactions, variables affecting rate of reaction, definition of reaction rate. Kinetics of homogeneous reactions – concentration dependent term of a rate equation, temperature dependent term	9

	of a rate equation, theories of reaction – collision theory, transition theory, Arrhenius equation. Analysis of experimental reactor data, evaluation of rate equation, integral and differential analysis for constant and variable volume system	
4	Ideal reactors- Design for homogeneous systems, batch, stirred tank and tubular flow reactor, design equations for batch, continuous stirred tank and tubular flow reactors, size comparison of reactors. Elementary ideas of non-ideal reactor performance, residence time distribution curves E, F and C. The residence time distribution as a factor of performance; residence time functions.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub-divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course, students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the manufacturing methods for heavy chemicals, fertilizers, organic chemicals and polymers	K1,K2
CO2	Explain the importance of physical, chemical and physico – chemical transformations of the material in process industries.	K1,K2
CO3	Solve problems using the integral method and differential method of analysis.	K2,K3
CO4	Compute the volume of Batch, CSTR and PFR reactors using design equations	K2,K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3	2										
CO4	3	2	2									

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Dryden's outlines of chemical technology for the 21 st century	Gopal Rao, M. and Sittig, M	Affiliated East West Press, New Delhi	2010
2	Chemical reaction engineering	Levenspiel, O	Wiley India Pvt. Ltd., New Delhi.	3 rd 2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Text book of Chemical Technology - Vol.I	Shukla, S.D. Pandey, G.N		1977
2	Elements of chemical reaction engineering	Scott Fogler, H.	Prentice-Hall of India, New Delhi.	4 th 2005
3	Shreve's chemical process industries	George T. Austin	McGraw- Hill Book Co Inc., New York	5 th 1984

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-2	https://youtu.be/xKCYYZiqxu0
3-4	https://nptel.ac.in/courses/103/106/103106139/

SEMESTER S6
PRINCIPLES OF INDUSTRIAL MANAGEMENT

Course Code	OESFT614	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3: 0: 0: 0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	OE

Course Objectives:

1. To impart knowledge about various tools and techniques of Industrial Engineering.
2. To facilitate students to acquire knowledge about inventory management, lean manufacturing, agile manufacturing, enterprise resource planning and thus inculcate the skills needed to apply these principles in an organization.
3. To get acquainted with quality management practices

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Scope and functions of Industrial Engineering - Types of production (batch, flow and unit), Roles of line supervisors and production managers. Product Development and Design: Objectives - Quality and cost considerations - Human factors in design - Detailed design & prototyping - Functionality & manufacturability - Standardization, simplification and variety reduction - Concurrent engineering. Plant layout and Material handling: Types of plant layout - Principles of material handling - Material handling equipment – Types, selection and application.	10

2	Production Planning and Control: Aggregate production planning, materials requirement planning - Inventory Management: EOQ models, discount models, P system, Q system, reorder level – Selective inventory control techniques - JIT - Supply chain and management. Break down, preventive and predictive maintenance. Lean Manufacturing (LM): Basic elements – Tools - Concept of wastes - stages of 5S and waste elimination - Need for LM – Toyota Production System	8
3	Agile manufacturing: Definition, business need, conceptual frame work, characteristics and generic features - Approaches to enhance agility in manufacturing - Managing people in agile organization. Enterprise resource planning (ERP): Concept of Enterprise, ERP Overview - Integrated information system - ERP implementation – Benefits, challenges, success and failure factors - Business Process Reengineering (BPR), Customer relationship management (CRM).	8
4	Quality Management: Quality, quality planning, quality control, quality assurance, quality management – TQM, ISO, Six Sigma and Quality circle (Brief description only). Statistical Quality Control - Process capability - Causes of variation in quality- Control charts for $\bar{x}$ and R – Acceptance sampling. Reliability Engineering - Causes of failures - Bath tub curve - System reliability - Life testing.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Implement various tools and techniques in industrial engineering	K3
CO2	Apply inventory control techniques for materials management	K3
CO3	Identify the framework of lean and agile manufacturing	K2
CO4	Identify core and extended modules of enterprise resource planning	K2
CO5	To be conversant with important terms for quality management in organizations	K2
CO6	Implement different quality control techniques	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	-	-	-	-	-	-	-	-	-
CO2	2	3	2	-	-	-	-	-	-	-	-	-
CO3	3	2	2	-	-	-	-	-	-	-	-	-
CO4	3	2	2	-	2	-	-	-	-	-	-	-
CO5	3	2	2	-		-	-	-	-	-	-	-
CO6	2	2	3	-	2	-	-	-	-	-	-	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Engineering & Production Management	M. T. Telsang	S Chand	3rd Edition, 2018
2	Production and operations management	R. Paneerselvam	PHI	3rd Edition, 2012
3	Operations Management: Theory and Practice	B. Mahadevan	Pearson	3rd Edition, 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Engineering and Management – A new perspective	Philips E. Hicks	McGraw Hill	2nd Edition, 1994
2	Statistical Quality Control	Montgomery	Wiley Eastern	6th Edition, 2010
3	Lean and Agile Manufacturing:	S. R. Devadasan, V.	PHI Learning	1st Edition,

	Theoretical, Practical and Research Futurities	M. Sivakumar, R. Muruges and P. R. Shalij		2012
4	Operations Management: Processes and Supply Chains	L. J. Krajewski, M. K. Malhotra, S. K. Srivastava and L. P. Ritzman	Pearson Education	12th Edition, 2019

SEMESTER S6
FUNCTIONAL SAFETY

Course Code	PESFT635	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives

1. Learn the foundational concepts of functional safety and relevant standards (IEC 61508, ISO 26262, IEC 61511) for E/E/PE systems.
2. Develop skills to assess risks and determine Safety Integrity Levels (SIL) using methods like HAZOP, Bow Tie Analysis, and LOPA.
3. Gain proficiency in designing, implementing, and maintaining Safety Instrumented Systems (SIS), including failure management, safety lifecycle, and verification processes.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Deterministic approach and Probabilistic approach, Introduction to functional safety, Overview, History and application of relevant standards (IEC 61508, ISO 26262, IEC 61511), Scope of IEC 61508 in Electrical/Electronic/Programmable Electronics (E/E/PE)., Random Hardware failures, Systematic Failures, Common cause and Common Mode failures, Safety Instrumented Function, Safety Instrumented System, Safety Integrity Level, Systematic Capability, Probability of Failure on Demand (PFD), Probability of Failure per Hour (PFH), Process Safety Time (PST).	8

2	IEC 61511 General Principles, Risk reduction and layers of protection, Evacuation Procedures Internal Emergency Plan (IEP), Emergency Services External Emergency Plan (EEP), SIS Safety life-cycle phases, Risk Assessment methods, Risk Reduction Factor, Tolerable risk, Inductive and Deductive approaches, HAZOP, Bow Tie Method, Prevention safeguard and Protection safeguard, Exercise on HAZOP and Bow Tie. Allocation of Safety functions to the protection layer, Low demand Mode, High Demand mode, Continuous Mode, Determination of SIL for every SIF, Risk assessment Matrix, SIL Allocation/Determination of SIF using Risk graph and LOPA Method, Exercise on QRA.	8
3	SIS Safety Requirement Specifications, Independency of the protective layers, Demand rate and operational modes, SIF structure redundancies, Workshop reliability and acceptable level of spurious trip. Design and realization of SIS, SIS Safety plan, Verification and validation plan, Safety requirement specification, SIS Failure Management, Failure types, Fail-Safe Failures, Dangerous failures, Safe Failure Fraction, Hardware Fault Tolerance (HFT), Prior in use requirements, Analysis and Fault detection – Basic Architecture, Choice safety, Choice availability, Mixed choice.	10
4	PFD Calculation, Failure Rate of Electronic and Mechanical components, MTTF, MTDF, MRT, MTTR, MTBF, PFD of SIF, Reliability block diagram, Series and parallel System. Formula for PFD and PFH, Operating or system software, Utility Software, Application software, Software Safety lifecycle: V Cycle, Software design Documentation, Installation and validation, Nominal Tests, Outside Limit tests, Performance tests, Interface tests, Operation and Maintenance of SIS, Proof tests, Impact of periodic tests, Test Coverage/depth, Importance of test depth in PFD Calculation, Bypass Management, Process demand, Trip test, Spurious trip, Failures and process states, Decommissioning, Verification and audits, Functional Safety Assessment (FSA), Independence Level	10

**Course Assessment
Method (CIE: 40 marks,
ESE: 60 marks)**

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

Continuous Internal Evaluation Marks (CIE):

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand and apply functional safety standards (IEC 61508, ISO 26262, IEC 61511) to E/E/PE systems.	K3
CO2	Perform risk assessments and determine Safety Integrity Levels (SIL) using HAZOP, Bow Tie, and LOPA methods.	K2
CO3	Design and specify requirements for Safety Instrumented Systems (SIS) with redundancy and failure management.	K2
CO4	Calculate safety metrics (PFD, PFH) and conduct verification, validation, and maintenance of SIS.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	2	2	2					2
CO2	3	3	3	3	2	2	3					2
CO3	3	3	3	3	3	2	3	2			2	2
CO4	3	3	3	3	3	2	3	2			2	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Safety Critical Systems Handbook	David J Smith Kenneth GL Simpson	Butterworth-Heinemann	2011
2	IS/IEC 61511 Functional Safety — Safety Instrumented Systems For The Process Industry Sector	-	Indian Standard	2003
3	Fundamentals of Industrial Instrumentation and Process Control, 2nd Edition	William C. Dunn	McGraw-Hill Education	2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Reliability Engineering,	Srinath L.S.	Affiliated East-West Press, New Delhi.	2005
2	Reliability Engineering,	Rao S.S	Pearson Education.	2014

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=ryuJTouFXjU
2	https://www.youtube.com/watch?v=loXgscmKcc4
3-4	https://www.youtube.com/watch?v=z_cZz7avHO8

SEMESTER S6

CHEMICAL PROCESS SAFETY

Course Code	PBSFT604	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	PBL

Course Objectives

1. To give basic idea about the fundamentals of chemical process safety
2. To provide basic knowledge about toxicity and Industrial Hygiene.
3. To impart ideas on source models, explosion protections and testing and commissioning of process plants

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: safety programs- Accident and Loss statistics(Simple Calculations)- Acceptable Risk- Nature of Accident process- Design principles- Conceptual design and detailed design- Flow diagrams- Equipment Specifications- Reliability and safety in designing- Inherently safer design- Engineered Safety. Process and plant hazards: Hazards of pressure- Causes for overpressures- Flow deviations- Effects of leakages/releases- hazards of temperature- Cryogenic fluids- Vibration and noise- Human errors-Ergonomics Toxicity-Effects of Toxicants on Biological Organisms- Dose versus response curve- Threshold limit values- Permissible Exposure Levels Industrial Hygiene: Properties of chemicals- OSHA: Process Safety Management- EPA: Risk Management Plan- Material Safety Data Sheet- Estimating work exposure to toxic vapours- Estimating the vaporization rate of a liquid- Estimating worker exposure during vessel filling operations- Control Techniques: Respirators- Ventilation.	11
2	Fires and Explosions: Fire triangle- Flammability characteristics of liquids and vapours- Flammability Diagram- Ignition energy- Auto ignition- Auto	11

	oxidation- Ignition sources- Explosions-Detonations and Deflagration, Confined explosions, Blast damage resulting from overpressure, TNT equivalency- Energy of chemical, mechanical explosions, Vapour cloud explosions, Boiling liquid expanding vapour explosions(BLEVE)- Inerting- Vacuum purging- Pressure purging- Design methods to prevent electrostatic ignitions- Relaxation- Bonding and Grounding- Dip pipes- Sprinkler systems	
3	Explosion Protection: Containment- Detection- Control- use of fail safe devices- venting- isolation- suppression. Fire Protection: fireproofing- water deluge- water spray systems- emergency scrubbing. Safety for confined space entry- Lockout/Tagout Procedures- Personnel protective systems Relief Valves: Conventional pressure relief valve- Balanced pressure relief valve- Pilot operated pressure relief valve- Temperature activated relief valve system Rupture disc- Rupture pin- Breather vents for storage tanks- Explosion vents- Flame arrestors.	11
4	Scrubbers- Flares- Destructive Testing- Non-destructive testing methods- Thermochemical screening- Colour coding for pipelines and cylinders- Condition monitoring- Control valves- safety valves- bypass valves. Commissioning- Pre-commissioning- start-up and post commissioning- Planning for major emergencies- identification of major hazard situations and assessment of risk, works organisation for major emergencies, Emergency control centre, Assembly points- Evacuation for personnel. Accident Investigations: Learning from accidents- layered investigations- investigation process- investigation summary- aids for diagnosis- aids for recommendations.	11

Course Assessment Method
(CIE: 60 marks , ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. - Each question carries 6 marks. (4x6 = 24 marks)	40

Project based learning (8Hours)

1. Direct the students to select a simple chemical manufacturing process. The faculty may ensure that it is suitable for the course. The process should have been analysed for inherent safety, design for safety, etc.
2. Students should prepare the analysis report before completion of the I module.
3. Students should use appropriate software for the simulation of the process.
4. Design the process modifications if any.
5. Apply design principles in the process.
6. Prepare a report.
7. Demonstrate/present the results.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand fundamentals of chemical process safety	K1,K2,K6
CO2	Gain idea about the plant hazards, source models, explosion protection and testing and commissioning of chemical processes	K1,K2,K4
CO3	Understand explosion protection methods	K2,K4
CO4	Prepare accident reports	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2		2								
CO2	3	2		2								
CO3	3	2		2								
CO4	3											

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical Process Safety: Fundamentals with Applications	Daniel A. Crowl, Joseph F. Louvar	-	(3rd Edition).
2	Chemical Process Industry Safety	K S N Raju	Tata Mc Graw Hill Education	1st Edition

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Accident Prevention Manual for Industrial Operations:		Vol. I & II NSC Chicago	
2	Industrial Environment and its Evolution and Control	NIOSH,		
3	King's Safety in the Process Industries	Ralph King and Ron Hirst	Arnold, London	1998
4	Dangerous properties of industrial materials,	Sax N Irvin	Reinhold, New York	1957

Module	Video Link(NPTEL)
1	http://www.aiche.org/ccps/resources/publications
2	http://surejrajan.wixsite.com/surejrajan

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S6

DESIGN THINKING AND PRODUCT DEVELOPMENT

(Common to Group C & Group D)

Course Code	GZEST605	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	60
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To guide students through the iterative stages of design thinking, including empathizing with users, defining problems, ideating solutions and developing Proof of Concepts (PoC) and technical feasibility studies.
2. To promote the development of critical thinking skills by engaging students in integrative inquiry, where they ask meaningful questions that connect classroom knowledge with real-world applications.
3. To equip students with the ability to involve in product design considering the sustainability, inclusivity, diversity and equity aspects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of design thinking and product development: Overview of stages of product development lifecycle; Design thinking -Definition-Design thinking for product innovation; Bringing social impact in ideation- Identifying societal needs-understanding multi-faceted issues-community engagement and empathetic design- technological innovation meeting societal needs; Understanding and Bridging the divide using Human Centered Design (HCD); Designing for inclusivity in product development- embracing user diversity - Long term impact - sustainability encompassing environmental, economic and social dimensions; Technology Readiness Level in the Innovation Life-cycle; Performing a self-check on innovative ideas - Originality of idea- understanding innovation landscape -	6

	patentability - understanding the economic landscape - Unique Selling Proposition (USP) - Repeatability and Manufacturability - Sustainability - Leveraging business models for comprehensive analysis	
2	Empathize: Design thinking phases; Role of empathy in design thinking; Methods of empathize phase - Ask 5 Why/ 5 W+H questions; Empathy maps - Things to be done prior to empathy mapping - Activities during and after the session; Understanding empathy tools - Customer Journey Map - Personas. Define: Methods of Define Phase: Storytelling, Critical items diagrams, Define success.	6
3	Ideation : Stages of ideation; Techniques and tools - Divergent thinking tools - Convergent thinking tools - Idea capturing tools; Cross-industry inspiration; Role of research in ideation - Market research - consumer research - leveraging research for informed ideation; Technological trends - navigating the technological landscape - Integrating emerging technologies; Feasibility studies - technical, economic, market, operational, legal, and ethical feasibility; Ideation session- techniques and tips. Proof of Concept (PoC) : Setting objectives; Risk assessment; Technology scouting; Document and process management; Change management; Knowledge Capture; Validating PoC; Story telling in PoC presentation	6
4	Design: Navigating from PoC to detailed design; Developing Specification Requirement Document (SRD); Design for manufacturability; Design to cost; Pre-compliance; Design Failure Mode and Effects Analysis (DFMEA); Forecasting future design changes. Prototyping: Alpha prototypes; Beta prototypes; Transition from design to prototype; Goals and expectations for Alpha and Beta prototypes; Effective strategies for maintaining timeline in prototyping; Testing and refining Alpha prototypes; Transitioning to Beta prototypes. Pilot build: Definition and purpose of a pilot build; setting objectives; Identification and selection of manufacturing partner for pilot build; Testing procedures in pilot build; Scaling from pilot build to full-scale production / implementation.	6

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Empathize to capture the user needs and define the objectives with due consideration of various aspects including inclusivity, diversity and equity	K5
CO2	Ideate using divergent and convergent thinking to arrive at innovative ideas keeping in mind the sustainability, inclusivity, diversity and equity aspects.	K6
CO3	Engage in Human Centric Design of innovative products meeting the specifications	K5
CO4	Develop Proof of Concepts (PoC), prototypes & pilot build of products and test their performance with respect to the Specification Requirement Document.	K4
CO5	Reflect on professional and personal growth through the learnings in the course, identifying areas for further development	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2		2	3	3	3	2	2		3
CO2	3	2	3		2	3	3	3	2	2		3
CO3	3	2	3		2	3	3	2	2	2		3
CO4	3	2	2		3	3	3	2	2	2		3
CO5	3					3	3	2	2	2		3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Product Sense: Engineering your ideas into reality	Dr. K R Suresh Nair	NotionPress.com	2024
2	Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation	Tim Brown	HarperCollins Publishers Ltd.	2009
3	Design Thinking for Strategic Innovation	Idris Mootee	John Wiley & Sons Inc.	2013

Sample Assignments:

1. Evaluate and prepare a report on how the aspects including inclusivity, diversity and equity are taken into consideration during the empathize and define phases of the Miniproject course.
2. Evaluate and prepare a report on how the aspects including sustainability, inclusivity, diversity and equity are taken into consideration during the ideate phase of the Miniproject course.
3. Evaluate and prepare a report on how User-Centric Design (UCD) is used in the design and development of PoC of the product being developed in the Miniproject course.
4. Prepare a plan for the prototype building of the product being developed in the Miniproject course.
5. Report on the activities during the empathize phase including the maps & other materials created during the sessions.
6. Report on the activities during the define phase including the maps & other materials created during the sessions.

SEMESTER S6

RELIABILITY ENGINEERING

Course Code	OESFT611	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Recall probability fundamentals and describe different hazard models, simple and complex systems and tools for assessing reliability.
2. Define probability rules& theorems and exemplify standard distribution functions and simple & complex systems and reliability assessing techniques.
3. Compare different hazard models, systems and compute the reliability of complex systems using different tools.
4. Determine reliability, MTTF, MTBF from failure data, implement methods for reliability improvement and differentiate the reliability assessing techniques.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Reliability: Definition; Probability Concept; Addition of Probabilities; Complimentary Events; Kolmogorov Axioms. Failure Data Analysis: Introduction, Mean Failure Rate, Mean Time to Failure (MTTF), Mean Time between Failures (MTBF), Graphical Plots, MTTF in terms of Failure Density, MTTF in Integral Form.	8
2	Hazard Models: Introduction, Constant Hazard; Linearly Increasing Hazard, The Weibull Model, Density Function and Distribution Function, Reliability Analysis, Important Distributions and their Choice, Standard Deviation and Variance. Conditional Probability: Introduction, Multiplication Rule, Independent Events, Vern Diagram, Hazard Rate as conditional probability, Bayes Theorem.	10

3	System Reliability: Series. Parallel and Mixed Configurations, Complex Systems, Logic Diagrams, Markov Models. Reliability Improvement & Repairable Systems: Redundancy, Element, Unit and standby Redundancy, Optimization; Reliability – cost trade- off, Introduction to Repairable Systems, Instantaneous Repair Rate, MTTR, Reliability and Availability Functions, Important Applications.	10
4	Fault-Tree Analysis and Other Techniques: Fault-tree Construction, Calculation of Reliability, Tie- set and Minimal Tie-set. Maintainability and Availability: Introduction, Maintenance Planning, Reliability and Maintainability trade – off.	8

**Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recall probability fundamentals and describe different hazard models, simple and complex systems and tools for assessing reliability.	K1
CO2	Define probability rules & theorems and exemplify standard distribution functions and simple & complex systems and reliability assessing techniques.	K2
CO3	Compare different hazard models, systems and compute the reliability of complex systems using different tools.	K3
CO4	Determine reliability, MTTF, MTBF from failure data, implement methods for reliability improvement and differentiate the reliability assessing techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2				2						2
CO2	3		2									3
CO3	3	3			3		2			2		2
CO4	3	3		2			2					3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Reliability Engineering	Srinath L.S	Affiliated East-West Press, New Delhi.	(2005)
2	Reliability Engineering	Govil A.K	Tata Mc-Graw Hill, New Delhi.	(1983)
3	Reliability Engineering	Rao S.S	Pearson Education	(2014)

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Reliability Engineering	Balagurusamy L	Tata Mc-Graw Hill New Delhi	(2002)
2	Reliability in Engineering Design	Kapur K.C Lamberson L.R	Wiley Publications	(2009)

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-4	https://onlinecourses.nptel.ac.in/noc21_ce41

SEMESTER S6

SUSTAINABLE MANUFACTURING

Course Code	OESFT612	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. This course provides an overview of the Sustainability through Green Manufacturing Systems; various methodologies and its application in improving the eco-efficiency are focused.
2. Students will also learn about the commonly used Sustainable manufacturing tools such as Environmentally Conscious Quality Function Deployment (ECQFD) and Life Cycle Assessment (LCA).

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Concept of sustainability, manufacturing operations, resources in manufacturing. Concept of triple bottom line, environmental, economic and social dimensions of sustainability. Relation between green, lean and sustainable manufacturing.	9
2	Environmental conscious- quality function deployment-R3 and R6 cycles- Environmental impact assessment methods - CML, EI 95 and 99, ISO 14001, EMS and PAS 2050 standards, environmental impact parameters. Sustainability assessment-concept models and various approaches, product sustainability and risk assessment-corporate social responsibility.	10
3	Life cycle analysis-Remanufacture and disposal, tools for LCA, optimization for achieving sustainability in manufacturing, Environmentally Conscious Quality Function Deployment (ECQFD) Extended Producer Responsibility - Value analysis, Analysis for carbon footprint-software packages for sustainability analysis	9

4	Indicators of sustainability - Environmental, Economic, Societal and Business indicators - Concept Models and Various Approaches. Lifecycle of products/services, from the development, to the end of life stages	8
----------	---	----------

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Students will be able to have a basic knowledge about sustainable manufacturing	K2
CO2	Students will be able to use the tools required for implementing sustainable manufacturing	K2
CO3	Students will be able to do calculations required for implementing sustainable manufacturing	K3
CO4	Manufacturing Students will know about practical applications of sustainable manufacturing	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					2	3	3				3
CO2	3					2	3					3
CO3	2	3		3		3	3					3
CO4	2				2	2	3	2			3	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Handbook of Sustainable Manufacturing	G. Atkinson S. Dietz E. Neumayer	Edward Elgar Publishing Limited	2007
2	An Introduction to Sustainable Development	Rogers P.P Jalal K.F Boyd I A	Earth scan London	2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Development for the 21st Century: Sustainable Development Perspectives	D. Rodick	UN New York	2007
2	Sustainable Development Indicators in Ecological Economics	P. Lawn	Edward Elgar Publishing Limited	
3	The Economics of Sustainable Development	S. Asefa	W.E. Upjohn Institute for Employment Research	2005

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc23_me138/unit?unit=18&lesson=19
2	https://onlinecourses.swayam2.ac.in/aic19_ge05/unit?unit=7&lesson=13
3-4	https://onlinecourses.swayam2.ac.in/aic19_ge05/unit?unit=7&lesson=14

SEMESTER S6

CONSTRUCTION SAFETY MANAGEMENT

Course Code	OESFT 613	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2:30
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide basic knowledge on accidents and their impacts on construction; safety legislations and rules to be followed in construction; site safety practices to be followed during construction practices; human factors; ergonomics and cost of accidents and make aware on role of various parties for site safety management.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Accidents in Construction - Nature and Causes of Accidents - Impact of Accidents - Evolution of Safety Concepts - Overview Construction Safety - Current Situation – Organizational Aspect - Behavioural Aspect - Important Safety Rules - Accident Reporting - Storage of Materials - Atmosphere in Confined Place - Prevention from Drowning - Fire Prevention and Protection - First Aid and Medical Care.	9
2	Site Safety Management - Workplace and Equipment - Structures and Equipment - Working Platforms - Safety Organizations - Safety in Construction - Operations Planning For Safety – Excavation - Blasting Tunnelling - Building Works – Scaffolding - Lifting - Use of Electricity - Safety in the Use of Construction Equipment - Psychology of Construction Workers - Rights and Obligation of Parties - Health of Equipment Operators - Vehicles - Cranes - Lifting Gears - Temporary Power Supply	8
3	Safety and Economy - Direct Costs of Accidents - Indirect Cost of Accidents - Cost of Safety Programs - Safety Cost Optimization - Psychological Aspect and Ergonomics – Carelessness – Related Physical Factor - Other Factors - The Shop Environment and Safe Behaviour - Job Stress and its Effect -	9

	Biomechanics and Ergonomics - Human Factors in Construction Safety - Employee Selection – Placement - Motivation: Awareness and Training	
4	Personal Protection - Eye Protection - Finger, Arm and Hand Protection - Foot and Leg Protection - Noise Safeguard - Head Protection - Safety Belt Safety Legislation in Construction Industry - Safety Codes Applicable to Construction Industry - ILO Standards - OSHA Regulations -Health and Safety Provision in India - Contract Conditions on Safety in Civil Work Projects Safety Management Role of Various Parties – Employers – Designers – Supervisors - Manufacturers/Dealers - Workers/employees - Motivating management - Contractual provisions	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recognize the safety issues at different stages of construction activity.	K2
CO2	Understand the safety requirements in various construction operations and frame guidelines to ensure safety at construction site.	K2
CO3	Interpret the safety requirements in the operation and handling of different tools, equipment and vehicles in framing guidelines to ensure safety at construction site.	K3
CO4	Summarize the legal provisions with respect to the health and welfare of workers at construction site.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					2	3	3				3
CO2	3					3						3
CO3	3					3					2	3
CO4	3					3		3		2		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Construction Safety Management	Vaid, K.N.	NICMAR Publication	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Safety Management	Grimaldi John V Simonds R.H	All India Traveller Book Seller	Fifth Edition

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I	https://onlinecourses.swayam2.ac.in/nou24_ce10/unit?unit=1&lesson=2
II	https://onlinecourses.swayam2.ac.in/nou24_ce10/unit?unit=30&lesson=55
III	https://onlinecourses.swayam2.ac.in/nou24_ce10/unit?unit=1&lesson=4
IV	https://onlinecourses.swayam2.ac.in/nou24_ce10/unit?unit=6&lesson=9

SEMESTER S6
PRINCIPLES OF INDUSTRIAL MANAGEMENT

Course Code	OESFT614	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	OE

Course Objectives:

1. To impart knowledge about various tools and techniques of Industrial Engineering.
2. To facilitate students to acquire knowledge about inventory management, lean manufacturing, agile manufacturing, enterprise resource planning and thus inculcate the skills needed to apply these principles in an organization.
3. To get acquainted with quality management practices

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Scope and functions of Industrial Engineering - Types of production (batch, flow and unit), Roles of line supervisors and production managers. Product Development and Design: Objectives - Quality and cost considerations - Human factors in design - Detailed design & prototyping - Functionality & manufacturability - Standardization, simplification and variety reduction - Concurrent engineering. Plant layout and Material handling: Types of plant layout - Principles of material handling - Material handling equipment – Types, selection and application.	10
2	Production Planning and Control: Aggregate production planning, materials requirement planning - Inventory Management: EOQ models, discount models, P system, Q system, reorder level – Selective inventory control	8

	techniques - JIT - Supply chain and management. Break down, preventive and predictive maintenance. Lean Manufacturing (LM): Basic elements – Tools - Concept of wastes - stages of 5S and waste elimination - Need for LM – Toyota Production System	
3	Agile manufacturing: Definition, business need, conceptual frame work, characteristics and generic features - Approaches to enhance agility in manufacturing - Managing people in agile organization. Enterprise resource planning (ERP): Concept of Enterprise, ERP Overview - Integrated information system - ERP implementation – Benefits, challenges, success and failure factors - Business Process Reengineering (BPR), Customer relationship management (CRM).	8
4	Quality Management: Quality, quality planning, quality control, quality assurance, quality management – TQM, ISO, Six Sigma and Quality circle (Brief description only). Statistical Quality Control - Process capability - Causes of variation in quality- Control charts for $\bar{x}$ and R – Acceptance sampling. Reliability Engineering - Causes of failures - Bath tub curve - System reliability - Life testing.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Implement various tools and techniques in industrial engineering	K3
CO2	Apply inventory control techniques for materials management	K3
CO3	Identify the framework of lean and agile manufacturing	K2
CO4	Identify core and extended modules of enterprise resource planning	K2
CO5	To be conversant with important terms for quality management in organizations	K2
CO6	Implement different quality control techniques	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	-	-	-	-	-	-	-	-	-
CO2	2	3	2	-	-	-	-	-	-	-	-	-
CO3	3	2	2	-	-	-	-	-	-	-	-	-
CO4	3	2	2	-	2	-	-	-	-	-	-	-
CO5	3	2	2	-		-	-	-	-	-	-	-
CO6	2	2	3	-	2	-	-	-	-	-	-	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Engineering & Production Management	M. T. Telsang	S Chand	3rd Edition, 2018
2	Production and operations management	R. Paneerselvam	PHI	3rd Edition, 2012
3	Operations Management: Theory and Practice	B. Mahadevan	Pearson	3rd Edition, 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Engineering and Management – A new perspective	Philips E. Hicks	McGraw Hill	2nd Edition, 1994
2	Statistical Quality Control	Montgomery	Wiley Eastern	6th Edition, 2010
3	Lean and Agile Manufacturing: Theoretical, Practical and Research Futurities	S. R. Devadasan, V. M. Sivakumar, R. Muruges and P. R. Shalij	PHI Learning	1st Edition, 2012
4	Operations Management: Processes and Supply Chains	L. J. Krajewski, M. K. Malhotra, S. K. Srivastava and L. P. Ritzman	Pearson Education	12th Edition, 2019

SEMESTER S6

QUALITY ENGINEERING AND MANAGEMENT

Course Code	OESFT615	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. The course aims to impart basic knowledge about various aspects of Quality Engineering and Management.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Quality Management- A conceptual frame work, Strategic Quality Management, Benchmarking. Quality system Standards, Bureau of Indian Standards, Agmark Grading and standardization, Quality council of India, International Organization for Standardization, Conformance to Specifications, Quality Assurance, Quality Circles, Quality audits, ISO 14000, Customer Operations Performance Centre (COPC).	9
2	W. Edwards Deming's Contribution to TQM, Juran's Contribution to TQM, Crosby's contribution to TQM, Ishikawa's contribution to TQM, Comparing the Quality Gurus, Total Productive Maintenance (TPM).	8
3	Products and services, Classification of services, Service Quality, Measuring Service Quality, Prevention costs, Appraisal Costs, Internal and External failure costs, Cost of quality models, India's Quality Journey so far, Quality management in India, Quality related priorities of Indian companies, Case studies	9
4	Meaning of Six sigma, The seven magnificent Quality tools, Introduction of experimental design, Taguchi Method in Experimental Design, Concept, Application of QFD, and Case Study. Quality control-its introduction and benefits, Variation in processes: factors, process capability & Its analysis,	10

	control charts for variables and attributes, Establishing & interpreting control charts, Concept of Acceptance Sampling, sampling by attributes, single and double sampling plans, inspections by samples, AQL, LTPD, consumers and producer's risk, construction and use of operating characteristic curves, use of standard sampling tables and related IS, sampling by variables, Continuous sampling plan, vendor ratings.	
--	--	--

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recall statistical fundamentals and describe the basic concepts, principles and tools of Quality	K2
CO2	Exemplify the principles of Quality Management, the seven tools of SQC, the tools of Quality Systems.	K3
CO3	Determine the hindrances for implementation of Quality management principles, application of seven tools of SQC and the tools of Quality Systems.	K2
CO4	Differentiate the costs of quality and outline the processes of development of Quality Policy in an organization	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3						3
CO2	3		2		2	3		2				3
CO3	3					3						3
CO4	3					3						3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Quality Without Tears,	Croby, Philip B.,	McGraw Hill Book Company New York,	2001,
2	Quality Management	Kanishka Bedi,	Oxford University Press, New Delhi.	2008
3	Quality Management	Donna C S Summers	Pearson	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	TQM	Oakland S John,	Butterwort-Heinemann.	2000,
2	What is Total Quality Control? The Japanes Way	Ishikava, K	Prentice-Hall, Englewood Cliffs, N.J. (USA).	1995

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I-IV	https://onlinecourses.swayam2.ac.in/nou20_cs12/preview

SEMESTER S6
CHEMICAL PROCESS SAFETY

Course Code	OESFT616	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives

1. To give basic idea about the fundamentals of chemical process safety
2. To provide basic knowledge about toxicity and Industrial Hygiene.
3. To impart ideas on source models, explosion protections and testing and commissioning of process plants

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: safety programs- Accident and Loss statistics(Simple Calculations)- Acceptable Risk- Nature of Accident process- Design principles- Conceptual design and detailed design- Flow diagrams- Equipment Specifications- Reliability and safety in designing- Inherently safer design- Engineered Safety. Process and plant hazards: Hazards of	9

	pressure- Causes for overpressures- Flow deviations- Effects of leakages/releases- hazards of temperature- Cryogenic fluids- Vibration and noise- Human errors-Ergonomics Toxicity-Effects of Toxicants on Biological Organisms- Dose versus response curve- Threshold limit values- Permissible Exposure Levels Industrial Hygiene: Properties of chemicals- OSHA: Process Safety Management- EPA: Risk Management Plan- Material Safety Data Sheet- Estimating work exposure to toxic vapours- Estimating the vaporization rate of a liquid- Estimating worker exposure during vessel filling operations- Control Techniques: Respirators- Ventilation.	
2	Fires and Explosions: Fire triangle- Flammability characteristics of liquids and vapours- Flammability Diagram- Ignition energy- Auto ignition- Auto oxidation- Ignition sources- Explosions-Detonations and Deflagration, Confined explosions, Blast damage resulting from overpressure, TNT equivalency- Energy of chemical, mechanical explosions, Vapour cloud explosions, Boiling liquid expanding vapour explosions(BLEVE)- Inerting- Vacuum purging- Pressure purging- Design methods to prevent electrostatic ignitions- Relaxation- Bonding and Grounding- Dip pipes- Sprinkler systems	9
3	Explosion Protection: Containment- Detection- Control- use of fail safe devices- venting- isolation- suppression. Fire Protection: fireproofing- water deluge- water spray systems- emergency scrubbing. Safety for confined space entry- Lockout/Tagout Procedures- Personnel protective systems Relief Valves: Conventional pressure relief valve- Balanced pressure relief valve- Pilot operated pressure relief valve- Temperature activated relief valve system Rupture disc- Rupture pin- Breather vents for storage tanks- Explosion vents- Flame arrestors.	9
4	Scrubbers- Flares- Destructive Testing- Non-destructive testing methods- Thermochemical screening- Colour coding for pipelines and cylinders- Condition monitoring- Control valves- safety valves- bypass valves. Commissioning- Pre-commissioning- start-up and post commissioning- Planning for major emergencies- identification of major hazard situations and assessment of risk, works organisation for major emergencies, Emergency control centre, Assembly points- Evacuation for personnel. Accident Investigations: Learning from accidents- layered investigations- investigation process- investigation summary- aids for diagnosis- aids for recommendations.	9

Course Assessment Method
(CIE: 60 marks , ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. - Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand fundamentals of chemical process safety	K1,K2,K6
CO2	Gain idea about the plant hazards, source models, explosion protection and testing and commissioning of chemical processes	K1,K2,K4
CO3	Understand explosion protection methods	K2,K4
CO4	Prepare accident reports	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2		2								
CO2	3	2		2								
CO3	3	2		2								
CO4	3											

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical Process Safety: Fundamentals with Applications	Daniel A. Crowl, Joseph F. Louvar	-	(3rd Edition).
2	Chemical Process Industry Safety	K S N Raju	Tata Mc Graw Hill Education	1st Edition

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Accident Prevention Manual for Industrial Operations:		Vol. I & II NSC Chicago	
2	Industrial Environment and its Evolution and Control	NIOSH,		
3	King's Safety in the Process Industries	Ralph King and Ron Hirst	Arnold, London	1998
4	Dangerous properties of industrial materials,	Sax N Irvin	Reinhold, New York	1957

Module	Video Link(NPTEL)
1	http://www.aiche.org/ccps/resources/publications
2	http://surejrajan.wixsite.com/surejrajan

OESFT617

SEMESTER S6

ENVIRONMENTAL ENGINEERING AND INDUSTRIAL HYGIENE LAB

Course Code	PCSFL607	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. Estimate the different pollutants in the atmosphere using air sampling equipment.
2. Compute heat stress in industries.
3. Determine the noise level in industries.
4. Determine the illumination level in working environment.
5. Interpret the different body movements.

Expt. No.	Experiments
1	Study of Personal Protective equipments

2	Study of occupational diseases with photographic models.
3	Study of different body movements
4	Demonstration of Air sampling equipment.
5	Sampling and estimation of dusts using high volume sampler
6	Sampling and estimation of dust using personal sampler
7	Measurement of Noise
8	Measurement of illumination
9	Vision testing
10	Lung function testing.
11	Measurement of thermal stress.
12	Study of industrial fatigue

**Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Estimate the different pollutants in the atmosphere using air sampling equipment.	K3
CO2	Compute heat stress in industries.	K3
CO3	Determine the noise level in industries.	K2
CO4	Determine the illumination level in working environment.	K2
CO5	Interpret the different body movements.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3					2					3
CO2	3	3										3
CO3	3	2										3
CO4	3											3
CO5	3											2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High)

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Basic Principles of Industrial Hygiene	Vivek K Manikandan H	Notion Press Chennai	1 st Edition 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction To Environmental Engineering	Mackenzie L. Davis David A. Cornwell	McGrawHill Inc	2017
2	Basic Concepts of Industrial Hygiene	Ronald Scott	CRC Press	1997

Video Links (NPTEL, SWAYAM...)	
Exp No.	Link ID
1-12	https://onlinecourses.swayam2.ac.in/nou24_es06/unit?unit=27&lesson=32

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S6

RELIABILITY AND COMPUTATIONAL LABORATORY

Course Code	PCSFL608	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives

1. Proficiency in Safety and Fire Engineering Software Tools
2. Advanced Computational Techniques for Safety Engineering
3. Quantitative Risk Assessment and Reliability Engineering

Expt. No.	Experiments
1	Application of BIM (Building Information Modelling) software for designing and drafting building fire protection systems.
2	Use of spread sheets in computing and graphically representing data in safety and fire engineering.
3	Use of statistical software for safety application –regression, ANOVA, statistical tests of significance.
4	Applications of consequence modelling software in quantitative risk assessment (QRA)
5	Scientific computing using MATLAB to solve safety engineering problems.
6	Application of system simulation in solving plant safety problems.
7	Fire and smoke modelling using open-source software.
8	Monte Carlo simulation, Application to uncertainty propagation
9	Monte Carlo Simulation, Application in resolving numerical integrals (Trivial Integration, Simple Integration and Double integration)
10	Exercises in reliability engineering to find out MTTR, repair rate and Lifetime Distribution

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Design and draft building fire protection system by Building Information Modelling.	K3
CO2	Analyse data in safety and fire engineering using spread sheets and statistical software.	K3
CO3	Estimate quantitative risk using software packages.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	-	-	-	-	-	-	-	-	-
CO2	3	3	2	2	-	2	-	-	-	-	-	2
CO3	3	3	-	-	-	-	-	3	3	3	-	-

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Techniques for safety management - A systems approach	Petersen D	ASSE	1998
2	Reliability Evaluation of Engineering Systems	Roy Billinton and Ronald N. Allan	B. S. Publications,	2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Reliability and Maintainability Engineering	Charles E. Ebeling,	Tata McGraw Hill,	2000.
2	Probabilistic Risk Assessment for Engineering and Scientists: A Systems Approach	Komamoto and Henley	IEEE Press,	1995

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc23_mg98/unit?unit=59&lesson=60
2	https://onlinecourses.nptel.ac.in/noc23_mg98/unit?unit=39&lesson=43

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.

- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 7

SAFETY & FIRE ENGINEERING

SEMESTER S7

ENERGY MANAGEMENT AND CONSERVATION

Course Code	PESFT741	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Calculate the efficiency of various thermal utilities.
2. Design suitable energy monitoring system to analyze and optimize the energy consumption in an organization.
3. Design suitable systems for heat recovery and co-generation to improve thermal efficiency.
4. Relate energy audit methods learnt to identify the areas deserving tighter control to save energy expenditure.
5. Explain to the employees of the organization about the need and the methods of energy conservation

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Energy Scenario :Commercial and Non-commercial energy, primary energy resources, commercial energy production, final energy consumption, Indian energy scenario, Sectoral energy consumption (domestic, industrial and other sectors), energy needs of growing economy, energy intensity, long term energy scenario, energy pricing, Energy security, energy conservation and its importance, energy strategy for the future, Energy Conservation Act 2001 and its features. Basics of Energy its various forms and conservation :Electricity basics - Direct Current and Alternative Currents, electricity tariff, Thermal Basics-fuels, thermal energy contents of fuel, temperature and pressure, heat	10

	capacity, sensible and latent heat, evaporation, condensation, steam, moist air and humidity and heat transfer.	
2	Evaluation of thermal performance: calculation of heat loss - heat gain, estimation of annual heating & cooling loads, factors that influence thermal performance, analysis of existing buildings setting up an energy management programme and use management - electricity saving techniques Energy Management & Audit: Definition, energy audit, need, types of energy audit. Energy management (audit) approach-understanding energy costs, Bench marking, energy performance, matching energy use to requirement, maximizing system efficiencies, optimizing the input energy requirements, fuel and energy substitution, energy audit instruments and metering	9
3	Energy Monitoring and Targeting: Defining monitoring & targeting, elements of monitoring & targeting, data and information-analysis, techniques - energy consumption, production, cumulative sum of differences (CUSUM). Energy Management Information Systems (EMIS)	8
4	Energy Efficiency in Thermal Utilities and systems: Energy efficiency in thermal utilities like boilers, furnaces, pumps and fans , compressors, cogeneration (steam and gas turbines), heat exchangers ,lighting system, Motors belts and drives, refrigeration system. Heat Recovery and Co-generation: Heat recovery from ventilation, air co-generation of heat and electricity, heat recovery and bottoming cycles.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Calculate the efficiency of various thermal utilities.	K3
CO2	Design suitable energy monitoring system to analyze and optimize the energy consumption in an organization.	K3
CO3	Design suitable systems for heat recovery and co-generation to improve thermal efficiency.	K3
CO4	Relate energy audit methods learnt to identify the areas deserving tighter control to save energy expenditure.	K4
CO5	Explain to the employees of the organization about the need and the methods of energy conservation	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2		2			3					
CO2	3		3		3				2		2	
CO3	3		3		3		3		3			
CO4	3			3	2						3	3
CO5	3					3		3		3		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Energy Conservation in Process Industry,	Kenny W.F.	Academic Press.	(2012).
2	Energy Engineering and Management	Amlan Chakrabarti	Prentice Hall India.	(2011).
3	Energy Management Principles	Smith C.B. Parmenter K.	Elsevier	(Second edition). (2015).

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hand outs,		Bureau of energy efficiency, New Delhi.	
2	Energy Management Hand Book.	Turner W. C Steve Doty	The Fairmount Press Inc	(Sixth edition), (2007)

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
II	https://onlinecourses.swayam2.ac.in/nou24_es10/unit?unit=6&lesson=8

SEMESTER S7

INDUSTRIAL PSYCHOLOGY

Course Code	PESFT742	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Recognize the nature, scope, and challenges of Industrial Psychology
2. Interpret the psychological factors that influence individual differences in behaviour at work
3. Explain the concepts of motivation and job satisfaction
4. Summarize information required to sustain employability

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Industrial Psychology: Nature and Development of Industrial/Work Psychology - Historical background- Scientific Management, Time and Motion Study, Hawthorne Studies, World War I & II. Scope & Challenges: Current status.	8
2	People at Work: Individual Differences: Personality, Intelligence, Emotional Intelligence, Creativity & Innovation, Perception & Attitudes. Motivation- N-Ach, Expectancy, Equity & Goal Setting Theories, Modern Approach to Motivation: Employee Predisposition, Expectations, Goals, & Incentives - Job Satisfaction- Job Characteristic Theory (Diagnostic Model), Measuring Job Satisfaction - Psychometric Testing at Work- Cognitive Abilities, Personality, Emotional Intelligence.	10

3	Employee Selection- Techniques, Fair Employment Practices. Recruitment- Biographical Information, Interviews, References & Letters of Recommendation. Job Analysis- Types, Newer Developments.	8
4	Performance Assessment: Evaluation & Appraisal- Objective & Subjective Techniques, Bias in Appraisals, Development Plans. Organizational Training- Types of Training, Psychological Issues. Career Development & Planning.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recognize the nature, scope, and challenges of Industrial Psychology	K2
CO2	Interpret the psychological factors that influence individual differences in behaviour at work	K3
CO3	Explain the concepts of motivation and job satisfaction	K2
CO4	Summarize information required to sustain employability	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3		3				3
CO2	3					3		2	2	2		3
CO3	3				2	3		2				3
CO4	3					3			2	3		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Psychology and Work Today: An Introduction to Industrial and Organizational Psychology.	Schultz, D. Schultz, S. E.	Pearson Education: New Delhi	7th Edition. . (2013).
2	Industrial Psychology.	Aamodt, M.G.	Cengage Learning: Delhi.	(2013).
3	Work Psychology.	Matthewman, L., Rose, A. Hetherington, A.	Oxford University Press: India.	(2009).

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Work in the 21st Century: An Introduction to Industrial and Organizational Psychology.	Landy, F. J. Conte, J. M.	Wiley India: New Delhi.	2nd Edition. (2010).
2	Psychology and Work Today.	Schultz, D. Schultz, S. E.	Pearson Education: New Delhi.	(2002).

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I	https://onlinecourses.nptel.ac.in/noc24_hs96/preview
II	https://onlinecourses.swayam2.ac.in/cec24_ed27/preview
III-IV	https://onlinecourses.nptel.ac.in/noc24_hs125/preview

SEMESTER S7

ENTREPRENEURSHIP AND SMALL BUSINESS ENTERPRISES

Course Code	PESFT743	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Recognize different types of entrepreneurial ventures
2. Interpret opportunity and risk analysis
3. Summarize the strategies for valuing their own company, and how venture capitalist and angel investors use valuations in negotiating milestones, influence and control
4. Determine correct marketing mix and how to position the company in the market by using analytical tools
5. Explain how organizations operate and their process matrices

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Market Research: Introduction to Entrepreneurship, Profile of the Entrepreneur, Market Gap /Opportunity Analysis, Market Research Methods, Defining the Focal Market: Market Segmentation, Industry analysing – Research / Competitive Analysis.Types of Companies and Organizations: Company/ Organization Types, Legal Aspects, Taxation, Government Liaison, Building the Team, Mergers and Acquisitions, import and export nuances.	10
2	Business Finance: Shares and Stakes, Valuation, Finance Creation (Investors / Financers), Revenue Plans and Projections, Financial Ratios, Business Lifecycle, Break Even, Balance Sheets, game theory.	8
3	Marketing: Marketing Basics, Marketing Strategy and Brand Positioning, Plans and Execution Techniques, Marketing Analytics, Online Marketing, Product Life Cycle. Sales: Understanding Sales, Pitching Techniques, Sales	8

	strategies, Inside Sales v/s Outside Sales, RFP	
4	Operations Management and HR: Operational Basics, Process Analysis, Productivity, Quality Start-ups: Start-up Basics, Terms, Start-up Financing, Start-up Incubation, Getting Listed	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recognize different types of entrepreneurial ventures	K2
CO2	Interpret opportunity and risk analysis	K2
CO3	Summarize the strategies for valuing their own company, and how venture capitalist and angel investors use valuations in negotiating milestones, influence and control	K3
CO4	Determine correct marketing mix and how to position the company in the market by using analytical tools	K3
CO5	Explain how organizations operate and their process matrices	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3						3
CO2	3					3		3				3
CO3	3				2	2		3		2		3
CO4	3					2		3		2	3	3
CO5	3	2									3	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Transformational leadership: Value based management for Indian Organizations (Ed.).	Bhargava, S	New Delhi: Response-Sage	. (2003)
2	Overcoming the Unseen Forces That Stand in the Way of True Inspiration	Ed Catmull. Creativity, Inc.:		
3	The Startup Playbook: Secrets of the Fastest-Growing Start-ups from their Founding Entrepreneurs	David Kidder.		

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Entrepreneurship: Starting, developing, and managing a new enterprise	Hisrich, R. D. & Peters, M. P.	New York: McGraw-Hill.	(5th Ed.). (2001).
2	Small business and industry: A handbook for entrepreneurs.	Verma, J. C., & Singh, G.	New Delhi: Response-Sage.	(2002).
3	Financial Management: Theory and Practice,	Prasanna Chandra	Tata McGraw Hills,	6th Edn. (2004).

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I –IV	https://onlinecourses.nptel.ac.in/noc24_mg93/preview

SEMESTER S7

DISASTER PREPAREDNESS AND EMERGENCY PLANNING

Course Code	PESFT744	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand the types, causes, effects, and analysis of natural and man-made disasters.
2. Explore the principles and practices of disaster preparedness, response, and the role of various stakeholders.
3. Analyze the rehabilitation, reconstruction, and recovery processes after a disaster, including long-term planning.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction on Disaster, Different Types of Disaster : A) Natural Disaster: such as Flood, Cyclone, Earthquakes, Landslides etc B) Man-made Disaster: such as Fire, Industrial Pollution, Nuclear Disaster, Biological Disasters, Accidents (Air, Sea, Rail & Road), Structural failures (Building and Bridge), War & Terrorism etc. Causes, effects and practical examples for all disasters. Risk and Vulnerability Analysis, Risk : Its concept and analysis, Risk Reduction, Vulnerability : Its concept and analysis, Strategic Development for Vulnerability Reduction	9
2	Disaster Preparedness and Response, Disaster Preparedness: Concept and Nature, Disaster Preparedness Plan, Prediction, Early Warnings and Safety Measures of Disaster., Role of Information, Education, Communication, and Training, Role of Government, International and NGO Bodies. Role of IT in Disaster Preparedness, Role of Engineers on Disaster Management.,	9

3	Disaster Response: Introduction, Disaster Response Plan, Communication, Participation, and Activation of Emergency Preparedness Plan, Search, Rescue, Evacuation and Logistic Management Role of Government, International and NGO Bodies, Psychological Response and Management (Trauma, Stress, Rumor and Panic), Relief and Recovery, Medical Health Response to Different Disasters	9
4	Rehabilitation, Reconstruction and Recovery: Reconstruction and Rehabilitation as a Means of Development., Damage Assessment, Post Disaster effects and Remedial Measures., Creation of Long-term Job Opportunities and Livelihood Options, Disaster Resistant House Construction, Sanitation and Hygiene, Education and Awareness, Dealing with Victims' Psychology, Long-term Counter Disaster Planning, Role of Educational Institute	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Classify the different types of natural and man-made disasters, and analyze their causes, effects, and practical examples.	K3
CO2	Examine the concepts of risk, vulnerability, and strategies for risk reduction and vulnerability mitigation.	K2
CO3	Evaluate the principles and practices of disaster preparedness, response, and the role of various stakeholders, including the use of information technology.	K5
CO4	Assess the rehabilitation, reconstruction, and recovery processes after a disaster, including damage assessment, long-term planning, and the role of educational institutions.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2		2	2		2	2	3	2
CO2	3	2	2	2		3	2					
CO3	2			3	2	2	2	2				
CO4	3	2	3	2	2	3	2	2	3	2	3	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Disaster Management	Dr. Mrinalini Pandey	Wiley India Pvt. Ltd	-
2	Disaster Science and Management	Tushar Bhattacharya	McGraw Hill Education (India) Pvt. Ltd	-
3	Disaster Management : Future Challenges and Opportunities	Jagbir Singh	K W Publishers Pvt. Ltd	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Disaster Management	J. P. Singhal	Laxmi Publications.	-
2	Biodiversity, Environment and Disaster Management	Shailesh Shukla, Shamna Hussain	Unique Publications	-

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I	https://www.youtube.com/watch?v=sSX9u5yIK6o
II	https://www.youtube.com/watch?v=HnuCLqwAGYU
III-IV	https://www.youtube.com/watch?v=rhCbxPeBEdM

SEMESTER S7

TOTAL QUALITY MANAGEMENT

Course Code	PESFT746	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Recall statistical fundamentals and describe the basic concepts, principles and tools of TQM.
2. Exemplify the principles of TQM, the seven tools of SQC, the tools of TQM and Quality Systems.
3. Determine the hindrances for implementation of TQM principles, application of seven tools of SQC and the tools of TQM
4. Differentiate the costs of quality and outline the processes of development of a TQM organization

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Definition of Quality, Dimensions of Quality, Quality Planning, Quality costs - Analysis Techniques for Quality Costs, Basic concepts of Total Quality Management, Historical Review, Principles of TQM, Leadership – Concepts, Role of Senior Management, Quality Council, Quality Statements, Strategic Planning, Deming Philosophy, Barriers to TQM Implementation.	8
2	TQM Principles - Customer satisfaction – Customer Perception of Quality, Customer Complaints, Service Quality, Customer Retention, Employee Involvement – Motivation, Empowerment, Teams, Recognition and Reward, Performance Appraisal, Benefits, Continuous Process Improvement – Juran Trilogy, PDCA Cycle, 5S, Kaizen, Supplier Partnership – Partnering, sourcing, Supplier Selection, Supplier Rating, Relationship Development, Performance Measures – Basic Concepts, Strategy, Performance Measure.	10

3	Statistical quality control - The seven tools of quality, Statistical Fundamentals – Measures of central Tendency and Dispersion, Population and Sample, Normal Curve, Control Charts for variables and attributes, Process capability, Concept of six sigma, New seven Management tools.	9
4	TQM tools - Benchmarking – Reasons to Benchmark, Benchmarking Process, Quality Function Deployment (QFD) – House of Quality, QFD Process, Benefits, Taguchi Quality Loss Function, Total Productive Maintenance (TPM) – Concept, Improvement Needs. Need for ISO 9000 and Other Quality Systems, ISO 9000:2000 Quality System – Elements, Implementation of Quality System, Documentation, Quality Auditing	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recall statistical fundamentals and describe the basic concepts, principles and tools of TQM.	K2
CO2	Exemplify the principles of TQM, the seven tools of SQC, the tools of TQM and Quality Systems.	K3
CO3	Determine the hindrances for implementation of TQM principles, application of seven tools of SQC and the tools of TQM	K3
CO4	Differentiate the costs of quality and outline the processes of development of a TQM organization	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3							2				3
CO2	2					2			2			2
CO3	2					3						3
CO4	2				2		2	2			3	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Total Quality Management,	Dale H.Besterfiled Inc.	Pearson Education.	2003 (Indian reprint 2004).
2	Total Quality Management,.	James R.Evans & William M.Dale H.Besterfiled,	Pearson Education	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Management and Control of Quality,	Lidsay.,	South-Western (Thomson Learning).	(5th Edition) (2002).
2	Quality Management – Concepts and Tasks,	Narayana V. and Sreenivasan, N.S.	New Age International.	1996.
3	Total Quality Management for Engineers,	Zeiri,	Wood Head Publishers.	1991.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I -IV	https://onlinecourses.swayam2.ac.in/nou24_me09/preview

SEMESTER S7

SAFETY IN PRODUCT DESIGN

Course Code	PESFT747	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand the fundamentals and challenges of successful product development and design processes.
2. Learn product planning methods, customer needs identification, and techniques for concept generation and selection.
3. Apply tools like DFMEA/PFMEA, product architecture strategies, and industrial design principles to enhance product quality.
4. Develop skills in prototyping, and product development economics to optimize design efficiency and cost-effectiveness.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Characteristics of successful product development, design and development of products, duration, and cost of product development, the challenges of product development. Development Processes and Organizations: Generic development process, concept development: the frontend process, adopting the generic product development process, the AMF development process, product development organizations, the AMF organization.	8
2	Product planning: Product planning process, identify opportunities, evaluate and prioritize projects, allocate resources and plan timing, complete pre	8

	project planning, reflect all the results and the process Identifying customer needs: Gather raw data from customers, interpret raw data in terms of customer needs, organize the needs into a hierarchy, establish the relative importance of the needs and reflect on the results and the process. Concept Generation: Activities of concept generation, need for systems level thinking, TRIZ and its comparison with brainstorming and lateral thinking, TRIZ tools Ideality and IFR, problem formulation and functional analysis, use of 40 principles to solve contradiction, use of S-curves and technology evolution trends.	
3	Concept selection: Overview of methodology, concept screening, and concept scoring, Pugh matrix and its application. Concept testing: Define the purpose of concept test, choose a survey population, choose a survey format, communicate the concept, measure customer response, interpret the result, reflect on the results and the process, Failure Mode Effect Analysis (DFMEA and PFMEA). Product architecture: implications of the architecture, establishing the architecture, variety and supply chain considerations, platform planning, related system level design issues. Industrial design: Assessing the need for industrial design, the impact of industrial design, industrial design process, managing the industrial design process, assessing the quality of industrial design.	9
4	Design for X (DFX): Design for manufacturing: Definition, estimation of manufacturing cost, reducing the cost of components, assembly, supporting production, impact of DFM on other factors, design for assembly, service and quality. Prototyping: Prototyping basics, principles of prototyping, technologies, planning for prototypes Product development economics: Elements of economic analysis, base case financial mode, sensitive analysis, project trade-offs, influence of qualitative factors on project success, qualitative analysis.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the characteristics used for product design and development.	K3
CO2	Assess the customer requirements in product design.	K2
CO3	Evaluate and select product concepts using systematic approaches such as the Pugh Matrix, DFMEA, and PFMEA.	K2
CO4	Identify various aspects of design such as industrial design, design for manufacture, assembly, service and quality and product architecture.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										
CO2	2	3	2							3		2
CO3	2	3	3	2						2		3
CO4	2	3	2	3	2							3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Product Design and Manufacturing	A K Chitale and R C Gupta	PHI, New Delhi	6th Edition, 2003
2	Product Design and Development	Karl.T.Ulrich and Steven D Eppinger Irwin,	McGraw -Hill,	5th Edition, 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Design	George E Deiter,	McGraw-Hill	5th Edition, 2012
2	Product Design for Manufacture and Assembly,	Boothroyd G, Dewhurst P and Knight W,	Marcel Dekker, New York,	2nd Edition, 2002.
3	And Suddenly the Inventor Appeared: TRIZ, The theory of Inventive Problem Solving,	G Altshuller, H Altov, Lev Shulyak	Technical Innovation Centre,	2nd Edition, May 1996.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I	https://www.youtube.com/watch?v=5eBGRnBdy0o
II	https://www.youtube.com/watch?v=HD9V4NzUg7U
III-IV	https://www.youtube.com/watch?v=pGJ0ZZPbgZw

SEMESTER S7

HUMAN FACTORS ENGINEERING

Course Code	PESFT748	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Interpret human information processing ability and the parameters influencing it.
2. Exemplify information receptors and visual and auditory displays.
3. Calculate physical work load, and energy consumption for various activities.
4. Design workspace and arrange components in work space.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Human factors - objectives and approach. Systems thinking - human – machine systems, characteristics of systems, system reliability. Human beings as information processors- information theory, displaying information, coding of information, characteristics of good coding system, compatibility, types of compatibility, perception, memory, decision making, attention, age and information processing, mental workload and its measurement.	8
2	Process of seeing, visual capabilities, accommodation, visual acuity, contrast sensitivity, factors affecting visual acuity and contrast sensitivity, adaptation, colour discrimination, perception. Design of hard copy and VDT screens. Graphic representations – symbols, objectives and criteria for selection, perceptual principles of symbolic design. Codes – dimension, colour. Design of dynamic information displays, uses of dynamic information, design of quantitative visual displays, design of qualitative visual displays, design of signal and warning lights, recommendations regarding signal and warning lights, representational displays, head-up displays.	10

	Hearing, nature and measurement of sound, complex sound, anatomy of ear, conversion of sound waves to sensations, masking. Auditory displays, detection of signals, relative discrimination and absolute identification of auditory signals, sound localization, principles of auditory display, cutaneous senses, tactual displays, substitutes for hearing and seeing, olfactory senses and displays.	
3	Physical work - muscle physiology, work physiology, measures of physiological strain, physical work load, work efficiency, energy consumption, grades of work, factors affecting energy consumption, controlling energy expenditure, strength and endurance, measurement of strength, factors affecting strength. Manual materials handling – lifting tasks, carrying tasks, pushing tasks, limits of MMH tasks, reducing risks of MMH overexertion. Motor skills – biomechanics of human motion, types of body movements, range of movements, classes of motor movements, Speed of movements – reaction time, movement time, accuracy of movements. Human control of systems – compatibility, spatial compatibility, movement compatibility. Supervisory control. Controls devices – functions of control, factors in control design. Principles of hand tool and device design.	10
4	Workplace design – anthropometry, static dimensions, dynamic dimensions, principles in the application of anthropometric data. Work spaces - work-space envelopes for sitting and standing personnel, out-of-reach and clearance requirements. Design of work surfaces. Science of seating - general principles of seat design. VDT workstations. Arrangement of components within a physical space – principles of arranging components, methodologies for arranging components, types and uses of various data, link diagrams, general location of various controls and displays within work space, specific arrangements of controls and displays within work space, spacing of control devices. General guidelines in designing individual workplaces.	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Interpret human information processing ability and the parameters influencing it.	K2
CO2	Exemplify information receptors and visual and auditory displays.	K3
CO3	Calculate physical work load, and energy consumption for various activities.	K3
CO4	Design workspace and arrange components in work space.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3		2		3		3
CO2	3					3						3
CO3	3	2			2	3	2	2				3
CO4	3					3	2					3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human factors in engineering & design	Sanders, M.M. and McCormick, E.J.	McGraw-Hill, New York.	(Seventh edition). (1993).
2	A guide to ergonomics of manufacturing.	Martin Helander.	Tata McGraw Hill, New Delhi.	(1996).
3	Introduction to ergonomics.	Bridger, R.S.	CRC Press	(Third edition) (2008).

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to human factors and ergonomics for engineers	Mark R. Lehto and James R. Buck.	Lawrence Erlbaum Associates, Taylor & Francis Group, New York	.(2008).

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I-IV	https://onlinecourses.nptel.ac.in/noc24_mg108/preview

SEMESTER S7

SAFETY IN HYDROCARBON INDUSTRIES

Course Code	PESFT745	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Hazards and risks prevailing in petrochemical industries and emergency preparedness of petrochemical industries
2. Measure to improve health and safety in petrochemical industries statutory and regulatory requirements of petrochemical industries.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Simplified flow diagrams of a typical refinery – distillation unit, catalytic cracker, reformer, treating unit (hydro forming, gas purification, Sulphur recovery, lubricating oil unit) Simplified flow diagrams of Petrochemical Industry – steam cracking, butadiene extraction, ethane recovery, butyl rubber polymerization. Potential fire hazards in petroleum and petrochemical industries (ignition by local sources, spark, flame, hot surface, ignition of oil mists and fumes.). Storage tank farms of petroleum and petrochemical industries – Identification of Hazards, Type of Tanks, Design, Layout	9

2	Fire prevention measures including lightning protection. Fire protection arrangements in large tank farms, Design concepts of various fixed fire protection systems like Foam- Water Systems, Halogen & DCP systems. Lock out procedures. Salient features of codes / standards: NFPA, API, OISD and SHELL. Petroleum and Petroleum products – Fuels- Petroleum solvents – Lubricating oils – Petroleum wax, greases – Miscellaneous products	9
3	On and off shore oil operation – Construction of Installation – Pipe line Construction –Maintenance and repair activities – Safety and associated hazards Fire protection facilities in Oil Refineries, Depots & Terminals-Transportation of petroleum and petrochemical products (safety considerations, statutory considerations). Design and Construction requirements for cross country hydrocarbon pipelines. Liquefied Petroleum Gas (LPG) Bottling Plant Operations. Design Philosophies. Operating Practices- Safety and Fire Protection in bottling plants. Transportation of Bulk Petroleum Products. Storage and Handling of Bulk Liquefied Petroleum Gas.	9
4	Drilling oil – Technique and equipment - On- Shore and Off- shore drilling. Classification of wells. Drilling method. Rotary drilling. Drilling equipment. Ground and offshore structures for drilling. Offshore platforms and drilling vessels. Drilling mud – functions, classification and properties. Blow-off, well kicks, Blow out preventer. Shallow gas. Directional drilling. Well killing procedure. Emergency shutdown, Work position –Working condition – safety and associated hazards- lighting and its effects Petroleum Extraction and transport by sea – Oil field products – Operation – Transport of crude by sea – Crude oil hazards. Petroleum product storage and transport –Storage equipment –Precaution –Tank cleaning Methods of Rescue & Fire Fighting. Petroleum and Natural Gas (Safety in Offshore Operations) Rules, 2008.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Interpret the various processes employed in petroleum refining and petrochemical industries	K3
CO2	Estimate the firewater requirement for a petroleum refinery and petroleum depots	K3
CO3	Recognise the hazards involved in on-shore and off-shore drilling and state control measures	K2
CO4	Describe the various sources of ignition in process industries and the fire prevention systems employed	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											2
CO2	3				2							3
CO3	3					3	3	3		2		2
CO4	3						2				2	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Dryden's outlines of chemical technology for the 21st century.	Gopal Rao, M. and Sittig, M (Eds).	Affiliated East West Press, New Delhi.	(third edition). (2010).
2	Lee's loss prevention in the process industries.	Sam Mannan (Editor).	Butterworth-Heinemann Ltd., UK.	(Fourth edition). (2012).
3	Risk analysis for prevention of hazardous situations in petroleum and natural gas engineering.	Davorin Matanovic. Nedilika Gaurina– Medjimurec. and Katarina Simon. (Editors).	Engineering Science Reference, Hershey PA.	(2014).

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Risk management with applications from the offshore petroleum industry.	Aven, T. and Vinnem, J.E.	Springer-Verlag, U.K.	(2007)
2	Fire Protection Hand book	NFPA	NFPA	. (2008).
3	OISD Standards published by Oil Industry Safety Directorate.			

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I -IV	https://onlinecourses.nptel.ac.in/noc24_ch41/preview

SEMESTER S7

ADVANCED SAFETY ENGINEERING AND MANAGEMENT

Course Code	PESFT751	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. Describe various methods to improve the safety and reliability of safety systems
2. Solve simple problems related to layer of protection analysis (LOPA) by adding additional independent protection layers (IPL).
3. Explain the concept of behaviour-based-safety and its implementation
4. Interpret the various threats and vulnerability in process industries in order to devise strategies to counter attacks.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Domino incident investigation – technique, logic diagram, input requirements, output, example. Unavailability analysis of protective systems – technique, logic diagram, input requirements, example. Reliability analysis of automatic control systems - PES safety system development logic diagram system analysis, calculation of fractional dead time, application, strengths and weaknesses. Introduction to MORT analysis, Sneak analysis.	9
2	Layer of Protection Analysis (LOPA) – Overview of relevant standards and guidelines, risk tolerance criteria. Preparation of LOPA – LOPA methodology, the LOPA team. Scenario development – components, inherently safe considerations. Initiating causes / effects – identification,	10

	estimation of frequencies. Independent protection layers – IPL criteria, allocation of IPL credit – basic process control systems, operator response, pressure relief device, safety instrumented system, safety instrumented function. Safety integrity level (SIL) assignment, Interpreting LOPA results, risk decisions and making recommendations.	
3	Security for chemical process industries - Assessments and regulatory environment, methods for assessing security vulnerability, emerging security regulations, government development and industry activities that relate to security for process facilities. Strategies and counter measures – prevention of intentional releases and theft of chemical releases at process facilities. Site security for process industries – Essential elements – threat analysis, security counter measures, mitigation and emergency response. Specific security measures – information security, cyber security, physical security, policies and procedures, training, mitigation and response, inherently safer processes. Case study.	9
4	Behaviour- Based Safety (BBS) - Fundamentals of BBS, Observation and feedback process – people based safety, BBS experience, Checklist for critical behaviour, Outcomes of BBS work, Implementation problems in BBS, Praxis six cell analysis model. Behavioral safety Observation Process. Managers role in developing BBS culture. BBS steering committee. Main steps of True BBS approach. Safety Management Systems: SHEMS, OHSAS 18001 and OSHA's PSM – Policy, planning, training, implementation, management control and review.	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub-divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course, students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe various methods to improve the safety and reliability of safety systems	K2
CO2	Solve simple problems related to layer of protection analysis (LOPA) by adding additional independent protection layers (IPL).	K4
CO3	Explain the concept of behaviour-based-safety and its implementation	K2
CO4	Interpret the various threats and vulnerability in process industries in order to devise strategies to counter attacks.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3	3		2	3						2	2
CO3	3					2		3		3		2
CO4	3					3						2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	. The Behaviour – based safety process: Managing involvement for an injury-Free Culture	Thomas R. Krause	John Wiley & Sons.	. (Second edition). (1996).
2	Guidelines for Chemical Process Quantitative Risk Analysis,	Centre for Chemical Process Safety. AIChE:	Centre for Chemical Process Safety. AIChE:	Second edition. (2000).
3	Lee's Loss Prevention in the Process Industries.	Sam Mannan (Editor).	Butterworth-Heinemann Ltd., UK.	(Fourth edition). (2012).
4	Layer of Protection Analysis – Simplified Process Risk Assessment.	Centre for Chemical Process Safety, AIChE:	Centre for Chemical Process Safety, AIChE:	(2001).

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Site Security Guidelines for the U.S Chemical Industry .	ACC:	American Chemistry Council, Washington DC	(2001)
2	. Behaviour based safety in organisations – a practical guide.	Kaila, H.L	IK International, New Delhi.	(2010).

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc23_mg98/unit?unit=17&lesson=21
2	https://onlinecourses.nptel.ac.in/noc23_mg98/unit?unit=24&lesson=28
3-4	https://onlinecourses.nptel.ac.in/noc23_mg98/unit?unit=87&lesson=90

SEMESTER S7

FLUID POWER SAFETY

Course Code	PESFT752	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Define the laws pertaining to hydraulics & pneumatics and describe the elements of hydraulic & pneumatic systems.
2. Classify the different components of hydraulic & pneumatic systems: pumps, compressors, control valves and actuators.
3. Determine the different components required for a specified hydraulic & pneumatic application.
4. Draw circuits to illustrate the function of various components of hydraulic & pneumatic systems and outline simple circuits for specific applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Hydraulics- Pascal's Law- Conservation of energy- Pressure, Work and Power-Principles of Power Hydraulics, Pressure and Flow Measurements- Bernoulli's Principle- Hydraulic symbols- Advantages. Hydraulic fluids, Properties Piping and Seals- Reservoirs. Actuators- Cylinders, Rams, Hydraulic Motors. Pumps- Gear, Vane and Piston types- Fixed and variable flow. Testing of Actuators & Pumps- Safety Precautions	10
2	Directional Control- Check valve, Pilot – operated, Two- way and Four -way valves- Rotary valves. Pressure Control – Relief valves- Different functions.	8

	Volume control- Methods and Types. Testing of Control Valves and Safety precautions	
3	Pneumatic Systems: Introduction: Production of compressed air, Air Receives, Accumulators, Dry and oil free compressed air. Pneumatic control: Components, Types of Cylinders, Control Valves- Direction, Pressure and Flow, Air Motors and Pneumatic Symbols. Maintenance & Safety: Compressors & Accessories.	9
4	Accessories- Accumulators, Pressure Switches. Fluid Power Systems, Simple circuits- Hydraulic, Pneumatic, Hydro pneumatic and Electrohydraulic. System Maintenance and Safety.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

At the end of the course students should be able to:

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Oil hydraulic Systems– Principles & Maintenance	Majumdar.S.R.	Mcgraw Hill Education.	,(28th edition) .(2017)
2	Pneumatic Systems – Principles & Maintenance	Majumdar. S. R.	Mcgraw Hill Education.	(1st edition) .(2017)
3	Fluid Power with applications	Esposito, Anthony.	Pearson Education Asia	,(7th edition) (2008)

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	. Hydraulic Machines	Jagadish Lal	Metropoliton Books	, (1994)
2	Industrial Hydraulics.	J.Pippenger & T. Hicks.	McGraw Hill.	(1980).

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I -IV	https://www.youtube.com/watch?v=_V4qOEQFprU

SEMESTER S7

EXPLOSIVES TECHNOLOGY AND SAFETY

Course Code	PESFT753	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Explain the chemistry of explosives and the mechanisms of burning
2. Describe the concepts of shock, detonation and initiation
3. Interpret scaling in design and analysis of explosive devices
4. Summarize the importance of safety in explosives operations, storage and transportation

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Chemistry of Explosives – Chemical reactions – Categories of explosives by chemical type – Use forms of Explosives, Propellants, and Pyrotechnics. Mechanics of Burning – Burning model – Geometry shape of grains – Calculating the state of the gas – interior ballistics.	8
2	Sound, shock and Detonation – Sound waves – shock waves – Detonation waves – Explosive output tests. Initiation and initiators – Initiation theory and criteria – Initiation sensitivity testing – Non electric initiators – Hot-wire initiators – exploding bridge wire detonators – Slapper detonators.	10
3	Scaling in Design and Analysis – Geometric similarity – Accelerating metal with explosives – Shock waves in air – Shock waves in water – Craters from explosives – Conical – shaped chargers. Off-the - Shelf explosive Devices – Linear explosive products – Mechanical / explosive devices.	9

4	Classification, Transportation and Storage of Explosives – Explosives classification – Transportation of explosives – Storage of explosives. Explosive Facilities and Explosives Operations – Explosive facilities – Explosive operations – Good work practices – Maintenance – Explosive waste – Spills and general cleaning – Explosive handling – Testing and firing of explosives – Licenses, permits and penalties.	9
----------	---	----------

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the chemistry of explosives and the mechanisms of burning	K2
CO2	Describe the concepts of shock, detonation and initiation	K2
CO3	Interpret scaling in design and analysis of explosive devices	K3
CO4	Summarize the importance of safety in explosives operations, storage and transportation	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3						2					2
CO3	3		3									3
CO4	3					3	2	2		2		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to the Technology of Explosives.	Paul Cooper and Stanley R. Kurowski.	Wiley – VCH, New York.	(1996).

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemistry and Technology of Explosives, Vol. 1	T. Urbanski	Pergamon Press	1964

SEMESTER S7

SAFETY IN DESIGN OF STRUCTURES

Course Code	PESFT754	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Interpret the concepts of RC design and Safe design of RC flexural members
2. Safe Design RC columns
3. Explain the concepts of steel structural design and safe design of different types of joints in steel structures
4. Safe design structural steel tension and compression members and beams

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Concrete structures: Structural forms – loads – serviceability, strength and structural safety – design basis – design codes. Material properties: Properties in compression – tensile strength – reinforcing steels for concrete. Flexural design of R.C. beams: Bending of homogeneous beams – reinforced concrete beam behaviour – design for tension reinforcement in rectangular reinforced concrete beam – introduction to design aids – rectangular beams with tension and compression reinforcement.	8
2	Edge supported R.C. Slabs: Types of slabs – design of one-way slabs – temperature and shrinkage reinforcement. R.C. Columns: Design of short columns – axial compression – lateral ties and spirals – compression plus bending in rectangular columns – Code provisions for design of short columns.	8
3	Structural steel materials and specifications: Rolled steel sections- types of structural steel – specifications. Structural Fasteners: Types of riveted and bolted joints –strength of a riveted joint –design of riveted joints for axially	10

	loaded members – welded joints – advantages and disadvantages of welded joints – types of welds and their symbols – design of fillet weld – design of butt weld – design of plug and slot weld. Design of bolted joint.	
4	Structural steel tension member: Net sectional area – permissible stresses – design of axially loaded tension member – lug angle – tension splice. Compression member: Strength of an axially loaded compression member – effective length – maximum slenderness ratio. Beams: Design procedure for laterally supported and unsupported beams – built up beams – plate thickness – simple beam end connectors.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Interpret the concepts of RC design and Safe design of RC flexural members.	K2
CO2	Safe Design RC columns	K4
CO3	Explain the concepts of steel structural design and safe design of different types of joints in steel structures	K2
CO4	Safe design structural steel tension and compression members and beams	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3			2	2					2
CO2	3	3	3	2		3						3
CO3	3	2	3	2		3						3
CO4	3	3	3			2						3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design of Concrete Structures	Nilson, A. H.	McGraw Hill Companies Inc.	(14th ed.). (2009).
2	Design of Steel Structures I. K. International Publishing House. (2012).	Bhavikatti, S. S.		(3rd ed.).
3	Design of Steel Structures.	Ramamrutham, S.	Dhanpat Rai Publications.	(2004).

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Reinforced Concrete Design	Pillai, S. U. Menon, D.	Tata McGraw Hill, New Delhi.	(3rd ed.). (2009).
2	Limit State Design of Reinforced Concrete	Varghese, P. C.	Prentice Hall India, New Delhi.	(2nd ed.) (2002)
3	Design of Steel Structures.	Dayaratnam	S. Chand	, (1996).

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I -IV	https://onlinecourses.nptel.ac.in/noc24_ce99/preview https://onlinecourses.nptel.ac.in/noc24_me152/preview

SEMESTER S7

SAFETY IN TEXTILE INDUSTRY

Course Code	PESFT756	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. hazards, occupational diseases and control measures specific to textile industries
2. health and safety measures applicable for textile process industries

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to process flow charts of i) short staple spinning, ii) long staple spinning, iii) viscose rayon and synthetic fibre, manufacturer, iv) spun and filament yarn to fabric manufacture, v) jute spinning and jute fabric manufacture-accident hazard, guarding of machinery and safety precautions in opening, carding, combing, drawing, flyer frames and ring frames, doubles, rotor spinning, winding, warping, softening/spinning specific to jute.	10
2	Accident hazards i) sizing processes- cooking vessels, transports of size, hazards due to steam ii) Loom shed – shuttle looms and shuttleless looms iii) knitting machines iv) non-woven.	8
3	Scouring, bleaching, dyeing, punting, mechanical finishing operations and effluents in textile processes. Health hazards in textile industry related to dust, fly and noise generated-control measures relevant occupational diseases, personal protective equipment-health and welfare measures specific to textile industry, Special precautions for specific hazardous work environments.	10

4	Relevant provision of factories act and rules and other statues applicable to textile industry – Effluent treatment and waste disposal in textile industry.	8
----------	---	----------

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks \(8x3 =24marks\)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recognize the hazards present in textile processing activities	K2
CO2	Interpret the accident hazards, guarding of machinery and safety precautions for textile industry	K3
CO3	Identify the occupational hazards and associated diseases involved in textile industry	K2
CO4	Choose appropriate personal protective equipment	K3
CO5	Articulate and administer health and welfare measures specific to textile industry	K3
CO6	State relevant provisions of acts and rules applicable to textile industry	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											2
CO2	3					3		3		3		3
CO3	3											2
CO4	3											
CO5	3		2			3	3	3		3	2	3
CO6	3					3	3	3		2		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	“Hand book of textile testing and quality control”	Groover and Henry DS,		
2	Safety in Textile Industry”		Thane Belapur Industries Association, Mumbai.	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Quality tolerances for water for textile industry		”, BIS	
2	“A technology of textile processing”, Vol.I, Textile Fibres	Shenai, V.A.		
3	Water supplies and the treatment and disposal of effluent”	Little, A.H.,“		

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I -IV	https://onlinecourses.nptel.ac.in/noc24_te10/preview

SEMESTER S7

INTELLECTUAL PROPERTY RIGHTS

Course Code	PESFT757	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Recall important international Conventions on intellectual properties and understand their implications on IP laws
2. Describe the provisions of laws on patent, copyright, trademarks, designs, confidential information and semiconductor IC layout-design
3. Explain the legal provisions with case laws
4. Analyse the provisions and the rationale behind decisions in important cases and apply law to fictitious cases

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Concept of property vis-a-vis intellectual property – Overview of important international Conventions – Paris Convention 1883 – Berne Convention 1886 – WIPO – WIPO Copyright Treaty 1996 – Hague Agreement 1925 – TRIPS Agreement – WTO.	8
2	Patents Act 1970 – invention – inventive step – new invention – product patent – process patent – general conditions for conferring patent right – inventions not patentable – pipeline protection – procedure for obtaining patent – provisional and complete specification – priority dates – publication and examination – search for anticipation by previous publication and prior claim – Controller and his powers – opposition – grant of patents – term of patent – patents of addition – register of patents – lapse and restoration of patents – compulsory licence – revocation of patents – rights of patentee –	10

	infringement of patent right – remedies – Patent agent.	
3	Copyright Act 1957 – meaning – general principles – author of copy right – registration of copyright – term of copyright – Performer’s right – Assignment and licence – compulsory licence – infringement of copyright – remedies – moral right. Copyright establishments.	8
4	Trade Marks Act 1999 – meaning – registration – duration – collective mark – certification trade mark – infringement and remedies – assignment and transmission of trademarks – passing off. Design Act 2000 – meaning – author and proprietor of design – registration – copyright in registered designs – cancellation of registration – piracy. Confidential information. Semiconductor Integrated Circuits Layout-design Act 2000 – definitions – provisions of registration of certain layout designs – registration – opposition of registration – effect of registration – infringement – assignment and transmission – appeal.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recall important international Conventions on intellectual properties and understand their implications on IP laws	K2
CO2	Describe the provisions of laws on patent, copyright, trademarks, designs, confidential information and semiconductor IC layout-design	K2
CO3	Explain the legal provisions with case laws	K2
CO4	Analyse the provisions and the rationale behind decisions in important cases and apply law to fictitious cases	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3							2				3
CO2	2					2		2		2		3
CO3	2							2				3
CO4	3					3		3				3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Intellectual Property Laws Bare Acts,.		Universal Law Publishing Co	. (2014).
2	Principles of Intellectual Property	Gopalakrishnan, N. S. and Agitha, T. G.	Eastern Book Co.	(2nd ed.). (2014)

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Intellectual Property: Trademarks, copyrights, patents and trade secrets	Deborah, E. Bouchoux	Cengage.	(3rd ed.) (2012).

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I -IV	https://onlinecourses.swayam2.ac.in/cec24_lw13/preview

SEMESTER S7

FOOD SAFETY AND SANITATION

Course Code	PESFT758	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Recognise the food additives and food contaminants and their chemical and toxicological properties
2. Recognise the effects of pests on food and the various methods for controlling them
3. Explain the national and international regulations for biosafety.
4. Interpret the environmental, social and ethical implications of biotech applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Quality attributes of foods, size and shape, colour and gloss, texture – visual and objectively measurable attributes. Aroma of foods – Introductory ideas, formation and chemistry. Introduction to sensory evaluation of foods and beverages. Food safety, food additives and food contaminants, their chemical, technological and toxicological aspects, Food laws – development and enforcement. Prevention of Food Adulteration Act and Food Regulations. ISO 9000 series and HACCP. Codex Alimentarius protocols for export.	9
2	Principles of food commodity storage, Insect pests – their biology and food preference. Effects of pests on food commodities. Pesticide classification and chemistry. Pesticide formulations. Pesticide appliances. Insect growth regulators, bio pesticides and grain protectants. Fumigants, Sanitation in food processing / handling units. Ballooning techniques. Irradiation and	9

	other physical methods of control. Pesticide and health hazards. Safety devices, pesticide residues in foods, residue analysis and decontamination. Concept of organic foods.	
3	The legal and socio-economic impacts of biotechnology – Public education of the processes of biotechnology involved in generating new forms of life for informed decision making – Biosafety regulation and national and international guidelines. r-DNA guidelines – Challenges for the Indian biotechnological research and industries – Ethical implications of biotechnological products and techniques.	9
4	Experimental protocol approvals – Levels of containment – Environmental aspects of biotech applications – Use of genetically modified organisms and their resistance in environment – Special procedures for r-DNA based product production – Social and ethical implications of biological weapons – Good safety practices – GLP standards – Lab contaminants – PI, PII, PIII guidelines.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recognise the food additives and food contaminants and their chemical and toxicological properties	K2
CO2	Recognise the effects of pests on food and the various methods for controlling them	K2
CO3	Explain the national and international regulations for biosafety.	K2
CO4	Interpret the environmental, social and ethical implications of biotech applications	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	2									3		2
CO3	2					2	3	3		3		3
CO4	2					2	2	3		2		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Food Sanitation.	Norman G. Marriott and Robert B. Gravani.	. Springer.	5th edition (2006).
2	Biotechnology and Safety Assessment.	Thomas, J. A. and Fuchs, R.L.	Academic Press.	(3rd Edition). (2002).
3	Food Safety Handbook.	Ronald H. Schmidt and Gary E. Rodrick.	Wiley	1st edition. (2002).

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Bioethics and Biosafety.	Sateesh, M.K.	I.K. International Pvt Ltd., New Delhi.	(2008).

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I -IV	https://onlinecourses.swayam2.ac.in/aic20_ge07/preview https://onlinecourses.swayam2.ac.in/aic22_ge05/preview

SEMESTER S7

MARITIME SAFETY

Course Code	PESFT755	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide an in-depth understanding of maritime safety regulations, standards, and best practices.
2. To explore the technical and engineering aspects of safety systems, equipment, and procedures on board ships.
3. To critically analyze safety challenges, risk assessment techniques, and human factors in maritime operations.
4. To develop problem-solving skills related to designing and managing safety systems in compliance with international maritime standards.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Maritime Safety, Definition, importance, and scope of maritime safety. Historical development and evolution of maritime safety practices, Detailed study of IMO conventions: SOLAS, MARPOL, STCW, ISPS, and MLC. Major maritime disasters throughout history, Lessons learned from accidents, Overview of the International Safety Management (ISM) Code. The role of Flag States, Port State Control, and Classification Societies.	8
2	Introduction to FSS Code, Purpose of FSS code, General, International Shore Connections, Personnel Protection, Fire Extinguishers, Fixed gas fire extinguishing Systems, Fixed foam fire extinguishing systems, Low-location	8

	lighting systems, Fixed emergency fire pumps, Arrangement of means of escape , Fixed deck foam systems, Inert gas systems. Fire Test Procedures (FTP), Introduction to Fire Test Procedures, Purpose of FTP, Relevant SOLAS regulations and standards, Type approval, Types of fire tests, Non-combustibility test, Smoke and toxicity test, Test for "A", "B" and "F" class divisions, Test for fire door control systems, Test for surface flammability, Test for vertically supported textiles and films, Test for upholstered furniture, Test for bedding components, Test for fire-restricting materials for high-speed craft, Test for fire-resisting divisions of high-speed craft.	
3	Navigation and Collision Avoidance, Principles of Navigation, Celestial navigation, Electronic navigation systems (ECDIS, GPS, radar), Chartwork and publications, Collision Avoidance Rules, International Regulations for Preventing Collisions at Sea (COLREGs), Maneuvering characteristics of ships, Collision avoidance techniques, Search and Rescue (SAR) Operations, SAR principles and procedures, Maritime search patterns, SAR equipment and technology. Marine Structures and Stability, Ship Hydrostatics and Stability, Buoyancy, stability, and trim, Inclining experiment, Damage stability and survivability, Marine Materials and Corrosion, Properties and selection of marine materials, Corrosion mechanisms and prevention, Fatigue analysis and design, Offshore Structures, Types of offshore structures (fixed, floating, and subsea), Environmental loads and design considerations, Structural integrity and inspection.	10
4	Marine Pollution and Environmental Protection, Marine Pollution Sources and Types, Oil spills, chemical pollution, and solid waste, Marine pollution prevention and control, International Maritime Pollution Treaties, MARPOL Convention and its annexes, Port State Control, Environmental Impact Assessment (EIA), Principles and methodology of EIA, Environmental management systems (EMS). Maritime Security and Counter-Terrorism, Maritime Security Threats, Piracy, armed robbery, and terrorism , Maritime domain awareness, International Maritime Security Conventions, ISPS Code and its requirements, Maritime security measures, Counter-Terrorism Measures, Detection and prevention of terrorist acts, Security drills and exercises	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand key maritime safety conventions and regulations.	K3
CO2	Apply the FSS and FTP Codes for fire safety on ships.	K2
CO3	Use navigation principles and collision avoidance techniques.	K2
CO4	Assess marine pollution control and security measures.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2					2					
CO2	3	3	2	2			2					
CO3	2	3	3		2			2				
CO4	2	2		3		3	2	2				

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	SOLAS Consolidated Edition	International Maritime Organization (IMO)	International Maritime Organization (IMO)	2020
2	Handbook of Marine Surveying	Thomas Ask	The Nautical Institute	2017
3	The Maritime Engineering Reference Book: A Guide to Ship Design, Construction, and Operation	Anthony F. Molland	Butterworth-Heinemann	2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Maritime Safety	Paul A. Stott	Routledge	2020
2	Marine Fire Prevention, Firefighting, and Fire Safety	David L. Dittman	Schiffer Publishing Ltd.	2016
3	Marine Environmental Protection and Pollution Control	J. William B. Knox and Donna Fletcher Knox	Springer	2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I	https://www.youtube.com/watch?v=qODE59L-VfM
II	https://www.youtube.com/watch?v=nxRRrOsS5pE
III-IV	https://www.youtube.com/watch?v=OOnyzgH6UXY

SEMESTER S7

SUSTAINABILITY IN ENGINEERING

Course Code	OESFT721	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Explain sustainable development and different environmental agreements and protocols
2. Discuss real time activities causing environmental issues and different methods to use renewable energy resources
3. Explain local and global environmental issues
4. Differentiate between carbon emissions for regular and sustainable cities and explain different practices to move industries towards sustainability
5. Discuss different renewable energy resources and explain methods to implement green technology

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Sustainable Engineering- Sustainable development, concepts of sustainable development: three pillar model, egg of sustainability model, Atkisson's pyramid model, prism model, principles of sustainable development, sustainable engineering, threats for sustainability. Environmental Ethics and Legislations – Environmental ethics and education, multilateral environmental agreements and protocols, enforcement of environmental laws in India – The Water Act, The Air Act, The Environment Act.	9
2	Environmental Issues- Solid waste, impact of solid waste on natural resources, zero waste concept and three R concept, waste to energy technology: thermo-chemical conversion, biochemical conversion. Global Environmental Issues- Resource degradation: deterioration of water	8

	resources, land degradation, air pollution, climate change and global warming, ozone layer depletion, carbon footprint, carbon trading.	
3	Tools for Sustainability - Environmental management System (EMS), concept of ISO14000, life cycle assessment (LCA): basic components, advantages, disadvantages, case study. Environmental impact assessment (EIA), environmental auditing, bio mimicking, case studies. Sustainable Habitat - Concept of green building, green building materials, green building certification and rating: green rating for integrated habitat assessment (GRIHA), leadership in energy and environmental design (LEED) rating, energy efficient buildings, sustainable cities, sustainable transport, sustainable pavements, case studies in sustainability engineering: Green building, sustainable city, sustainable transport system. Sustainable Industrialization and Urbanization – Sustainable urbanization, industrialization, material selection, pollution prevention, industrial ecology, industrial symbiosis, poverty reduction.	10
4	Renewable energy resources- Conventional and non- conventional forms of energy, solar energy, fuel cells, wind energy, small hydro plants, biogas systems, biofuels, energy from ocean, geothermal energy, conservation of energy. Green technology and Green Business: Sustainable business, green technology, green energy, green construction, green transportation, green chemistry, green computing.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain sustainable development and different environmental agreements and protocols	K2
CO2	Discuss real time activities causing environmental issues and different methods to use renewable energy resources	K2
CO3	Explain local and global environmental issues	K2
CO4	Differentiate between carbon emissions for regular and sustainable cities and explain different practices to move industries towards sustainability	K3
CO5	Discuss different renewable energy resources and explain methods to implement green technology	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3	3	2		2		3
CO2	3					2	3			3		3
CO3	3					3	3			2		3
CO4	3	2			2	2	3	2				2
CO5	3				3	3	3			3		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Sustainable Engineering.	R. L. Rag and Lekshmi Dinachandran Remesh.	PHI Learning Pvt. Ltd.,.	2nd Edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Sustainability Engineering: Concepts, Design and Case Studies,	D. T. Allen and D. R. Shonnard.	Prentice Hall,.	1st Edition 2011
2	Engineering applications in sustainable design and development,	A.S. Bradley, A. O. Adebayo, P.Maria.	Cengage learning,	1st Edition, 2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I-IV	https://onlinecourses.nptel.ac.in/noc24_ce61/preview https://onlinecourses.nptel.ac.in/noc24_me144/preview https://onlinecourses.nptel.ac.in/noc24_ge49/preview https://onlinecourses.nptel.ac.in/noc24_mg118/preview

SEMESTER S7

SAFETY IN DESIGN OF STRUCTURES

Course Code	OESFT722	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Interpret the concepts of RC design and Safe design of RC flexural members
2. Safe Design RC columns
3. Explain the concepts of steel structural design and safe design of different types of joints in steel structures
4. Safe design structural steel tension and compression members and beams

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Concrete structures: Structural forms – loads – serviceability, strength and structural safety – design basis – design codes. Material properties: Properties in compression – tensile strength – reinforcing steels for concrete. Flexural design of R.C. beams: Bending of homogeneous beams – reinforced concrete beam behaviour – design for tension reinforcement in rectangular reinforced concrete beam – introduction to design aids – rectangular beams with tension and compression reinforcement.	8
2	Edge supported R.C. Slabs: Types of slabs – design of one-way slabs – temperature and shrinkage reinforcement. R.C. Columns: Design of short columns – axial compression – lateral ties and spirals – compression plus bending in rectangular columns – Code provisions for design of short columns.	8
3	Structural steel materials and specifications: Rolled steel sections- types of structural steel – specifications. Structural Fasteners: Types of riveted and bolted joints –strength of a riveted joint –design of riveted joints for axially	10

	loaded members – welded joints – advantages and disadvantages of welded joints – types of welds and their symbols – design of fillet weld – design of butt weld – design of plug and slot weld. Design of bolted joint.	
4	Structural steel tension member: Net sectional area – permissible stresses – design of axially loaded tension member – lug angle – tension splice. Compression member: Strength of an axially loaded compression member – effective length – maximum slenderness ratio. Beams: Design procedure for laterally supported and unsupported beams – built up beams – plate thickness – simple beam end connectors.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Interpret the concepts of RC design and Safe design of RC flexural members.	K2
CO2	Safe Design RC columns	K4
CO3	Explain the concepts of steel structural design and safe design of different types of joints in steel structures	K2
CO4	Safe design structural steel tension and compression members and beams	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3			2	2					2
CO2	3	3	3	2		3						3
CO3	3	2	3	2		3						3
CO4	3	3	3			2						3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design of Concrete Structures	Nilson, A. H.	McGraw Hill Companies Inc.	(14th ed.). (2009).
2	Design of Steel Structures I. K. International Publishing House. (2012).	Bhavikatti, S. S.		(3rd ed.).
3	Design of Steel Structures.	Ramamrutham, S.	Dhanpat Rai Publications.	(2004).

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Reinforced Concrete Design	Pillai, S. U. Menon, D.	Tata McGraw Hill, New Delhi.	(3rd ed.). (2009).
2	Limit State Design of Reinforced Concrete	Varghese, P. C.	Prentice Hall India, New Delhi.	(2nd ed.) (2002)
3	Design of Steel Structures.	Dayaratnam	S. Chand	, (1996).

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I-IV	https://onlinecourses.nptel.ac.in/noc24_ce99/preview https://onlinecourses.nptel.ac.in/noc24_me152/preview

SEMESTER S7

MANUFACTURING STRATEGY

Course Code	OESFT723	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand manufacturing and operations strategy concepts for world-class competitiveness.
2. Learn to develop and apply strategies for quality management and world-class manufacturing.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Manufacturing output, Operations Systems, Operations Strategy, Functional strategy within context of a firm, Functional dominance within corporate strategy, Concept of world class manufacturing organization, 6 Ps of Manufacturing, Skinners' view and Hayes and Wheelwright framework of Manufacturing Strategy, Alternative paradigm of manufacturing strategy, Some generic manufacturing strategies	8
2	Developing a manufacturing strategy, Understanding markets, The concept of order winners and qualifiers, Basic Characteristics and Specific Dimensions of Order Winners and Qualifiers. Developing an Operations Strategy: Methodology	8
3	Developing an Operations Strategy: Roth and Miller Classification, Enlightened View of Manufacturing, Manufacturing Strategy Taxonomy, Total Quality Management and Manufacturing Excellence, Deming's approach to Quality, Business Excellence Awards.	10
4	Critical success factors for World Class Manufacturing, Value Added	10

	Engineering, Total Employee Involvement, HR theories for Operations Strategy, Flexible Manufacturing system, Toyota production System, World Class Manufacturing and India	
--	--	--

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of manufacturing output, operations systems, and strategies, including world-class manufacturing concepts and frameworks.	K3
CO2	Analyze the process of developing manufacturing and operations strategies, focusing on order winners, qualifiers, and their role in market competitiveness.	K2
CO3	Gain knowledge of key models and classifications related to operations strategy, such as Roth and Miller's classification, Total Quality Management (TQM), and process choice.	K2
CO4	Explore the critical success factors of world-class manufacturing, including value-added engineering, total employee involvement, and systems like the Toyota Production System.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2		2	2		2	2	3	2
CO2	3	3	2	2		3	2		2	2	3	2
CO3	2	3	3	3	2	3	2	2	2	2	3	3
CO4	2	2	2	2	2	3	2	2	3	2	3	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Manufacturing Operations Strategy (2)	Alex Hill & Terry Hill Palgrave	Macmillan	-
2	Total Quality Management	Dale H Besterfield and others	Pearson	-

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Manufacturing- Missing Link in Corporate Strategy	W Skinner	Harvard Press,	1969
2	The Toyota Way	J K Liker	Tata McGraw Hill	-

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I	https://www.youtube.com/watch?v=sSX9u5yIK6o
II	https://www.youtube.com/watch?v=HnuCLqwAGYU
III-IV	https://www.youtube.com/watch?v=rhCbxPeBEdM

SEMESTER S7

TOTAL QUALITY MANAGEMENT

Course Code	OESFT724	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Recall statistical fundamentals and describe the basic concepts, principles and tools of TQM.
2. Exemplify the principles of TQM, the seven tools of SQC, the tools of TQM and Quality Systems.
3. Determine the hindrances for implementation of TQM principles, application of seven tools of SQC and the tools of TQM
4. Differentiate the costs of quality and outline the processes of development of a TQM organization

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Definition of Quality, Dimensions of Quality, Quality Planning, Quality costs - Analysis Techniques for Quality Costs, Basic concepts of Total Quality Management, Historical Review, Principles of TQM, Leadership – Concepts, Role of Senior Management, Quality Council, Quality Statements, Strategic Planning, Deming Philosophy, Barriers to TQM Implementation.	8
2	TQM Principles - Customer satisfaction – Customer Perception of Quality, Customer Complaints, Service Quality, Customer Retention, Employee Involvement – Motivation, Empowerment, Teams, Recognition and Reward, Performance Appraisal, Benefits, Continuous Process Improvement – Juran	10

	Trilogy, PDSA Cycle, 5S, Kaizen, Supplier Partnership – Partnering, sourcing, Supplier Selection, Supplier Rating, Relationship Development, Performance Measures – Basic Concepts, Strategy, Performance Measure.	
3	Statistical quality control - The seven tools of quality, Statistical Fundamentals – Measures of central Tendency and Dispersion, Population and Sample, Normal Curve, Control Charts for variables and attributes, Process capability, Concept of six sigma, New seven Management tools.	9
4	TQM tools - Benchmarking – Reasons to Benchmark, Benchmarking Process, Quality Function Deployment (QFD) – House of Quality, QFD Process, Benefits, Taguchi Quality Loss Function, Total Productive Maintenance (TPM) – Concept, Improvement Needs. Need for ISO 9000 and Other Quality Systems, ISO 9000:2000 Quality System – Elements, Implementation of Quality System, Documentation, Quality Auditing	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recall statistical fundamentals and describe the basic concepts, principles and tools of TQM.	K2
CO2	Exemplify the principles of TQM, the seven tools of SQC, the tools of TQM and Quality Systems.	K3
CO3	Determine the hindrances for implementation of TQM principles, application of seven tools of SQC and the tools of TQM	K3
CO4	Differentiate the costs of quality and outline the processes of development of a TQM organization	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3							2				3
CO2	2					2			2			2
CO3	2					3						3
CO4	2				2		2	2			3	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Total Quality Management,	Dale H.Besterfiled Inc.	Pearson Education.	2003 (Indian reprint 2004).
2	Total Quality Management,.	James R.Evans & William M.Dale H.Besterfiled,	Pearson Education	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Management and Control of Quality,	Lidsay.,	South-Western (Thomson Learning).	(5th Edition) (2002).
2	Quality Management – Concepts and Tasks,	Narayana V. and Sreenivasan, N.S.	New Age International.	1996.
3	Total Quality Management for Engineers,	Zeiri,	Wood Head Publishers.	1991.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I -IV	https://onlinecourses.swayam2.ac.in/nou24_me09/preview

SEMESTER S7

SAFETY IN ENGINEERING INDUSTRY

Course Code	OESFT725	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To study workplace hazards in manufacturing and engineering industry
2. To control hazards appropriately and use proper personal protective equipment.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	General safety rules, principles, maintenance, Inspections of turning machines, boring machines, milling machine, planning machine and grinding machines, CNC machines, Woodworking machinery, types, safety principles, electrical guards, work area, material handling, inspection, standards and codes- saws, types, hazards.	8
2	Design aspects of machine guarding, Guarding during maintenance, Zero Mechanical State(ZMS), Definition, Policy for ZMS – guarding of hazards - point of operation protective devices, machine guarding, types, fixed guard, interlock guard, automatic guard, trip guard, electron eye, positional control guard, fixed guard fencing- guard construction- guard opening. Selection and suitability: lathe-drilling-boring-milling-grinding-shaping-sawingshearingpresses- forge hammer-flywheels-shafts-couplings-gears-sprockets wheels and chains-pulleys and belts-authorized entry to hazardous installations-benefits of good guarding systems.	10

3	Gas welding and oxygen cutting, resistance welding, arc welding and cutting, common hazards, personal protective equipment, training, safety precautions in brazing, soldering and metalizing – explosive welding, selection, care and maintenance of the associated equipment and instruments – safety in generation, distribution and handling of industrial gases-colour coding – flashback arrestor – leak detection-pipe line safety-storage and handling of gas cylinders. Cold working, power presses, point of operation safe guarding, auxiliary mechanisms, feeding and cutting mechanism, hand or foot-operated presses, power press electric controls, power press set up and die removal, inspection and maintenance-metal sheers-press brakes.	9
4	Hot working safety in forging, hot rolling mill operation, safe guards in hot rolling mills – hot bending of pipes, hazards and control measures. Safety in gas furnace operation, cupola, crucibles, ovens, foundry health hazards, work environment, material handling in foundries, foundry production cleaning and finishing foundry processes. Heat treatment operations, electro plating, paint shops, sand and shot blasting, safety in inspection and testing, dynamic balancing, hydro testing, valves, boiler drums and headers, pressure vessels, air leak test, steam testing, safety in radiography, personal monitoring devices, radiation hazards, engineering and administrative controls, Indian Boilers Regulation	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify and interpret the hazards present in metal working machinery, wood working machinery, welding, gas cutting, hot and cold metal working operations	K2
CO2	Demonstrate safe practices in heat treatment operations.	K3
CO3	Evolve safe operating procedures in hazardous inspection processes.	K3
CO4	Select and use suitable personal protective equipment.	K3
CO5	Design appropriate guards for machines to protect humans from mechanical hazards.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					2		2				
CO2	2									3		2
CO3	2	2										3
CO4	2				2					2		3
CO5	3		3			2	2			3		

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Safety Management,	John V. Grimaldi and Rollin H. Simonds	All India Travelers Book seller, New Delhi,	1989
2	Safety in Industry	” N.V. Krishnan	Jaico Publishery House	, 1996.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	“Accident Prevention Manual”		NSC, Chicago	Third Edition 2008.
2	Occupational safety Manual		BHEL, Trichy,	1988.
3	Indian Boiler acts and Regulations,		Government of India.	
4	Safety in the use of wood working machines,		HMSO, UK	1992.
5	Health and Safety in welding and Allied processes,	Welding Institute, UK,	High Tech. Publishing Ltd., London,	1989.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I -IV	https://onlinecourses.nptel.ac.in/noc23_mg98/unit?unit=17&lesson=18

SEMESTER S7

RENEWABLE ENERGY ENGINEERING

Course Code	OESFT726	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory (OE)

Course Objectives:

1. The course is aimed at imparting basic knowledge on the different types of renewable energy resources and utilization of these energy effectively so as to minimize the consumption of non-renewable energy to a greater extend.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	The Energy Scenario- Commercial energy sources -World's production and reserves- India' Production and reserves. Solar Energy collectors: Solar thermal collectors -Flat plate collectors –Solar concentrators (parabolic trough, parabolic dish, Central Tower Collector) – Solar Air Heaters. Solar thermal electric power generation -Thermal Energy storage, sensible heat storage, latent heat storage , Thermo chemical storage , photovoltaic system for power generation , Solar pond -Solar Cells-Types of solar cells , principle of working and performance characteristics, Production process- Block diagram only Applications- Solar space heating and cooling of buildings, solar pumping, solar cooker, solar still, solar drier, solar refrigeration and air-conditioning, heliostat, solar furnace.	11
2	Wind Energy- classification of wind turbines and power performance curve, Energy in wind, calculation of energy content, Power coefficients, Betz limit theory, tip speed ratio, solidity of turbine' power control strategies, Basic	7

	principles of Wind Energy Conversion Systems (WECS), Classification of WECS, Parts of WECS	
3	Ocean Energy – Devices for Wave Energy conversion, Ocean Thermal Energy Conversion (OTEC): Principle of OTEC system, Methods of OTEC power generation – Open Cycle (Claude cycle), Closed Cycle (Anderson cycle) and Hybrid cycle (block diagram description of OTEC); Geothermal energy: Introduction, hot dry rock resources, magma resources, vapour and liquid dominated systems, binary cycle, advantages and disadvantages	10
4	Bio Mass Energy- Biomass conversion technologies –Bio Gasification, Bio ethanol, Bio Diesel, Biogas production from waste biomass, factors affecting biogas generation Bio Gas -KVIC and Janata model. Hydrogen Energy – various routes for production of Hydrogen energy.	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain solar energy collectors, storages, solar cell characteristics and applications	K2
CO2	Explain the different types of wind power machines and control strategies of wind turbines	K2
CO3	Explain the ocean energy and conversion devices and different Geothermal sources	K2
CO4	Explain biomass energy conversion devices. Calculate the Net Present value and payback period	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	-	-	-	-	-	-	-
CO2	3	2	-	-	-	-	-	-	-	-	-	-
CO3	3	2	-	-	-	-	-	-	-	-	-	-
CO4	3	2	-	-	-	-	-	-	-	-	-	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Solar Energy: Principles of Thermal Collection and Storage	S P Sukhatme , J K Nayak	Mc Graw Hill	2015
2	Fundamentals of renewable energy sources	Tiwari G N, Ghosal M K	Alpha Science International Ltd	2007
3	Sustainable Energy Choosing among options	Jefferson W Tester et.al	PHI	2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Renewable energy resources and emerging technologies	D.P. Kothari	Prentice Hall of India Pvt. Ltd	2011
2	Fundamentals and Applications of Renewable Energy	Mehmet KanoğluYunus A. Çengel John M. Cimbala	Mc Graw Hill	2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1 to 4	https://nptel.ac.in/courses/103103206

SEMESTER S7

MARITIME SAFETY

Course Code	OESFT727	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide an in-depth understanding of maritime safety regulations, standards, and best practices.
2. To explore the technical and engineering aspects of safety systems, equipment, and procedures on board ships.
3. To critically analyze safety challenges, risk assessment techniques, and human factors in maritime operations.
4. To develop problem-solving skills related to designing and managing safety systems in compliance with international maritime standards.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Maritime Safety, Definition, importance, and scope of maritime safety. Historical development and evolution of maritime safety practices, Detailed study of IMO conventions: SOLAS, MARPOL, STCW, ISPS, and MLC. Major maritime disasters throughout history, Lessons learned from accidents, Overview of the International Safety Management (ISM) Code. The role of Flag States, Port State Control, and Classification Societies.	8
2	Introduction to FSS Code, Purpose of FSS code, General, International Shore Connections, Personnel Protection, Fire Extinguishers, Fixed gas fire extinguishing Systems, Fixed foam fire extinguishing systems, Low-location lighting systems, Fixed emergency fire pumps, Arrangement of means of	8

	escape , Fixed deck foam systems, Inert gas systems. Fire Test Procedures (FTP), Introduction to Fire Test Procedures, Purpose of FTP, Relevant SOLAS regulations and standards, Type approval, Types of fire tests, Non-combustibility test, Smoke and toxicity test, Test for "A", "B" and "F" class divisions, Test for fire door control systems, Test for surface flammability, Test for vertically supported textiles and films, Test for upholstered furniture, Test for bedding components, Test for fire-restricting materials for high-speed craft, Test for fire-resisting divisions of high-speed craft.	
3	Navigation and Collision Avoidance, Principles of Navigation, Celestial navigation, Electronic navigation systems (ECDIS, GPS, radar), Chartwork and publications, Collision Avoidance Rules, International Regulations for Preventing Collisions at Sea (COLREGs), Maneuvering characteristics of ships, Collision avoidance techniques, Search and Rescue (SAR) Operations, SAR principles and procedures, Maritime search patterns, SAR equipment and technology. Marine Structures and Stability, Ship Hydrostatics and Stability, Buoyancy, stability, and trim, Inclining experiment, Damage stability and survivability, Marine Materials and Corrosion, Properties and selection of marine materials, Corrosion mechanisms and prevention, Fatigue analysis and design, Offshore Structures, Types of offshore structures (fixed, floating, and subsea), Environmental loads and design considerations, Structural integrity and inspection.	10
4	Marine Pollution and Environmental Protection, Marine Pollution Sources and Types, Oil spills, chemical pollution, and solid waste, Marine pollution prevention and control, International Maritime Pollution Treaties, MARPOL Convention and its annexes, Port State Control, Environmental Impact Assessment (EIA), Principles and methodology of EIA, Environmental management systems (EMS). Maritime Security and Counter-Terrorism, Maritime Security Threats, Piracy, armed robbery, and terrorism , Maritime domain awareness, International Maritime Security Conventions, ISPS Code and its requirements, Maritime security measures, Counter-Terrorism Measures, Detection and prevention of terrorist acts, Security drills and exercises	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand key maritime safety conventions and regulations.	K3
CO2	Apply the FSS and FTP Codes for fire safety on ships.	K2
CO3	Use navigation principles and collision avoidance techniques.	K2
CO4	Assess marine pollution control and security measures.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2					2					
CO2	3	3	2	2			2					
CO3	2	3	3		2			2				
CO4	2	2		3		3	2	2				

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	SOLAS Consolidated Edition	International Maritime Organization (IMO)	International Maritime Organization (IMO)	2020
2	Handbook of Marine Surveying	Thomas Ask	The Nautical Institute	2017
3	The Maritime Engineering Reference Book: A Guide to Ship Design, Construction, and Operation	Anthony F. Molland	Butterworth-Heinemann	2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Maritime Safety	Paul A. Stott	Routledge	2020
2	Marine Fire Prevention, Firefighting, and Fire Safety	David L. Dittman	Schiffer Publishing Ltd.	2016
3	Marine Environmental Protection and Pollution Control	J. William B. Knox and Donna Fletcher Knox	Springer	2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I	https://www.youtube.com/watch?v=qODc59L-VfM
II	https://www.youtube.com/watch?v=nxRRrOsS5pE
III-IV	https://www.youtube.com/watch?v=OOnyzgH6UXY

SEMESTER 8

SAFETY & FIRE ENGINEERING

SEMESTER S8

ENVIRONMENTAL ECONOMICS

Course Code	PESFT861	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3: 0: 0: 0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	PE

Course Objectives:

1. Explain the concepts of willingness to pay, public goods, property rights, market and nonmarket valuation techniques
2. Summarize the economics of pollution prevention.
3. Exemplify the economic instruments for environment protection
4. Interpret the optimal use of exhaustible resources

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Economics: Principles of Economics – Economics, Ecology and Ethics - Wealth, Welfare, Scarcity, Growth and Sustainability definitions – Concepts of Costs, Benefits, Opportunity costs , Social Costs –Marginal Costs and Marginal Benefits - Positive and Normative criteria for decision making - Consumer Choice theory –Supply and Demand– Economic Efficiency and Markets–Static and dynamic efficiency - market failures – property rights, externalities and environmental problems – Coase Theorem - Public Goods and Externalities - Free rider problem – Tragedy of the commons.	9
2	Environmental costs and benefits: Types of Economic value - Environmental Benefits and Environmental Costs - Valuing the Environment – Direct and indirect methods – Surrogate markets – Stated Preference and Revealed Preference methods- hedonic prices, travel cost models, contingent valuation, benefit transfer – economic valuation of ecosystem services- Assessment of Loss of Ecology - Valuation of Health impacts -	8

	Environmental accounting. Economics of pollution prevention: Economics of Environmental Quality- - Cost benefit analysis and Cost effectiveness analysis – Principles, methodology and Limitations – Discounting - Profitability of Pollution Prevention - Payback period – Present value estimation – Internal rate of return – Opportunity costs – Economic analysis of Pollution Prevention. Case studies.	
3	Economic instruments for environmental protection: Point vs. Nonpoint Sources - Stock vs. Fund Pollutants - Nature of Marginal abatement cost and that of Marginal damage cost -Efficient level of pollution, total cost of efficient level of pollution - Polluter pays Principle –Economic Optimum level of Pollution- Marginal Damage Functions – Marginal Abatement Costs - Allocation of Stock and Fund Pollutants - Economic analysis of Environmental Policy -Regulatory versus Economic Instruments – Decentralized Policies: Liability Laws, Property Rights, and Moral Suasion - Command-and-Control Strategies - Pigovian and Pollution Taxes – Incentive-Based Strategies: Emission Charges and Subsidies– Marketable permits – Emission trading – Non Compliance fees, bonds and deposit refunds –Evaluation of Instruments – Choice of instruments for Environmental policy.	10
4	Natural resource economics: Types, scarcity and classification of Natural Resources – Depletable and non renewable resources – Recyclable resources – Replenishable but depletable resources – Storable renewable resources – Renewable common property Resources– Economic Theory of Depletable Resources- Optimal Use of Exhaustible Resources- – Natural resources accounting - Economics of Forestry and fisheries exploitation –Trade and environment – Income Effects and Environmental Kuznets Curves – Race to the Bottom and Pollution Haven Hypothesis - Porter Hypothesis - Economics of Climate Change.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the concepts of willingness to pay, public goods, property rights, market and nonmarket valuation techniques	K2
CO2	Summarize the economics of pollution prevention.	K3
CO3	Exemplify the economic instruments for environment protection	K3
CO4	Interpret the optimal use of exhaustible resources	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3		2				2
CO2	3					2	3			2		3
CO3	3					2	3	2				3
CO4	3					2	3	2				3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Environmental Economics: An Introduction	Barry Field Martha Field	McGraw-Hill	
2	Environmental Economics	Kolstad Charles	Oxford University Press New York	(2011)
3	Environmental and Natural Resource Economics	Tom Tietenberg	Harper Collins College Publishers	5th Edition (2000)

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Environmental Economics – In theory and practice	Nick Hanley Jaison F. Shogren Ben White	Macmillan India Ltd, New Delhi	(1999)
2	Natural Resource and Environmental Economics	Perman R Y. Ma J. McGilvray M. Common	Pearson Education Harlow	3rd edition 2003
3	Environmental Economics for non-Economists – Techniques and policies for Sustainable Development	John Asafu Adjaye	World Scientific	2005

SEMESTER S8

RENEWABLE ENERGY ENGINEERING

Course Code	PESFT862	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. The course is aimed at imparting basic knowledge on the different types of renewable energy resources and utilization of these energy effectively so as to minimize the consumption of non-renewable energy to a greater extend.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	The Energy Scenario- Commercial energy sources -World's production and reserves- India' Production and reserves. Solar Energy collectors: Solar thermal collectors -Flat plate collectors –Solar concentrators (parabolic trough, parabolic dish, Central Tower Collector) – Solar Air Heaters. Solar thermal electric power generation -Thermal Energy storage, sensible heat storage, latent heat storage , Thermo chemical storage , photovoltaic system for power generation , Solar pond -Solar Cells-Types of solar cells , principle of working and performance characteristics, Production process- Block diagram only Applications- Solar space heating and cooling of buildings, solar pumping, solar cooker, solar still, solar drier, solar refrigeration and air-conditioning, heliostat, solar furnace.	11
2	Wind Energy- classification of wind turbines and power performance curve, Energy in wind, calculation of energy content, Power coefficients, Betz limit theory, tip speed ratio, solidity of turbine' power control strategies, Basic principles of Wind Energy Conversion Systems (WECS), Classification of WECS, Parts of WECS	7

3	Ocean Energy – Devices for Wave Energy conversion, Ocean Thermal Energy Conversion (OTEC): Principle of OTEC system, Methods of OTEC power generation – Open Cycle (Claude cycle), Closed Cycle (Anderson cycle) and Hybrid cycle (block diagram description of OTEC); Geothermal energy: Introduction, hot dry rock resources, magma resources, vapour and liquid dominated systems, binary cycle, advantages and disadvantages	10
4	Bio Mass Energy- Biomass conversion technologies –Bio Gasification, Bio ethanol, Bio Diesel, Biogas production from waste biomass, factors affecting biogas generation Bio Gas -KVIC and Janata model. Hydrogen Energy – various routes for production of Hydrogen energy.	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain solar energy collectors, storages, solar cell characteristics and applications	K2
CO2	Explain the different types of wind power machines and control strategies of wind turbines	K2
CO3	Explain the ocean energy and conversion devices and different Geothermal sources	K2
CO4	Explain biomass energy conversion devices. Calculate the Net Present value and payback period	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	-	-	-	-	-	-	-
CO2	3	2	-	-	-	-	-	-	-	-	-	-
CO3	3	2	-	-	-	-	-	-	-	-	-	-
CO4	3	2	-	-	-	-	-	-	-	-	-	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Solar Energy: Principles of Thermal Collection and Storage	S P Sukhatme , J K Nayak	Mc Graw Hill	2015
2	Fundamentals of renewable energy sources	Tiwari G N, Ghosal M K	Alpha Science International Ltd	2007
3	Sustainable Energy Choosing among options	Jefferson W Tester et.al	PHI	2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Renewable energy resources and emerging technologies	D.P. Kothari	Prentice Hall of India Pvt. Ltd	2011
2	Fundamentals and Applications of Renewable Energy	Mehmet KanoğluYunus A. Çengel John M. Cimbala	Mc Graw Hill	2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1 to 4	https://nptel.ac.in/courses/103103206

SEMESTER S8

PROJECT MANAGEMENT

Course Code	PESFT863	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Learn the basics of project management, key terms and definitions, and roles and responsibilities
2. Learn how to start, plan, and manage projects, including how to schedule tasks, manage time, and control costs
3. Learn how to evaluate and manage risk factors, and establish processes for managing uncertainty
4. Learn how to achieve project goals within constraints, and ensure customer satisfaction
5. Learn how to align project plans with an organization's business model and strategic goals

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Strategic Management and Project Selection - Project selection models, Project portfolio process, Analysis under uncertainty, Project organization, Matrix organization - Project Planning - Work Breakdown Structure, Systems integration, Interface coordination, Project life cycle, Conflict and negotiation.	9
2	Project Implementation - Estimating Project Budgets, Process of cost estimation, Scheduling: Network Techniques PERT and CPM, Risk analysis using simulation, CPM- crashing a project, Resource loading, levelling, and allocation.	9
3	Monitoring and Information Systems - Information needs and the reporting process, computerized PMIS, Earned value analysis, Planning-Monitoring-Controlling cycle, Project control: types of control processes, design of control systems, control of change and scope.	9

4	Project Auditing - Construction and use of audit report, Project audit life cycle, Essentials of audit and evaluation, Varieties of project termination, the termination process, Final Report, Project History.	9
----------	--	----------

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic terms of a project and how it is implemented	K2
CO2	Solve the intricacies of a project using Network Analysis	K3
CO3	Understand the project monitoring and controlling techniques	K2
CO4	Understand and Evaluate Audit reports and Final Reports for Projects	K2, K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					2		2				3
CO2	3	3			2				2			3
CO3	3					3		3			2	3
CO4	3							2				3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Project Management	R.Panneer selvam, P. Senthil Kumar	PHI	2010
2	Project Management A life cycle approach	Arun Kanada	PHI	2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Project Management – A Managerial Approach	Jack R. Meredith, and Samuel J. Mantel Jr.,	John Wiley and Sons,	2006
2	Project Management – A Systems Approach to Planning, Scheduling and Controlling	Harold Kerzner	John Wiley and Sons,	2006.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I-IV	https://onlinecourses.nptel.ac.in/noc24_mg75/preview
	https://onlinecourses.swayam2.ac.in/cec24_mg15/preview
	https://onlinecourses.nptel.ac.in/noc24_mg78/preview
	https://onlinecourses.nptel.ac.in/noc24_mg116/preview

SEMESTER S8

SAFETY IN ENGINEERING INDUSTRY

Course Code	PESFT864	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To study workplace hazards in manufacturing and engineering industry
2. To control hazards appropriately and use proper personal protective equipment.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	General safety rules, principles, maintenance, Inspections of turning machines, boring machines, milling machine, planning machine and grinding machines, CNC machines, Woodworking machinery, types, safety principles, electrical guards, work area, material handling, inspection, standards and codes- saws, types, hazards.	8
2	Design aspects of machine guarding, Guarding during maintenance, Zero Mechanical State(ZMS), Definition, Policy for ZMS – guarding of hazards - point of operation protective devices, machine guarding, types, fixed guard, interlock guard, automatic guard, trip guard, electron eye, positional control guard, fixed guard fencing- guard construction- guard opening. Selection and suitability: lathe-drilling-boring-milling-grinding-shaping-sawingshearingpresses- forge hammer-flywheels-shafts-couplings-gears-sprockets wheels and chains-pulleys and belts-authorized entry to hazardous installations-benefits of good guarding systems.	10

3	Gas welding and oxygen cutting, resistances welding, arc welding and cutting, common hazards, personal protective equipment, training, safety precautions in brazing, soldering and metalizing – explosive welding, selection, care and maintenance of the associated equipment and instruments – safety in generation, distribution and handling of industrial gases-colour coding – flashback arrestor – leak detection-pipe line safety-storage and handling of gas cylinders. Cold working, power presses, point of operation safe guarding, auxiliary mechanisms, feeding and cutting mechanism, hand or foot-operated presses, power press electric controls, power press set up and die removal, inspection and maintenance-metal sheers-press brakes.	9
4	Hot working safety in forging, hot rolling mill operation, safe guards in hot rolling mills – hot bending of pipes, hazards and control measures. Safety in gas furnace operation, cupola, crucibles, ovens, foundry health hazards, work environment, material handling in foundries, foundry production cleaning and finishing foundry processes. Heat treatment operations, electro plating, paint shops, sand and shot blasting, safety in inspection and testing, dynamic balancing, hydro testing, valves, boiler drums and headers, pressure vessels, air leak test, steam testing, safety in radiography, personal monitoring devices, radiation hazards, engineering and administrative controls, Indian Boilers Regulation	9

**Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify and interpret the hazards present in metal working machinery, wood working machinery, welding, gas cutting, hot and cold metal working operations	K2
CO2	Demonstrate safe practices in heat treatment operations.	K3
CO3	Evolve safe operating procedures in hazardous inspection processes.	K3
CO4	Select and use suitable personal protective equipment.	K3
CO5	Design appropriate guards for machines to protect humans from mechanical hazards.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					2		2				
CO2	2									3		2
CO3	2	2										3
CO4	2				2					2		3
CO5	3		3			2	2			3		

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Safety Management,	John V. Grimaldi and Rollin H. Simonds	All India Travelers Book seller, New Delhi,	1989
2	Safety in Industry	” N.V. Krishnan	Jaico Publishery House	, 1996.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	“Accident Prevention Manual”		NSC, Chicago	Third Edition 2008.
2	Occupational safety Manual		BHEL, Trichy,	1988.
3	Indian Boiler acts and Regulations,		Government of India.	
4	Safety in the use of wood working machines,		HMSO, UK	1992.
5	Health and Safety in welding and Allied processes,	Welding Institute, UK,	High Tech. Publishing Ltd., London,	1989.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I -IV	https://onlinecourses.nptel.ac.in/noc23_mg98/unit?unit=17&lesson=18

SEMESTER S8

MANUFACTURING STRATEGY

Course Code	PESFT866	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand manufacturing and operations strategy concepts for world-class competitiveness.
2. Learn to develop and apply strategies for quality management and world-class manufacturing.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Manufacturing output, Operations Systems, Operations Strategy, Functional strategy within context of a firm, Functional dominance within corporate strategy, Concept of world class manufacturing organization, 6 Ps of Manufacturing, Skinners' view and Hayes and Wheelwright framework of Manufacturing Strategy, Alternative paradigm of manufacturing strategy, Some generic manufacturing strategies	8
2	Developing a manufacturing strategy, Understanding markets, The concept of order winners and qualifiers, Basic Characteristics and Specific Dimensions of Order Winners and Qualifiers Developing an Operations Strategy: Methodology, Developing an Operations Strategy	8
3	Developing an Operations Strategy: Roth and Miller Classification, Enlightened View of Manufacturing, Manufacturing Strategy Taxonomy, Total Quality Management and Manufacturing Excellence, Deming's approach to Quality, Business Excellence Awards, Process Choice, Process Choice: 3 Dimensional View, Product Profiling	10
4	Critical success factors for World Class Manufacturing, Value Added	10

	Engineering, Total Employee Involvement, HR theories for Operations Strategy, Flexible Manufacturing system, Toyota production System, World Class Manufacturing and India	
--	--	--

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of manufacturing output, operations systems, and strategies, including world-class manufacturing concepts and frameworks.	K3
CO2	Analyze the process of developing manufacturing and operations strategies, focusing on order winners, qualifiers, and their role in market competitiveness.	K2
CO3	Gain knowledge of key models and classifications related to operations strategy, such as Roth and Miller's classification, Total Quality Management (TQM), and process choice.	K2
CO4	Explore the critical success factors of world-class manufacturing, including value-added engineering, total employee involvement, and systems like the Toyota Production System.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2		2	2		2	2	3	2
CO2	3	3	2	2		3	2		2	2	3	2
CO3	2	3	3	3	2	3	2	2	2	2	3	3
CO4	2	2	2	2	2	3	2	2	3	2	3	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Manufacturing Operations Strategy (2)	Alex Hill & Terry Hill Palgrave	Macmillan	-
2	Total Quality Management	Dale H Besterfield and others	Pearson	-

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Manufacturing- Missing Link in Corporate Strategy	W Skinner	Harvard Press,	1969
2	The Toyota Way	J K Liker	Tata McGraw Hill	-

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I	https://www.youtube.com/watch?v=sSX9u5yIK6o
II	https://www.youtube.com/watch?v=HnuCLqwAGYU
III-IV	https://www.youtube.com/watch?v=rhCbxPeBEeM

SEMESTER S8

SUPPLY CHAIN AND LOGISTICS MANAGEMENT

Course Code	PESFT867	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand the complexity and key issues in supply chain management.
2. Describe logistics networks, distribution planning, routing design and scheduling models.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Understanding the Supply chain, Achieving strategic fit in a supply chain. Supply chain drivers and metrics. Analysing and designing the supply chain network, factors affecting distribution network design, role of network design, models for designing regional network configuration. Impact of globalisation on supply chain networks.	8
2	Demand forecasting in supply chain, role of forecasting, components of forecasting, forecasting methods. Aggregate planning in supply chain, basic trade-offs in aggregate planning, linear programming in aggregate planning. Coordination in supply chain, impact of lack of coordination in the chain, obstacles to coordination, Managerial levers to improve coordination.	8
3	Managing economies of scale in a supply chain: Cycle inventory, Aggregating Multiple Products in a Single Order & Quantity Discounts. Managing uncertainty in a Supply chain – Safety inventory: Factors affecting the optimal level of product availability. Impact of supply uncertainty on safety inventory. Impact of aggregation on safety inventory. Factors affecting the optimal level of Product availability.	10

4	Logistics management and its components. Modes of transportation and their performance Characteristics & Transportation infrastructure policies. Design options for a transportation and logistics network. Trade-offs in transportation design, Tailored transportation. Role of sustainability in a supply chain. Sustainability connected supply chain drivers.	10
----------	--	-----------

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To analyse supply chains and design the supply chain network.	K4
CO2	To solve demand forecasting problems in the supply chain and enhance coordination in the network.	K4
CO3	To plan and manage inventories in the supply chain.	K4
CO4	To develop and plan transportation networks for supply chain considering sustainability also.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	1	2								
CO2	2	2		2						2		
CO3	2	2		2								
CO4	2	2	1	2								

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Supply Chain Management: Strategy, planning & Operation	Sunil Chopra and Dharam Vir Kalra	Pearson	7th edition, 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Designing and managing the supply chain concepts, strategies, and cases studies	David Simchi-Levi, Edith Simchi-Levi	McGraw Hill	4 th edition, 2022

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I	https://nptel.ac.in/courses/110106045
II	https://nptel.ac.in/courses/110106045
III	https://nptel.ac.in/courses/110106045
IV	https://nptel.ac.in/courses/110106045

SEMESTER S8

ECOLOGICAL ENGINEERING

Course Code	PESFT868	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Explain the concept connectedness of engineering industry and ecology
2. Summarize the environmental interactions during product use
3. Interpret the methodology of LCA
4. Exemplify metabolic and resource analysis

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Humanity and the Environment – The Ecology and Engineering Concept - Technological Change and Evolving Risk. The Relevance of Biological Ecology to Technology- The Status of Resources - Governments, Laws, and Economics	9
2	Industrial Product & Process Design and Operation - Choosing Materials - Designing for Energy Efficiency. Product Delivery -Environmental Interactions During Product Use - Design for End of Life.	9
3	An Introduction to Life-Cycle Assessment - The LCA Impact and Interpretation Stages - Streamlining the LCA Process. Using the Corporate Engineering Ecology Toolbox - Managing Ecological Engineering aspects in the Corporation- Indicators and Metrics - Services Technology and Environment	9
4	Ecology Management in Industrial Ecosystems - Metabolic and Resource Analyses - Systems, Analysis, Models and Scenario Development. Earth Systems Engineering and Management - The Future of Connectedness of Ecology and Engineering	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the concept connectedness of engineering industry and ecology	K2
CO2	Summarize the environmental interactions during product use	K3
CO3	Interpret the methodology of LCA	K3
CO4	Exemplify metabolic and resource analysis	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3	2	2				
CO2	2					2	3			3		2
CO3	2	3										3
CO4	2					2				2		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	. Industrial Ecology	Graedel, T.E. and Allenby, B. R	Prentice-Hall	(2nd edition). . (2003).

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Ecology and Sustainable Engineering.	Graedel, T.E. and Allenby, B. R.	Pearson	. (2010).

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I -IV	https://onlinecourses.nptel.ac.in/noc24_hs149/preview

SEMESTER S8

STATISTICAL AND COMPUTATIONAL METHODS

Course Code	PESFT865	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To equip the student with the expertise to computationally model real life problems and subsequently educate the student to solve these models with the help of the available statistical and numerical methods. The mathematical concepts and models deal with solving engineering problems using minimum available resources.
2. To apply classical, numerical and modern methods for solving engineering problems.
3. To get an awareness about the usefulness of numerical and statistical methods in solving different engineering problems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Motivation and Applications of statistical and numerical methods, Significant digits, Accuracy and precision; Truncation and round-off errors. Roots of nonlinear equations-Bisection method, Newton Raphson method Solution of system of linear equations-Gauss elimination, Gauss Jordan Gauss Siedel, LU decomposition. Numerical differentiation-Difference formula, Newton's forward, backward and divided difference method, Sterling's formula Numerical integration -Trapezoidal, Simpson's one-third rule, Simpson's three-eighth rule, Gauss Quadrature, Taylor's method, Euler method, Modified Euler method, Runge Kutta fourth order method for first order	9
2	Solving using Simplex method Big-M Method and Graphical method.Single variable optimization methods- Fibonacci search method. Multi-variable methods Hook-Jeeves pattern search method, Cauchy's (steepest descent)	9

	method. Introduction to network tree - Minimal Spanning Tree - Prim's Algorithm. Shortest path problems- Solution methods – Dijkstra's Method.	
3	Introduction to Fuzzy logic. Fuzzy sets and membership functions. Operations on Fuzzy sets. Optimization of Fuzzy Systems. Decision Making: Decision under certainty, risk and uncertainty: Decision trees, EMV method, EOL method, Maximin criterion, Minimax criterion. Sequencing Problem: Basic terminologies, Processing of n Jobs through 2, 3 and m machines.	9
4	Game Theory: Games with saddle points, Games without saddle points – 2 x 2 games, Graphical method for m x 2 and 2 x n games Queuing theory: Scope, terminology, classification Importance and applications, Performance Measures in Queuing Systems, Single-server exponential arrival and service times. Multi - server problems. Simulation: Monte Carlo simulation – Queuing simulation model- Generation of random numbers	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To apply decision theory under various conditions of certainty, risk, and uncertainty.	K3
CO2	To sequence and schedule jobs and projects	K3
CO3	Solve differentiation and integration numerically	K3
CO4	Apply different methods of Game Theory, Network Tree and Shortest Path.	K3
CO5	To solve problems using classical queuing theory models	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2		2		2						3
CO2	3	2										3
CO3	3	3		2								3
CO4	3	2			2							2
CO5	3	2										3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Numerical methods for Engineers	Steven C Chapra, Reymond P Canale	Mc Graw Hill	6th Edition, 2010
2	Numerical Methods for Engineers	Gupta S.K.	New Age International	1995
3	Numerical methods	E Balagurusamy	Mc Graw Hill Education	2017
4	Operations Research	H.A. Taha	Pearson	Eight Edition 2006
5	Operations Research	Paneerselvam R.	PHI	Third edition, 2023

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Operations Research	F. S. Hillier and G. J. Lieberman	McGraw Hill	Tenth edition, 2017
2	Discrete Event System Simulation	Banks, Carson, Nelson and Nicol	Pearson	Fifth edition, 2013
3	Applied Numerical Analysis	Gerald, C. F. and Wheatly P O	Wesley	6th Edition
4	Numerical Methods for Scientific and Engineering Computation	Jain, M. K., Iyengar, S. R. K. and Jain, R. K.	New Age Pvt. Pub, New Delhi	
5	Elementary Numerical Analysis	Conte, S. D. and De Boor, C.	Mc Graw Hill Publisher	
6	Applied Numerical Analysis	Krishnamurthy, E. V. & Sen, S. K.	East West Publication	
7	An introduction to numerical analysis	Suli & Mayers	Cambridge University Press	
8	Applied Numerical Analysis	Gerald, C. F. and Wheatly P O	Wesley	6th Edition
9	Numerical Methods for Scientific and Engineering Computation	Jain, M. K., Iyengar, S. R. K. and Jain, R. K.	New Age Pvt. Pub, New Delhi	
10	Operations Research	Kanti Swarup, P.K.Gupta and Man Mohan	Sultan Chand and Sons	20 th Revised Edition 2022
11	Operations Research – Principles and Practice	A. Ravindran, D. T. Phillips, J. J. Solberg	John Wiley and Sons.	2 nd Edition 2007

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I -IV	https://onlinecourses.swayam2.ac.in/nou24_hs38/preview https://onlinecourses.nptel.ac.in/noc24_cs107/preview https://onlinecourses.nptel.ac.in/noc24_cs106/preview

SEMESTER S8

ADVANCED SAFETY ENGINEERING AND MANAGEMENT

Course Code	OESFT831	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. Describe various methods to improve the safety and reliability of safety systems
2. Solve simple problems related to layer of protection analysis (LOPA) by adding additional independent protection layers (IPL).
3. Explain the concept of behaviour-based-safety and its implementation
4. Interpret the various threats and vulnerability in process industries in order to devise strategies to counter attacks.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Domino incident investigation – technique, logic diagram, input requirements, output, example. Unavailability analysis of protective systems – technique, logic diagram, input requirements, example. Reliability analysis of automatic control systems - PES safety system development logic diagram system analysis, calculation of fractional dead time, application, strengths and weaknesses. Introduction to MORT analysis, Sneak analysis.	9
2	Layer of Protection Analysis (LOPA) – Overview of relevant standards and guidelines, risk tolerance criteria. Preparation of LOPA – LOPA methodology, the LOPA team. Scenario development – components, inherently safe considerations. Initiating causes / effects – identification, estimation of frequencies. Independent protection layers – IPL criteria, allocation of IPL credit – basic process control systems, operator response, pressure relief device, safety instrumented system, safety instrumented	10

	function. Safety integrity level (SIL) assignment, Interpreting LOPA results, risk decisions and making recommendations.	
3	Security for chemical process industries - Assessments and regulatory environment, methods for assessing security vulnerability, emerging security regulations, government development and industry activities that relate to security for process facilities. Strategies and counter measures – prevention of intentional releases and theft of chemical releases at process facilities. Site security for process industries – Essential elements – threat analysis, security counter measures, mitigation and emergency response. Specific security measures – information security, cyber security, physical security, policies and procedures, training, mitigation and response, inherently safer processes. Case study.	9
4	Behaviour- Based Safety (BBS) - Fundamentals of BBS, Observation and feedback process – people based safety, BBS experience, Checklist for critical behaviour, Outcomes of BBS work, Implementation problems in BBS, Praxis six cell analysis model. Behavioral safety Observation Process. Managers role in developing BBS culture. BBS steering committee. Main steps of True BBS approach. Safety Management Systems: SHEMS, OHSAS 18001 and OSHA's PSM – Policy, planning, training, implementation, management control and review.	8

**Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub-divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course, students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe various methods to improve the safety and reliability of safety systems	K2
CO2	Solve simple problems related to layer of protection analysis (LOPA) by adding additional independent protection layers (IPL).	K4
CO3	Explain the concept of behaviour-based-safety and its implementation	K2
CO4	Interpret the various threats and vulnerability in process industries in order to devise strategies to counter attacks.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3	3		2	3						2	2
CO3	3					2		3		3		2
CO4	3					3						2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	. The Behaviour – based safety process: Managing involvement for an injury-Free Culture	Thomas R. Krause	John Wiley & Sons.	. (Second edition). (1996).
2	Guidelines for Chemical Process Quantitative Risk Analysis,	Centre for Chemical Process Safety. AIChE:	Centre for Chemical Process Safety. AIChE:	Second edition. (2000).
3	Lee's Loss Prevention in the Process Industries.	Sam Mannan (Editor).	Butterworth-Heinemann Ltd., UK.	(Fourth edition). (2012).
4	Layer of Protection Analysis – Simplified Process Risk Assessment.	Centre for Chemical Process Safety, AIChE:	Centre for Chemical Process Safety, AIChE:	(2001).

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Site Security Guidelines for the U.S Chemical Industry .	ACC:	American Chemistry Council, Washington DC	(2001)
2	. Behaviour based safety in organisations – a practical guide.	Kaila, H.L	IK International, New Delhi.	(2010).

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc23_mg98/unit?unit=17&lesson=21
2	https://onlinecourses.nptel.ac.in/noc23_mg98/unit?unit=24&lesson=28
3-4	https://onlinecourses.nptel.ac.in/noc23_mg98/unit?unit=87&lesson=90

SEMESTER S8

SUPPLY CHAIN MANAGEMENT

Course Code	OESFT832	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand the complexity and key issues in supply chain management.
2. Appreciate the design and network in supply chain management
3. To understand the role of coordination in supply chain management

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Understanding the Supply chain, Achieving strategic fit in a supply chain. Supply chain drivers and metrics. Analysing and designing the supply chain network, factors affecting distribution network design, role of network design, models for designing regional network configuration. Impact of globalization on supply chain networks.	8
2	Demand forecasting in supply chain, role of forecasting, components of forecasting, forecasting methods. Aggregate planning in supply chain, basic trade-offs in aggregate planning, linear programming in aggregate planning. Coordination in supply chain, impact of lack of coordination in the chain, Bullwhip effect, obstacles to coordination, Managerial levers to improve coordination.	8
3	Managing economies of scale in a supply chain: Cycle inventory, Aggregating Multiple Products in a Single Order & Quantity Discounts. Managing uncertainty in a Supply chain – Safety inventory: Factors affecting the optimal level of product availability. Impact of supply	10

	uncertainty on safety inventory. Impact of aggregation on safety inventory. Factors affecting the optimal level of Product availability.	
4	Transportation Networks and Sourcing- Role of transportation-modes and their performance – transportation infrastructure and policies - design options and their trade-offs –Tailored transportation. Sourcing – In-house or Outsource – 3rd and 4th PLs – supplier scoring and assessment.– building partnerships and trust –continuous replenishment and vendor-managed inventories – collaborative planning and replenishment.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To analyse supply chains and design the supply chain network.	K4
CO2	To solve demand forecasting problems in the supply chain and enhance coordination in the network.	K4
CO3	To plan and manage inventories in the supply chain.	K4
CO4	To develop and plan transportation networks for supply chain considering sustainability also.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	2								
CO2	2	2		2						2		
CO3	2	2		2								
CO4	2	2	2	2								

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Supply Chain Management: Strategy, planning & Operation	Sunil Chopra Dharam Vir Kalra	Pearson	7th edition 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Designing and managing the supply chain concepts, strategies, and cases studies	David Simchi-Levi Edith Simchi-Levi	McGraw Hill	4 th edition, 2022

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I	https://nptel.ac.in/courses/110106045
II	https://nptel.ac.in/courses/110106045
III	https://nptel.ac.in/courses/110106045
IV	https://nptel.ac.in/courses/110106045

SEMESTER S8

SAFETY IN MATERIAL HANDLING

Course Code	OESFT833	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand hazards involved in manual and mechanical material handling.
2. Expertise selection, testing, usage, inspection and maintenance of material handling equipment.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Preventing common injuries, lifting by hand, team lifting and carrying, handling specific shape machines and other heavy objects – accessories for manual handling, hand tools, jacks, hand trucks, dollies and wheel barrows – storage of specific materials - problems with hazardous materials, liquids, solids – storage and handling of cryogenic liquids - shipping and receiving, stock picking, dock boards, machine and tools, steel strapping and sacking, glass and nails, pitch and glue, boxes and cartons and car loading – personal protection – ergonomic considerations.	10
2	Fiber rope, types, strength and working load inspection, rope in use, rope in storage - wire rope, construction, design factors, deterioration causes, sheaves and drums, lubrication, overloading, rope fitting, inspection and replacement – slings, types, method of attachment, rated capacities, alloy chain slings, hooks and attachment, inspection.	8
3	Hoisting apparatus, types - cranes, types, design and construction, guards and limit devices, signals, operating rules, maintenance safety rules, inspection and inspection checklist – conveyors, precautions, types, applications.	8

4	Powered industrial trucks, requirements, operating principles, operators selection and training and performance test, inspection and maintenance, electric trucks, gasoline operated trucks, LPG trucks – power elevators, types of drives, hoist way and machine room emergency procedure, requirements for the handicapped, types- Escalator, safety devices and brakes, moving walks – man lifts, construction, brakes, inspection.	10
----------	--	-----------

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recognize the practical solutions to eliminate and/or minimize hazards in material handling	K2
CO2	Administer a crane and sling safety to operation.	K3
CO3	Disseminate the basic safety concepts and techniques in mechanical material handling.	K3
CO4	Understand the application of various types of industrial trucks	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3	2	2				3
CO2	3				2	3				2		3
CO3	3					3				2		3
CO4	3				2	3						3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Material Handling Equipment,.	Alexandrov, M.P.,	Mir Publishers, Moscow,	1981
2	Material Handling Equipments,	Rudenko N.,	Mir Publishers, Moscow,	1981.
3	“Conveyors and related Equipment”, Vol.I& II	Spivakosky,	Peace Pub. Moscow,	1982.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Accident Prevention Manual for Industrial Operations	NSC, Chicago,	NSC, Chicago,	Third Edition , 2008.
	Material Handling Hand Book - II	Reymond, A.Kulwice,	”, John Wiley and Sons, New York,	1985.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I-IV	https://www.safetyvideos.com/Materials_Handling_Safety_Training_Video_p/2.htm

SEMESTER S8

HAZARD CONTROL IN INDUSTRIES

Course Code	OESFT834	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. Describe the various manufacturing processes- hot & cold working of metals, metal cutting, welding & cutting, heat treatment methods, and material handling methods and the equipment and machineries employed in engineering industries.
2. Summarise the various hazards associated with different manufacturing processes, heat treatment methods and material handling methods.
3. Evaluate the various hazards, relate to the hazard control principles and determine the different hazard control measures.
4. Differentiate various hazards identified and suggested hazard control methods.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction - Classification of Engineering Industry –Manufacturing Processes - Hot Working-Foundry operations-furnace and equipment, health hazard, safe methods of operation. Forging operations, heat radiation, maintenance of machines, shop equipment and hand tools - safe work practice. Operations in hot and cold rolling mills.	9
2	Machinery safeguard-Point-of-Operation, Principle of machine guarding - breakdown of machine guarding - types of guards and devices. Cold Working-Safety in Power Presses, primary & secondary operations - shearing -bending - rolling – drawing. Metal Cutting- safety in turning, boring, milling, planning and grinding. Maintenance of machine tools - health hazards and prevention.	10

3	Welding and Cutting-Safety Precautions of Gas welding and Arc Welding, Cutting and Finishing. Gas Cylinders and Equipment's. Heat Treatment-Furnaces and Salt baths-operations and maintenance -safety in handling and storage of salts- disposal of effluents - health precautions, exposure to hazardous fumes, source of fumes, ventilation and fume protection.	9
4	Material Handling-Classification-safety consideration- manual and mechanical handling. Handling assessments and techniques- lifting, carrying, pulling, pushing, palletizing and stocking. Material Handling equipment-operation & maintenance. Maintenance of common elements-wire rope, chains slings, hooks, clamps.	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub-divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course, students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the various manufacturing processes- hot & cold working of metals, metal cutting, welding & cutting, heat treatment methods, and material handling methods and the equipment and machineries employed in engineering industries.	K1, K2
CO2	Summarise the various hazards associated with different manufacturing processes, heat treatment methods and material handling methods.	K2
CO3	Evaluate the various hazards, relate to the hazard control principles and determine the different hazard control measures.	K5
CO4	Differentiate various hazards identified and suggested hazard control methods.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					2			2	2		3
CO2	3					2			2	3		3
CO3	3					3	2		2	2		3
CO4	3					3			2	3		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Safety.	Ronald P. Blake.	Prentice Hall, New Delhi	(3rd edition) (2000)
2	National Safety Council	Chicago Accident prevention manual for industrial operations.		(1982)

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Health and Safety in Welding and Allied process	Balchin, N.C.	Jaico Publishers, New Delhi.	(2005).
2	. Manufacturing Engineering and Technology,	Kalpajian, S Schmid S.R	Pearson Education Asia	(Sixth edition)(2009)

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.swayam2.ac.in/nou24_ge81/unit?unit=10&lesson=46
2	https://onlinecourses.swayam2.ac.in/nou24_ge81/unit?unit=10&lesson=102
3-4	https://onlinecourses.swayam2.ac.in/nou24_ge81/unit?unit=10&lesson=47

SEMESTER S8

STRUCTURAL FIRE SAFETY

Course Code	OESFT835	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the behaviour of compartment fire at the pre-flashover and post-flashover stages.
2. To interpret the effect of fire on structural and non-structural materials used for construction.
3. To summarize the method of fire resistance test on structural and non-structural members, and the method of combustibility tests on building materials.
4. To calculate the fire resistance rating of a compartment and design it for a required fire resistance rating.
5. To determine the method of fire protection to structural members made of different materials and their repair methods if damaged due to fire.
6. To Implement the design concept of fire walls, fire screens, local barriers and fire doors in selecting them appropriately for a given fire scenario.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Compartment fire-factors controlling fire severity, ventilation controlled and fuel controlled fires. Pre-flashover compartment fire- growth period and definition of flashover, conditions necessary for flashover, fuel and ventilation conditions for flashover, factors affecting time to flashover, factors affecting growth. Post flashover compartment fire- regimes of burning, fully developed fire behaviour, temperatures achieved in fully developed fire, fire resistance and fire severity, methods of calculating fire resistance, projection of flames from burning compartments, spread of fire from compartments. Compartment temperature-time response at pre-flashover and post flashover periods; Equivalence of fire severity of compartment fire and furnace fire.	9
2	Effect of temperature on the properties of structural materials- Thermal data for concrete, steel, aluminium, masonry and wood; Mechanical properties of concrete, steel, aluminium, masonry and wood exposed to elevated	9

	temperatures(Material data); Constitutive stress-strain models for steel and concrete.; Fire protection of structural members made of steel, RCC and wood. Fire resistance test on structural elements-standard heating condition, types of furnaces, Indian standard test method, performance criteria, drawbacks to the fire resistance test. Approximate calculation of the fire resistance of a compartment; Method of arriving at the required fire resistance of structural members as per Indian Standard Specifications (BIS).	
3	Determination of combustibility of materials by fire tube method; Brief description on non-combustibility test as per BIS; Behaviour and performance of fabricated fire proof boards-Calcium silicate, Gypsum, Vermiculite, and Perlite boards. Behaviour of non-structural materials on fire- plastics, glass, textile fibers, geopolymers and other house hold materials. Classification of materials based on surface flame spread and their suitability. Reparability of fire damaged structures- Assessment of fire severity, Assessment of damage to concrete, steel, masonry and timber structures, Assessment of feasibility of repair; Repair techniques- repair methods to reinforced concrete Columns, beams and slabs, Repair to steel structural members, Repair to masonry structures.	9
4	Fire area- calculation of building fire area, subdivision of fire areas in Industrial, Residential and Public buildings; Fire separation between building-principle of calculation of safe distance. Design principles of fire resistant walls and ceilings; Fire resistant screens-solid screens and water curtains; Local barriers; Fire stopped areas-in roof, in partitioned fire areas, and in connecting structures; Fire doors- Low combustible, Non-combustible and Spark-proof doors; method of suspension of fire doors; Air-tight sealing of doors; Specification, test and performance criteria of Plate, Metal covered and rolling type fire doors as per BIS.	9

**Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the behaviour of compartment fire at the pre-flashover and post-flashover stages.	K2
CO2	Interpret the effect of fire on structural and non-structural materials used for construction.	K3
CO3	Summarize the method of fire resistance test on structural and non-structural members, and the method of combustibility tests on building materials.	K3
CO4	Calculate the fire resistance rating of a compartment and design it for a required fire resistance rating.	K3
CO5	Determine the method of fire protection to structural members made of different materials and their repair methods if damaged due to fire.	K2
CO6	Implement the design concept of fire walls, fire screens, local barriers and fire doors in selecting them appropriately for a given fire scenario.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3		2				2
CO2	3					3						2
CO3	3	3		2	2							3
CO4	3	2		2	2		2					2
CO5	3		2		2	2						2
CO6	3		3		2	2	2					3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design of buildings for fire safety..	Smith, E.E. and Harmathy, T.Z. (Editors),	ASTM Special Publication 685, American Society for Testing and Materials, Boston, U.S.A	(1979).
2	Designing of fire safety	Butcher, E. G. and Parnell, A. C.	. John Wiley and Sons Ltd., New York, U.S.A.	(1983).
3	Fire safety engineering design of structures	John A. Purkiss	Butterworth-Heinemann, Oxford, U.K	(2nd edn.) (2009)
4	Structural design for fire safety,	Andrew H Buchanan	John Wiley & Sons Ltd., England	. (2001).

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	An Introduction to Fire Dynamics,	Dougal Drysdale.	John Wiley & Sons Ltd., England.	(2nd edn.), (1999)
2	Principles of fire safety standards for building construction.	Roytman, M. Ya.	Amerind Publishing Co. Pvt. Ltd., New Delhi.	(1975)
3	Relevant Indian Standard Code of Practices(BIS)			

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I-IV	https://onlinecourses.swayam2.ac.in/nou24_ge81/preview https://onlinecourses.nptel.ac.in/noc24_mg110/preview

SEMESTER S8

DESIGN OF FIRE PROTECTION SYSTEMS

Course Code	OESFT836	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3: 0: 0: 0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	OE

Course Objectives:

1. To learn about the principle of fire extinguishment and the performance of extinguishing agent.
2. To understand the principle of different automatic fire detection systems and their application.
3. To learn about the different types of first aid and portable fire protection system; their operation and maintenance requirements.
4. To learn about the details and design principle of fixed firefighting systems with different extinguishers like water, CO₂, DCP and foam.
5. To learn about different types of fire alarm system and sprinkler systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fire protection Engineering - function of fire protection system, performance based fire protection system, prescriptive fire protection design. Principles of Fire Extinguishments - extinction of premixed flames, diffusion flames and burning metals, fire triangle, fire tetrahedron. Basic concept of fire fighting with water, carbon dioxide, powders, foams, inert gases halons. Need for halon replacement and halon substitutes, clean agent system design. Extinguishant performance - flame extinguishing concentration, inerting concentration, fire trials.	10
2	Fire detection - Need and importance of automatic fire detection system, principle of detection, classification of detectors. Heat detectors – fixed temperature, rate of rise, thermistor rate of rise and rate compensated type detectors. Smoke detectors - optical and ionization type, photo electric light scattering and light obstruction type detectors. Flame detectors - infra red and ultra violet detectors. Flammable gas detection - Pellistor and laser	8

	detectors. Testing of fire detection devices as per relevant Indian standards specifications. Comparison of detectors. Performance characteristics of detectors – fire sensitivity test. Lag time associated with fire detection. Design of detection system-design fire curve-plunge test-heat detection steady state and growing fires, Alpert equation, Power law fires. Smoke detection-modelling of smoke detector response, Radiant energy detection.	
3	First aid fire protection – fire bucket, sand bucket, fire blanket, fire pails & water barrels, hose reels. Description, working principle, method of operation of different types of portable fire extinguishers-water type, foam type and limitations, dry powder type, CO2 type, vapourizing liquid type. Placement of extinguishers. Extinguisher rating, Care, inspection, and maintenance of portable extinguishers as per relevant Indian standards specifications. Automatic water sprinkler system- risk categories, requirement and source of water supply, automatic pumps. Automatic sprinkler heads - Quartzoid type, fusible link type, and modern types. Mounting and protection of sprinkler heads. Sprinkler pipe works-standard and staggered lay out, hangers. Control valves for wet and dry installations. Deluge valve. Pre – action. Drenchers. High velocity and medium velocity spray system. Principles of water sprinkler system design as per relevant standards (ISI). Automatic fire sprinkler calculations-hydraulic calculations, area-density curve. Sprinkler design considerations for warehouse storage in a manufacturing plant. Fire protection of specialised occupancies-computer rooms-communication centre-airport.	10
4	: Fixed firefighting system using CO2, Dry chemical powder, and Foam - concept of total flooding and local application, advantages and disadvantages of each system. Basic system components. Design principles of fixed firefighting systems for total flooding and for local application as per relevant standards (ISI). Fire alarm system- classification of alarm system as per NBC. Manually operated system; Automatic alarm system. Component and features of Local system, Auxiliary system, Remote station system, Central station system and Proprietary system. False Alarm, Designing fire alarm audibility and visibility.	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the concept of fire extinguishment and able to choose the proper type of first aid and portable fire protection system appropriate to the given situation	K2
CO2	Understand the working principal of different types of fire detectors and will be able to select suitable fire detectors appropriate to the given situation	K2
CO3	Interpret the appropriate standards for the design, installation, inspection, testing and maintenance of fixed sprinkler systems as per IS specifications.	K3
CO4	Interpret the appropriate standards for the design, installation, inspection, testing and maintenance of fixed CO2, DCP and foam systems as per IS specifications	K3
CO5	Understand about the different types of fire alarm systems and about their requirements as per IS specifications.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					2						2
CO2	2						2			2		3
CO3	3		3				2					3
CO4	3		3				3					3
CO5	3					2						3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fire Safety in Buildings	Jain V.K.	New Age International (P) Ltd New Delhi	1996
2	Fire protection and prevention the essential handbook	Barendra Mohan Sen	UBS Publishers and Distributors Pvt. Ltd	2009
3	Principles of Fire Safety Engineering and Management- (Understanding Fire & Fire Protection)	A.K. Das		First edition 2014
4	Relevant IS Codes			

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Underdowns Practical Fire Precautions	Ron Hirst	Gower Publishing Company Ltd England	1989
2	SPFE handbook of fire protection Engineering	Morgan J. Hurley	Springer	5th Edition
3	N F P A - Fire Protection Hand Book			
4	Principles of Fire Safety Standards for Building Construction	M. Ya. Roytman	Amerind Publishing Co. Pvt. Ltd New Delhi	1975

SEMESTER S8

QUALITATIVE AND QUANTITATIVE APPROACH IN SAFETY

Course Code	OESFT837	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Develop Foundational Knowledge in Safety and Risk Assessment Techniques
2. Enhance Skills in Hazard Identification and Reliability Engineering
3. Foster Analytical and Quantitative Competence in Safety Engineering

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Safety domain terminologies, Hazard, Risk, Mishap, Incident, Hazard triangle, Risk Assessment & Control, Qualitative Perspective, Qualitative Perspective, Safety Engineering & Accident causing mechanisms, Safety quantification, Preliminary Hazard List, Preliminary Hazard Analysis, Hazard and operability study (HAZOP), Failure Modes and Effects Analysis (FMEA)- Identification of Failure Modes, Application of Hazard Identification Techniques, Fault Tree Analysis (FTA)- Construction, Gate by Gate method, Cut-set method, Importance measures.	8
2	Event Tree Analysis (ETA), Bowtie Tool, Common Cause Cut Sets, Cut-sets for Accident Scenarios, Identification of Safety Barriers, Risk Assessment, Consequence Assessment, Energy Control Model and Hazard Control Hierarchy, Safety Function Deployment, Ranking of Design Solutions: AHP approach, Quantification of Basic Events for Non-repairable Components, Hazard Rate, Exponential Distribution, Weibull Distribution.	8
3	Reliability engineering, Basic concepts – Evaluation of network Reliability / Unreliability – Series systems, Parallel systems, Series - Parallel systems, partially redundant systems – Types of redundancies - Evaluation of network Reliability / Unreliability using conditional probability method. Time Dependent Probability Basic concepts Failure, Fault, Error, Failure mode, Failure classification, Markovian models, Common cause and common	10

	mode failures, – Reliability functions $f(t)$, $F(t)$, $R(t)$, $h(t)$ – Relationship between these functions .	
4	Baths tubs curve – Exponential failure density and distribution functions - Expected value and standard deviation of Exponential distribution – Measures of reliability – MTTF, MTTR, MTBF – Evaluation of network reliability / Unreliability of simple Series, Parallel, Series-Parallel systems - Partially redundant systems - Evaluation of reliability measure – MTTF for series and parallel systems – Examples. Monte Carlo simulation, Application to uncertainty analysis, Simple integration, Trivial integration.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand Safety Domain Ontology and Risk Assessment Techniques	K2
CO2	Apply Hazard Identification and Safety Engineering Methods	K3
CO3	Conduct Event Tree Analysis and Use Bowtie Tool for Risk Management	K2
CO4	Implement Reliability Engineering Concepts and Network Reliability Evaluation	K3
CO5	Analyse Failure Data and Apply Reliability Measures	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											2
CO2	3											
CO3	3	3										
CO4	3	2										
CO5	3	3		2								

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	System Reliability Concepts by	V. Sankar,	Himalaya Publishing House,	2015.
2	Probabilistic Risk Assessment for Engineering and Scientists: A Systems Approach	Komamoto and Henley	IEEE Press,	1995
3	Techniques for safety management - A systems approach	Petersen D	ASSE	1998
4	Industrial Accident Prevention	Heinrich et al	McGraw Hill	1980

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Reliability Evaluation of Engineering Systems	Roy Billinton and Ronald N. Allan	B. S. Publications,	2007
2	Reliability Engineering	E. Balagurusamy,	Tata McGraw Hill,	2003.
3	Reliability and Maintainability Engineering	Charles E. Ebeling,	Tata McGraw Hill,	2000.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
I-II	https://onlinecourses.nptel.ac.in/noc23_mg98/unit?unit=32&lesson=33
III-IV	https://onlinecourses.nptel.ac.in/noc23_mg98/unit?unit=46&lesson=48